

ALGEBRA 1 BOOK

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Before beginning to learn the content in this book, you must have some basic prior knowledge of math. This includes how to add, how to subtract, how to multiply, how to divide, what the order of operations are (PEMDAS), what negative numbers are, how to do the four operations with them, what exponents are, what a decimal is and how to do the four operations with it, what a fraction is and how to do the four operations with it, and what a graph is. Once you can do all these things, you will be ready to begin learning Algebra.

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Chapter 0

Review

This section is named review because it is only for review. If you do not know how to add, subtract, multiply, and divide, this is not the place to learn it; this chapter does not go very in depth. This chapter reviews some basic things, but its main purpose is to reinforce things already learned and help find patterns.

0-1 The Operations

Before we can start learning about Algebra 1, we need to know some previous math. This is similar to how you cannot do Algebra 2 without knowing first Algebra 1. Section 1 of chapter 0 (0-1) will cover the basics of the four operations with various types of numbers.

The Four Operations (Figure 0.1)

	Add	Subtract	Multiply	Divide
What is it?	Combining two numbers to get another number.	Taking away a value of some number to get another number	Repeatedly adding a number to get a number	Seperating a number into equal parts to get a number.
Example	$2 + 5 = 7$ $5 + 2 = 7$	$7 - 5 = 2$ $5 - 2 = 3$	$2 \times 7 = 14$ $2(7) = 14$ $7 \cdot 2 = 14$	$14 \div 7 = 2$ $14 \div 2 = 7$
Notes	If you add two numbers, the order of which you do so doesn't matter. $2+5 = 5+2$ Addition equations can be rewritten as subtraction equations.	If you subtract two numbers, the order that you subtract the numbers matter $7-5 \neq 5-7$ Subtraction equations can be rewritten as addition equations.	$7+7=14$ or $2+2+2+2+2+2=14$ These two equations mean the same thing as the multiplication problem. Multiplication can also be expressed as a raised dot ($\cdot$) .	You can split 14 into 2 groups of 7. You can split 14 into 7 groups of 2. Division can also be expressed with a slash mark ($/$), fraction, or decimal. If you add, subtract, or multiply 2 whole numbers, you will get another whole number, but this isn't the case for division.

You should have your times tables memorized up to 12. If you do not, there is on the next page. It is very important to have this memorized and will make things a lot quicker when you do Algebra and math in general.

Multiplication tables up to 12 (Figure 0.2)

Factors	1	2	3	4	5	6	7	8	9	10	11	12
1	1	2	3	4	5	6	7	8	9	10	11	12
2	2	4	6	8	10	12	14	16	18	20	22	24
3	3	6	9	12	15	18	21	24	27	30	33	36
4	4	8	12	16	20	24	28	32	36	40	44	48
5	5	10	15	20	25	30	35	40	45	50	55	60
6	6	12	18	24	30	36	42	48	54	60	66	72
7	7	14	21	28	35	42	49	56	63	70	77	84
8	8	16	24	32	40	48	56	64	72	80	88	96
9	9	18	27	36	45	54	63	72	81	90	99	108
10	10	20	30	40	50	60	70	80	90	100	110	120
11	11	22	33	44	55	66	77	88	99	110	121	132
12	12	24	36	48	60	72	84	96	108	120	132	144

Note: Factors are numbers which are multiplied in an equation, 0 times anything is 0, and if you multiply any number by 1, the number you get is the same. The highlighted numbers are “square” numbers, or numbers that have a whole number square root.

There are a couple of other operations that you need to know about. Exponents are small numbers which are raised. Similar to how multiplication is repeated addition, exponents are repeated multiplication. The exponent tells how many times the number is multiplied by itself. Look at column 2 of the table below, figure 1.2 for more information on exponents. In other chapters you will learn more about properties of exponents

You will also need to know what a square root is. They are “the opposite” of exponents. The square root of a number is the number that, when multiplied by itself, equals the original number. You will learn about other types of roots in Chapter QW.

Exponents and square roots (Figure 0.3)

	Exponents	Square roots
What is it?	A small raised number which tells how many times a number needs to be multiplied by itself.	A symbol which asks for the square root of a number, or the number that when multiplied by itself (squared), will equal the original number.
Example	$3^3 = 27$	$\sqrt{9} = 3$ or -3
Note	3 to the third power is not equal to 3 times 3. 3 to the third power is equal to 27 because $3 \times 3 \times 3 = 27$. 3 has to be multiplied by itself 3 times.	9 divided by 2 is not equal to the square root of 9. The square root of a number squared equals the original number. In this case, 3 squared equals 9, because 3 times 3 equals 9.

Practice

1. $2 + 2 =$

2. $5 + 9 =$

3. $20 + 5 =$

4. $100 + 883 =$

5. $50 - 5 =$

6. $983 - 100 =$

7. $7 - 4 =$

8. $6 - 6 =$

9. $5 \times 5 =$

10. $2 \times 13 =$

11. $2 \times 27 =$

13. $12 \times 11 =$

14. $7 \times 9 =$

15. $10/2 =$

16. $15/5 =$

17. $42 / 6 =$

18. $100 \div 15 =$

19. $5555 / 5 =$

20. 0.5×21

21. $0.8 \times 0.7 =$

22. 100×0.001

0-2 The Order of Operations

You can use multiple different operations in a single equation. However, the order in which you do the operations is important. Take a look at this problem.

$6 + 3 - 4 \times 2$ is not equal to 10.

Why?

There is a rule when you use multiple operations in a single equation: you must do the the operations in a specific order. Even though $6 + 3$ is the first operation written in the equation, it is not the first thing you do. You may not necessarily solve all the operations in an equation from left to right. The order of operations can be remembered by using the acronym PEMDAS. This should be memorized.

P- Parenthesis

E- Exponents

M- Multiply

D- Divide

A- Add

S- Subtract

In an equation, the operations at the top (Parenthesis and Exponents) must be performed before the next 4. Parenthesis are "(" and ")", and any operation which is inside parentheses must be performed first.

Important!- The order which you multiply and divide do not matter so long as you do it after parenthesis and Exponents. This means that division can be done before multiplication or vice versa, but only if operations within parentheses or exponents have first been solved. The same is true for addition and subtraction; you can subtract before you add, or vice versa, but the other four operations must be done first.

Back to our original equation,

$$6 + 3 - 4 \times 2 = ?$$

We must first Multiply, since that comes before adding and subtracting in PEMDAS. Therefore, we will multiply 4×2 , which is 8, before doing anything else.

$$6 + 3 - 8 = ?$$

Now we only have addition and subtraction in our problem. It can be done in any order, but to make things simple I will do the equation from left to right. First off, $6 + 3 = 9$.

$$9 - 8 = ?$$

Lastly, we just have to subtract $9 - 8 = 1$. We have solved our problem. Now let's look at a problem with parenthesis.

$$(6 + 3 - 4)2 = ?$$

That this problem has the same numbers as the first one, but there are parenthesis. Most of the time numbers which are multiplied by something in parenthesis doesn't have a multiplication sign. Because of this, this problem is really just $(6 + 3 - 4) \times 2$. This problem is different, and the solution will be too. Normally we would multiply first, however anything in parenthesis must be done first. In this case, it is $6 + 3 - 4$, which is equal to 5.

$$5 \times 2 = 10$$

Now we just have 5 times 2 left, so our answer is 10. One more example:

$$(2 + 2^2)/2 = ?$$

First, we must do the operation in the parenthesis. However, the order which we do the operations in the parenthesis matters too. 2^2 must be done first because exponents comes before addition in the order of operations. $2^2 = 4$, and next we add the numbers in the parenthesis. $(2 + 4) = 6$.

$$6/2 = 3$$

Now we can divide, since we have solved all the operations in the parenthesis, and our final answer is 3.

Practice

1. $2 - 5 \times 6 =$

2. $10 \div 2 + 5 =$

3. $2^{-1} + 3 \times 4 / 2 =$

4. $(2 + 3) \times 11 =$

5. $2(5 + 3)/2 =$

6. $2 \times 5 \times 14 \div 10 =$

7. $15 + 24/2 - 23 =$

8. $1 + 2 \times 3 \div 3 =$

9. $1 \times 3 - 15/5 + 6 =$

10. $1 \times (10 - 2)/4 + 6 =$

11. $3^2 \times 3 + 1 =$

12. $[6(2 - 1)^3]/3 =$

13. $(2 \times 3^{1+2})/27 =$

14. $9 + 1 \times 2 =$

15. $10 \div 2 - 4 =$

0-3 Negative numbers

There are two types of numbers: Positive and negative numbers. Positive numbers are numbers which are greater than 0. Negative numbers are numbers which are less than zero. Zero is neither positive or negative. Negative numbers are written with a “-” symbol before the number. Look at the following problems.

$$6 - 0 = 6$$

$$6 - 1 = 5$$

$$6 - 2 = 4$$

$$6 - 3 = 3$$

$$6 - 4 = 2$$

$$6 - 5 = 1$$

$$6 - 6 = 0$$

$$6 - 7 = ?$$

The last problem of these is possible, and the answer is 1 less than 0. This is equal to negative 1, written as -1, because you are subtracting a number greater than the first number.

$$6 - 7 = 0 - 1 = -1$$

If you are getting a little confused by this, you can do this to help you.

$$6 - 7 = -(7 - 6) = -1$$

Another way to look at the problem is that when you have two numbers that are being subtracted, you can switch the order around and make the answer negative. This works because when you expand $-(7 - 6)$, you get $6 - 7$ (See chapter QW to learn more about expanding).

However, it is important not to just know what negative numbers are but how to do the four operations with them. Look at the table below (Figure 0.4) to see what rules there are when you do this.

The four operations with negative numbers (Figure 0.4)

	Addition	Subtraction	Multiplication	Division
Description	When you add a negative number, it is the same as subtracting a positive number.	When you subtract a negative number, it is the same as adding a positive number.	When you multiply an even number of negative numbers (even being a multiple of 2), the resulting number will be positive, and if you multiply an odd number of negative numbers, the resulting number will be negative.	When you divide 2 negative numbers, the answer will be positive, and if 2 numbers are divided, and one is positive and one is negative, the resulting number will be negative.
Examples	$2 + (-5) = 2 - 5 = -3$	$2 - (-5) = 2 + 5 = 7$	$3 \times (-5) = -15$ $(-3) \times (-5) = 15$	$(-10)/(-2) = 5$ $10/(-2) = -5$
Notes	Notice that the equations are equal to each other. Changing an equation to one that is easier to recognize can help you solve equations faster.	Notice that the equations are all equal. If you change an equation so it only has positive numbers, you will likely be able to solve it faster because you are more familiar with it.	The first equation has a negative result because one of the factors is negative, and 1 is an odd number, while the second equation has a positive result because 2 of the factors are negative, and 2 is an even number.	The first equation has a positive result because 2 negative numbers are being divided, while the second equation has a negative result, because only 1 number is negative.

This table, as many other things in this chapter, is important to remember. Let's try some problems.

$$\begin{aligned}
 &-3 + 2 \\
 &= 2 - 3 \\
 &= -(3 - 2) \\
 &= -1
 \end{aligned}$$

In the problem above, we changed the equation so it is easier to solve. However, if you can do $-3 + 2$ in your head, you can do that instead. Let's look at another one.

$$50 - 100 = -(100 - 50) = -50$$

Once you get used to this method, you will be able to solve many equations with negative numbers almost as fast as you could do equations without negative numbers.

Here is another problem:

$$33 + (-57) = 33 - 57 = -(57 - 33) = -24$$

Remember, adding a negative number is the same as subtracting a positive number, and subtracting a negative number is the same as adding a positive number. 2 negatives make a positive!

$$-(-5) = 5$$

Therefore, negative negative 5 is equal to positive 5, because there is an even number of negatives. Negative negative negative 5 is equal to negative 5.

Now let's do more problems with operations with negative numbers.

$$(-5) \times (-5) = 5 \times 5 = 25$$

Because there are 2, an even number, of negative numbers being multiplied, the answer will be positive. From this we can also learn that there are two square roots to a positive number, because a negative number squared is positive. The square roots of 25 are 5 and -5.

Finally, let's try a problem with division.

$$(-20) \div 5 = -(20 \div 5) = -4$$

This problem involves 2 numbers which are being divided, and one of them is negative, so the answer will be negative.

Practice

1. $6 - 5 =$

2. $5 - 5 =$

3. $5 - 6 =$

4. $(-3) \times 5 =$

5. $(-3) \times (-5)$

6. $-2 - 6 =$

7. $-6 - (-2) =$

8. $5 \times [-5 + (-4)] =$

9. $-12 / (-4) =$

10. $4 \times (-3) =$

11. $-11 - [-(-12)] =$

12. $- \{ - [-(-1)] \} =$

13. $- [(-35) / 7] =$

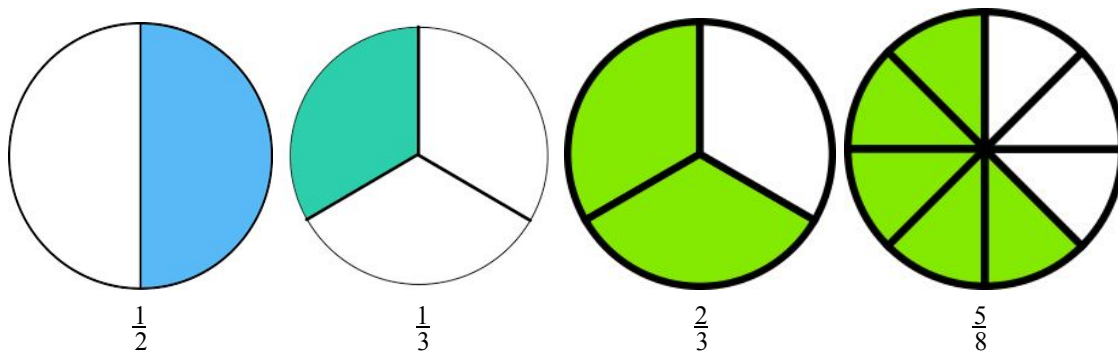
14. $-2 + 2 =$

15. $(-3) \times (-3) - 3 \times 3 =$

0-4 Fractions

So far, we have looked at two ways to write division, by using $\div$ or $/$. This section will teach you another way to write division, and how to do operations with them. An example of a fraction is $\frac{2}{3}$, which is equal to $2 \div 3$ and $0.6666\dots$ First off, there are 2 parts to a fraction: the numerator and denominator. The numerator is the number on the top of a fraction, while the denominator is on the bottom of the fraction. In the fraction $\frac{2}{3}$, the numerator is 2 and the denominator is 3. Fractions express part of a whole. Take a look at these circles.

Figure 0.5



In these circles, a portion is shaded. One way to look at fractions is that the denominator is how many equal sized pieces you cut, and the numerator is how many of those pieces you take from a circle. It helps, think of these circles as pizza. It may be easier to think that there are $\frac{5}{8}$ of a pizza left than 0.625 , the equivalent decimal. Fractions can also be represented by things other than circles, just the proportion must stay the same. Look at figure 0.6

It is also important to know that some fractions are equal to each other. For example, $\frac{2}{6} = \frac{1}{3}$.

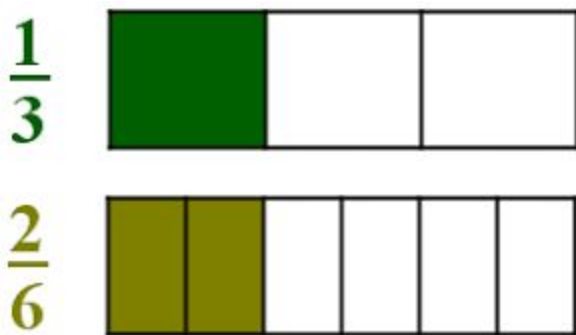


Figure 0.6

If you think about it, if you have to split a chocolate bar into 3 pieces and you take 1, you would get the same amount if a chocolate bar of the same size was split into 6 pieces, and then you took 2. This leads us into another very important concept with fractions. If you multiply or divide the numerator and denominator of a fraction by the same number, the fraction will have the same value. From this, you can prove that $\frac{1}{3} = \frac{2}{6}$. Using the rule above, $\frac{1}{3} = \frac{1 \times 2}{3 \times 2} = \frac{2}{6}$. Usually, it is actually better to "simplify" an

equation, or divide the top and bottom by a number, because it makes operations with fractions easier. This will be talked about more in the next section.

Fractions can be greater than 1. If the numerator is greater than the denominator and the denominator is greater than 0, then the fraction is greater than 1. $\frac{4}{3}$ or $\frac{909823}{7}$ are both greater than 1. There are two types of fractions: mixed fractions and improper fractions. Though you will generally only use improper fractions when doing operations, it is important to know what mixed fractions are and how to rewrite them into improper fractions. A mixed fraction is a whole number next to a fraction which is less than 1. An example is $3\frac{1}{5}$. This fraction is equal to $3 + \frac{1}{5}$. The other way to write a fraction greater than 1 (or less than -1) is an improper fraction, which are like the 2 examples shown earlier. The numerator is simply greater than the denominator, such as in $\frac{4}{3}$. To convert a mixed fraction into an improper fraction is by multiplying the denominator and the whole number, then adding the numerator. Therefore, $3\frac{1}{5}$ is equal to $\frac{16}{5}$ because 5, the denominator, times 3, the whole number, plus 1, the numerator, is equal to 16. We keep the original denominator the same, and 16 is the new numerator.. Let's get more practice with simplifying fractions and converting them.

$$\frac{105}{10} = \frac{105 \div 5}{10 \div 5} = \frac{21}{2}$$

In this problem, the top and bottom (numerator and denominator) were divided by 2 to get the fraction into the simplest form.

$$\frac{21}{7} = \frac{21 \div 7}{7 \div 7} = \frac{3}{1} = 3$$

In this problem, the top and bottom were divided by 7. We got $\frac{3}{1}$, and since any number divided by 1 stays the same, the final answer is 3.

Now for converting mixed fractions into improper fractions:

$$6\frac{3}{5} = \frac{5 \times 6 + 3}{5} = \frac{33}{5}$$

Finally, for a challenge, let's try changing an improper fraction into a mixed number.

$$\frac{19}{4} = \frac{19 \div 4}{4 \div 4} = (16 + 3) \div 4 = 4 + \frac{3}{4} = 4\frac{3}{4}$$

When we did $4 \div 4$, we got 1, we were left with $19 \div 4$, which is equal to 4 remainder 3, or $4 + \frac{3}{4}$, and so we get $4\frac{3}{4}$.

0-5 The operations with fractions

We can add, subtract, multiply, and divide with fractions, but it is very different from normal addition, subtraction, multiplication, and division. For example, adding and subtracting is usually easier than multiplying and dividing, but the opposite is usually true for fractions.

For example:

$$\frac{1}{2} + \frac{6}{5} \neq \frac{7}{7}$$

A common misconception for people who have never done operations with fractions is that you just add the numerators and add the denominators. That is just not true. The first thing you must do when you add and subtract fractions is make sure the fractions have the same denominator. In our example, our denominators are 2 and 5, which are not equal. What can we do to change this?

Back to our very important rule: if you multiply the top and bottom of an equation by the same number, the number stays the same. This is not just helpful for simplifying fractions, but also for adding and subtracting them.

We need to have the denominator of both numbers be the same. To make things easier, we want the lowest number that 2 and 5 is a factor of, which is 10. If you do not want to take the time to find the *lowest* number, just multiply the two numbers to find out which number should be the new denominator, though in most cases this is not the lowest possible number.

How do we change $\frac{1}{2}$ to have a denominator of 10, and how do we change $\frac{6}{5}$ to have a denominator of 10?

The answer is simple: we just need to multiply the top and bottom of $\frac{1}{2}$ by 5, and the top and bottom of $\frac{6}{5}$ by 2. By doing this, both fractions will get a denominator of 10 but the numbers are still equal to that of the original equation.

$$\frac{1 \times 5}{2 \times 5} + \frac{6 \times 2}{5 \times 2} = \frac{5}{10} + \frac{12}{10}$$

However, we are not done yet, though we've done the hard part. Now comes the easy part. Once you have 2 equations with the same denominator, you simply add the numerators and keep the denominator the same.

$$\frac{5}{10} + \frac{12}{10} = \frac{17}{10}$$

$\frac{17}{10}$ is our final answer. Watch out for answers that can be simplified, however. When we have an answer as a fraction, we want the lowest number in the numerator and denominator we can get while still keeping the number the same.

Next, subtraction. Subtraction is pretty much the same deal: we need to have the same denominators, and once we have done that we can subtract the numbers.

$$\frac{1}{2} - \frac{6}{5} = \frac{1 \times 5}{2 \times 5} - \frac{6 \times 2}{5 \times 2} = \frac{5}{10} - \frac{12}{10} = -\frac{7}{10}$$

As you can see, the process of subtraction and addition of fractions is very similar.

Multiplication is the easiest of the four operations with fractions. You simply multiply the numerators and multiply the denominators.

$$\frac{2}{3} \times \frac{4}{17} = \frac{8}{51}$$

No problems here.

Finally, division. Division is similar to multiplication, but you have to do one extra step. When you divide a number by a fraction, you multiply the 1st number by the reciprocal of the fraction. The reciprocal of a fraction is the fraction but you switch the numerator and denominator. Something important to know is that multiplying a fraction by its reciprocal is always 1, assuming the fraction doesn't have a denominator or numerator of 0. This will be helpful later on.

$$\frac{5}{4} \div \frac{5}{2} = \frac{5}{4} \times \frac{2}{5} = \frac{10}{20} = \frac{10 \div 10}{20 \div 10} = \frac{1}{2}$$

Notice that we had to simplify our answer at the end.

There are a couple other things that you need to know, too. If you have to do an operation with a fraction and a whole number, keep in mind that $4 = \frac{4}{1}$. If you ever run into this situation, just change the whole number to a fraction, and solve from there. This should make it easier.

The other thing you should know is something that will make multiplying 2 fractions easier. If the numerator of 1 fraction and the denominator of the other fraction are both divisible by a whole number besides 1, divide both numbers by that same number. you can do this to help make your multiplication easier. Here is an example where this really helps.

$$\frac{1000}{28} \times \frac{7}{500} = ?$$

If we just multiplied this out, it would take a while to simplify. With our new trick, here is what we can do. Notice that the numerator of the first fraction, 1000, and the denominator of the second fraction, 500, both are divisible by 500. Since both are, we can divide both 1000 and 500 by 500 and get 2 and 1. We can also divide 28 and 7 by 7, since 7 is a numerator, 28 is the denominator of the other fraction, and both are divisible by 7.

$$\frac{1000}{28} \times \frac{7}{500} = \frac{1000 \div 500}{28 \div 7} \times \frac{7 \div 7}{500 \div 500} = \frac{2}{4} \times \frac{1}{1} = \frac{2 \div 2}{4 \div 2} = \frac{1}{2}$$

This trick just helped us us solve this problem a whole lot faster.

Chapter 0 test

Solve the following problems.

1. $\sqrt{25} =$

2. $(-1)^3 =$

3. $7 + \frac{5}{2} \cdot -\frac{2}{5} =$

4. $(\sqrt{1/10} + 3/5)^2 =$

5. $5/12 - 7/4 =$

6. $18/7 \div 2 =$

7. $\frac{35}{21} =$

8. $4/(2)^4 - 21/8 =$

9. $5 \div 4 \times (-4) =$

10. $1/5 + 2/5 =$

11. $\frac{13}{6} - \frac{9}{3} =$

12. $(-1)^{10} =$

13. $5 \times 5 =$

14. $12 \times 8 \div 4 =$

15. $1/3 - 1/2 =$

16. $\frac{3 \cdot (4 + 1)}{4 + 1} =$

17. $(-2)(-8) =$

18. $\frac{12 \times 999}{999} =$

19. $1 + 2 \times 3 \div 4 - 5 \times 6 - 7 =$

20. $11 \times 11 \times \sqrt{9} =$

21. $-(\frac{9}{5} \div 5) =$

Chapter 1

Introduction to equations and inequalities

This chapter starts talking about variables and solving for them in equation, which is one of the main things in Algebra. This section will make sure that you understand what variables are, how to solve for them, and how to solve basic equations with them.

1-1 What are variables? Solving one step equations with addition and subtraction

Variable is a scary sounding word at first. In this section, we'll talk about what variables are.

Variables look like normal letters, such as a, b, c, x, y, z, etc. The most common ones are x, y, and z, and occasionally you will get a, b, and c. x and y are the most common because they are the axis of a graph (see chapter 2). You can assign any letter to a variable. First off, a variable is a number. That is essential to understand. While it looks like a letter, it is a number.

Here is a problem to help understand what the purpose of a variable is.

$$\square + 5 = 10$$

What goes in the blank? Something plus 5 equals 10. You should be able to figure out quickly that 5 goes in the blank. Understand that the blank is just a placeholder for a number which is unknown. This is basically the same thing as a variable. Now let's solve a world problem.

Greg has 50 water bottle total. He bought 32 water bottle on Wednesday. The rest of the water bottles were bought the day before on Tuesday. How many water bottles did Greg buy on Tuesday?

To solve this equation, we can make this:

$$\square + 32 = 50$$

How can we figure out how to solve this? I will give you a very important rule in algebra: If you add, subtract, multiply, or divide both sides of an equation by the same number, the equation is equal. For this equation, we can subtract both sides by 50 to find out what the unknown number is.

$$\square + 32 - 32 = 50 - 32$$

$$\square = 18$$

$$18 + 32 = 50$$

We ended with the right answer: the unknown value is 18. Notice how we subtracted 32 from both sides. Now, let's try the exact same problem but with a variable instead of a blank. When we plug the right answer into the original problem, we can also prove that this is the right answer. I will use x as the variable.

$$x + 32 = 50$$

Remember, x is just a number! Just like how the blank is a number, this is too. To figure out what x is, we have to do same thing: subtract both sides by 32.

$$x + 32 - 32 = 50 - 32$$

$$x = 18$$

Again, it is important to understand that x is a number, and in this specific case it equals 18. We are going to try a few more simple problems to make this clear.

$$a + 7 = 9$$

Something you need to remember is that when you solve an equation for a variable, which in this case is a , we must always isolate the variable by itself on one side of the equation. What this means is that if we must do whatever possible to have one side of the equation just have the variable. Look at the equation above. The Variable a is not by itself; 7 is also on the same side of the equation. To get rid of the 7, we must subtract both sides of the equation by 7.

$$\begin{aligned} a + 7 - 7 &= 9 - 7 \\ a &= 2 \end{aligned}$$

We can make things quicker by, instead of thinking about solving this by adding both sides or subtracting both sides by the same number, we can just move the number on the other side of the equation and make it the opposite sign. This does the same thing, but it is faster. This is called “transposing”, and it will make solving equations a lot faster.

We subtracted 7 on both sides of the equation because if we had subtracted any other number than 7, we would not have gotten rid of the 7 on the left side of the equation. For example, if we subtracted 5 from both sides of the equation, we would have gotten $a + 2 = 4$. This would not have gotten rid of the 7 completely. Likewise, if we subtracted 8 from both sides of the equation, we would have gotten $a - 1 = 1$. That is why we subtracted 7 from both sides. Let’s try another one.

$$y - 12 = -1$$

Think in your head, “how can I change this equation so that it is still equal but only y is on the left side of the equation?” We currently have y and -12 on the left side, so we need to get rid of that 12. To do this, we need to add 12 to both sides of the equation, because $-12 + 12 = 0$. Or, transpose, so just move the 12 to the other side, and make the sign the opposite, positive.

$$\begin{aligned} y - 12 + 12 &= -1 + 12 \\ y &= 11 \end{aligned}$$

$$\begin{aligned} 11 - 12 &= -1 \\ -1 &= -1 \end{aligned}$$

By adding 12 to both sides, we got y by itself on one side of the equation, and by doing this we were able to figure out what y was equal to. If you are not sure you solved an equation correctly, you can “boolean check” by plugging in (also known as substituting) whatever value you got for your variable back into the equation you started with.

1-2 Solving one and two step equations with the four operations

So far, we have only solved for variables with problems that only have simple addition and subtraction. However, many equations will also have multiplication or division. Let's try one with multiplication.

$$6 = 2x$$

Almost all the time when a variable is being multiplied by a number, there is no multiplication sign. Thus, $2x$ is actually 2 times x , the variable. $10x$ is 10 times x , and $\frac{5y}{4}$ is the same as 5 times y divided by four.

Back to our problem, we need to isolate x on the right side of the equation. In reality, the side of the equation that the variable is on doesn't really matter, so long as it is by itself. We need to change $2x$ to $1x$, or x , and the way to do this is by dividing both sides by 2, since $2x/2 = x$. Remember, you cannot just add, subtract, multiply, or divide one side of an equation by some number. What you do to one side, you must do to the other.

$$\begin{aligned}\frac{6}{2} &= \frac{2x}{2} \\ 3 &= x \\ x &= 3\end{aligned}$$

Once we divide both sides by 2, we got $3 = x$, so $x = 3$.

Finally, let's do a problem that we need to multiply both sides.

$$\frac{x}{2} = \frac{4}{3}$$

As I just said, we will need to multiply both sides by a number. Before we do that, note that $\frac{x}{2}$ is the same as $\frac{1}{2}x$. To change $\frac{x}{2}$ so that it is $1x$, we need to multiply both sides of the equation by 2.

$$\begin{aligned}\frac{2 \cdot x}{2} &= \frac{2 \cdot 4}{3} \\ x &= \frac{8}{3}\end{aligned}$$

By multiplying both sides of the equation by 2, we got $\frac{2x}{2} = \frac{8}{3}$, and since $2x/2 = x$, we got $x = 8/3$. Remember, to keep the equation equal, whatever you add, subtract, multiply, or divide one side of the equation by, you absolutely must do that same operation to the other side of the equation as well.

So far, we have tackled equations such as $x - 3 = 5$, which are "one step equations". It isn't always this easy. How would you solve this one?

$$2x + 5 = 3 - x$$

This is a little bit tougher. How can we isolate the variable on one side if it's on both? Remember how earlier we added, subtracted, multiplied, or divided both sides by the same number? We do that again, but with an extra step. First, let's subtract both sides by 5, or move 5 to the other side and change the sign.

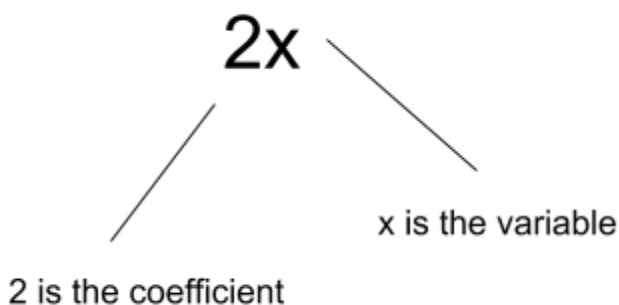
$$2x = -2 - x$$

What do we do next?

We have to add the same thing on both sides again. But it may not be what you think. Instead of adding a constant, or a number which is not multiplied by a variable, such as 5 or 3 to both sides, we add x to both sides. Think about this. Again, the rule is we can add, subtract, multiply, or divide the same number to both times. Like I said earlier in this chapter, a variable is a number, such as x . Therefore, we can add variables to both sides, because the rule state so.

$$2x + x = -2 - x + x$$

Something is a little different about this equation. How can we add $2x + x$? This is where combining like terms comes into play. You can add or subtract terms as long as they have the same variable. There are two parts to a variable. First is the coefficient, which is the "attached" number and is multiplied by the variable. The other part, the symbol that represents an unknown value, is the the variable.



When you add or subtract 2 variables, you must look at the coefficient. When you look at the $2x + x$, think of it as $(2 + 1)x = 3x$. See chapter QW to see why this works. Basically, when you need to add or subtract variables, add or subtract the coefficient, and just stick the variable to the end. Let's look at the equation again, and to make this easier I will show the coefficient, though generally if the coefficient is 1, it will not be shown.

$$\begin{aligned} 2x + 1x &= -2 - 1x + 1x \\ (2 + 1)x &= -2 + (-1 + 1)x \\ 3x &= -2 \end{aligned}$$

Now we just have to divide both sides by 3 to get x by itself and get our answer.

$$\begin{aligned} 3x/3 &= -2/3 \\ x &= -2/3 \end{aligned}$$

We get the final answer $x = -2/3$.

1-3 Solving multiple step equations with the four operations

In the last section we learned that we could solve equations by adding or subtracting a variable from both sides of the equation. However, you can't solve all equations like that. Look at the following equation.

$$\frac{1}{x} = 5$$

We must have just x on one side of the equation. What can we do to both sides of the equation to get $\frac{1}{x}$ to be x ? We just need to multiply both sides by x . $\frac{b}{a} \times a = b$, where a and b are numbers, and where a is a number not equal to 0. If we multiplied $\frac{2}{3x}$ by $3x$, we would have 2. This is because any number divided by itself (except 0) is 1. We can apply this to other equations similar to the one we are trying to solve.

$$\frac{1}{x} \cdot x = 5x$$

$$1 = 5x$$

$$x = \frac{1}{5}$$

When we multiply a fraction by a whole number, you can put the whole number as part of the numerator. Also, Do you remember the reciprocal we talked about earlier? It is when we flip the numerator and denominator, so we had the fraction $\frac{a}{b}$, the reciprocal is $\frac{b}{a}$. Another formula we can find out is that any fraction multiplied by its reciprocal is 1. Therefore, $\frac{a}{b} \times \frac{b}{a} = 1$. For example, $\frac{3}{2} \times \frac{2}{3} = \frac{6}{6} = 1$. You can also prove this by doing cross multiplication. See chapter 1 if you do not know what this is.

Look at this next problem:

$$\frac{x}{x+3} = 4$$

We need to just have $1x$ on the left side. First off, we multiply both sides by $x + 3$ to get rid of the denominator. Remember, $\frac{a}{a} = 1$, $\frac{ab}{a} = b$, where a and b are both real numbers not equal to 0, since any real number divided by itself (besides 0) is 1.

$$\frac{x(x+3)}{x+3} = 4(x+3)$$

$$x = 3x + 12$$

$$-2x = 12$$

$$x = -6$$

Here is another problem:

$$x + 3 = -\frac{3}{2}x + \frac{17}{4}$$

First, we can add $\frac{3}{2}x$ to both sides.

$$\frac{2}{2}x + \frac{3}{2}x + 3 = -\frac{3}{2}x + \frac{3}{2}x + \frac{17}{4}$$

$$\frac{5}{2}x + 3 = 17/4$$

Next, we can multiply both sides of the equation by 4 to get rid of the denominators.

$$\frac{5}{2}x \cdot 4 + 3 \cdot 4 = 17/4 \cdot 4$$

$$10x + 12 = 17$$

Subtract 12 from both sides:

$$10x = 5$$

And finally divide both sides by 10.

$$x = \frac{5}{10} = \frac{1}{2}$$

However, how would you solve this?

$$2(x + 2) = 8$$

According to the order of operations, you must do the operation in the parenthesis first. The problem is that we can't add $x + 2$ because we don't know what x is. While we could divide both sides by 2 to solve this, there is a very important property you need to know called the distributive property. Basically, it says that

$$1. \quad a(b + c) = ab + ac$$

And also that

$$2. \quad a(b - c) = ab - ac$$

For example, $5(2 + 3) = 10 + 15 = 25$. Alternatively, you could add what is inside the parentheses first to get $5 \times 5 = 25$. Either way, you get the right answer. For some situations, using the distributive property is more useful in case you have 2 terms inside the parenthesis that you cannot add or subtract together. You will use this property a lot in algebra, so it's important to remember it.

Using this property for this problem gives us

$$2x + 4 = 8$$

$$2x = 4$$

$$x = 2$$

Let's try one more with the distributive property:

$$(x + 2)(1/x + 2) = 11 + 2/x$$

If we use the distributive property, we get

$$(x + 2)(1/x) + (x + 2)(2) = 11 + 2/x$$

We can rewrite this again so that $1/x$ and 2 are in front.

$$(1/x)(x + 2) + 2(x + 2) = 11 + 2/x$$

Now we can use the distributive property *again*.

$$1/x \cdot x + 1/x \cdot 2 + 2x + 2 \cdot 2 = 11 + 2/x$$

Continuing to simplify,

$$1 + 2/x + 2x + 4 = 11 + 2/x$$

$$2/x + 2x = 6 + 2/x$$

$$2x = 6$$

$$x = 3$$

That was a lot of steps because we had to use the Distributive property twice. However, we can derive another formula using similar steps to what we did above.

$$\begin{aligned} (a + b)(c + d) \\ = (a + b)c + (a + b)d \\ = ac + bc + ad + bd \end{aligned}$$

As you can see, we used the distributive property again. What this equation really means is that

$$(a + b)(c + d) = ac + ad + bc + bd$$

Another way to look at it is



$$(a + b)(c + d) = ac + ad + bc + bd$$

This is still the distributive property, just we are doing it twice this time. We are distributing a into c and d , as well as distributing b into c and d . This is important to know as well, so you should memorize it.

1-4 Solving inequalities

An inequality is an expression which does not state that 2 things are equal, but that 2 things are unequal, and that one side of an inequality is greater than, less than, greater than or equal to, or less than or equal to the other side of the inequality.

A few of examples of equations are the following:

$$2 + 5 =$$

$$7 = 7$$

$$x + 2 = 7$$

In all 3 of these equations, there is an equal sign (=) which tells you that the left side of the equation is equal to the right side of the equation. Inequality signs, on the other hand, look different.

$\leq$ is less than or equal to

$\geq$ is greater than or equal to

$<$ is less than

$>$ is greater than

An important thing to remember is that the open end of the inequality side faces toward the bigger number. Here are a few examples of inequalities.

$$5 > 3$$

$$2 < 87$$

$$10 > 9$$

These are read as five is greater than 3, two is less than eighty-seven, and ten is greater than 9. You can solve equations for a variable, as we saw in the previous section. Likewise, you can also solve inequalities for variables. You solve for inequalities the exact same way you solve equations, except if you multiply or divide the equation by a negative number, you *must* flip the direction of the inequality sign. That is the most important thing to remember about inequalities. Another way to write this (solving for a) is

If $-a > b$, then $a < -b$

If $-a < b$, then $a > -b$

If $-a \geq b$, then $a \leq b$

If $-a \leq b$, then $a \geq b$

As you can see, when we multiplied the equations above so that a became positive, but we also made the inequality sign point toward the other number. So what does solving an inequality look like? Here is an example.

$$x + 5 > 7$$

This inequality says that some number plus 5 is greater than 7. Since you solve for inequalities the same way you solve equations, all you do is subtract 5 from both sides of the inequality.

$$x + 5 - 5 > 7 - 5$$

$$x > 2$$

We have gotten x by itself on one side of the inequality. The answer is $x > 2$. This means that any number that is greater than 2 would make this inequality true. Think about it: $1 + 5 < 7$, so 1 wouldn't satisfy the inequality because we are trying to get a number greater than 7. On the other hand, $99999999 + 5 > 7$, so 99999999 satisfies this inequality. Any number, no matter how big, will make this equation true, as long as $x > 2$. Something important to know is that $x = 2$ is NOT a solution to this inequality. Remember, $x + 5 > 7$. If we substituted $x = 2$ into this inequality, we would get $7 > 7$, which is not true because $7 = 7$. A number is not greater than or less than itself. However, 2.01 would work, because $2.01 > 2$ and $7.01 > 7$.

This next inequality should be easy.

$$x - 10 \leq 20$$

We have to find the range of numbers that, when 10 is subtracted from it, is less than 20. Since we solve inequalities the same way we solve equations,

$$x \leq 30$$

We got this answer by adding 10 to both sides of the inequality. Though you solve inequalities the same way you solve equation, the answer has a different meaning. An equation only has one answer. On the other hand, inequalities have an infinite number of solutions, and in this case, any number works as long as it's less than or equal to 30, and is not greater than 30 or equal to 30. While 2 did not satisfy the previous inequality we solved, 30 works for this one because the inequality sign is $\leq$, not $<$, meaning that the range of x values include 30.

A few more easy problems are worked out below.

$$\begin{aligned} 2x &\leq x + 1 \\ x &\leq 1 \end{aligned}$$

$$\begin{aligned} 20x &> 35 \\ x &> 7/4 \end{aligned}$$

$$\begin{aligned} a &\geq 10 + 27 \\ a &\geq 37 \end{aligned}$$

$$\begin{aligned} c &< 2c + 1 \\ 0 &< c + 1 \\ -1 &< c \\ c &> -1 \end{aligned}$$

This last problem here is especially important. First off, notice that when the we flipped the equation, we flipped the sign around. $-1 < c$ means the exact same thing as $c > -1$. Notice that in both cases the inequality sign has its open end facing toward c while the closed end faces toward -1 .

This problem is also important, however, because there is more than one way to solve it. When we solved it, we subtracted c from both sides, but we could have subtracted $2c$ from both sides. If we did that, we would've gotten the following.

$$\begin{aligned} c &< 2c + 1 \\ -c &< 1 \end{aligned}$$

It would seem that the next thing to do is multiply both sides of the equation by -1 . That is right, but because we are multiplying both sides of the equation by a negative number, we must change the direction of the inequality sign. Therefore, the final answer would be

$$c > -1$$

, and not $c < -1$.

If you look back to the equation when we first solved it, you will see we got $c > -1$ as the right answer. The absolute most important thing to remember in this section is that you solve an inequality the same way you solve an equation- by getting the variable by itself on one side of the equation or inequality- however, if you multiply or divide both sides of an inequality by a negative number, you must change the direction of the inequality sign. Here are a few more problems demonstrating this.

$$x + 5 \geq 2x + 10$$

$$-x \geq 5$$

$$x \leq -5$$

$$-3w > 23$$

$$w < -(23/3)$$

$$r - 2r \leq -24$$

$$-r \leq -24$$

$$r \geq 24$$

As you can see, inequalities are very similar to equations. Here are a few problems worked out just for practice.

$$\frac{x}{5} + 2 < 10$$

$$x + 10 < 50$$

$$x < 40$$

$$x + 10 \geq 2x - 25$$

$$-x \geq -35$$

$$x \leq 35$$

$$2(x + 5) > 5x - 27$$

$$2x + 10 > 5x - 27$$

$$-3x > -37$$

$$x < 37/3$$

$$5h/3 + 1/3 \leq -5/6$$

$$10h + 2 \leq -5$$

$$10h \leq -7$$

$$h \leq -(7/10)$$

It shouldn't take too long to get good at solving inequalities because you solve them the same way as equations.

Chapter 1 Test

Solve each equation for the variable.

1. $x + 5 = 6$

2. $2x + 1/3 = 7/6$

3. $5/b + 3 = 4$

4. $2x - 5 = 3x + 6$

5. $x/2 + 3/2 = 5/4$

6. $a - 5a > -16$

7. $2x/x + 1 = 4/x$

8. $z - 6 = -z + 6$

9. $100(y + 10) = 500y$

10. $\frac{y-5}{3} = 9 + y$

11. $2x - 5 \leq -10x + 19$

13. $\frac{4x+4}{3} = 8 - 4x$

14. $5v + 3(2v - 8) = 39$

15. $2x/3 \geq 5x/4 - 3$

Chapter 2

Graphing linear equations

Graphs are a way of clearly showing a pattern. This chapter will introduce you to the simplest type, a linear equation. This will teach how to graph a linear equation, how to transform a linear equation into point-slope form, what the slope and y-intercept is, and how to graph a linear equation given two points.

2-1 The linear equation

A linear equation has two variables: x and y . It is written in the form

$$y = mx + b$$

This is a general form of the equation. Only x and y are variables in this equation, and m and b are constants, or terms which are not multiplied by a variable. That being said, any linear equation can be made by replacing m or b with a constant. For example, the following are all linear equations:

$$y = x + 3$$

$$y = -5x$$

$$y = \frac{1}{2}x - 47$$

Looking at each individual part of the equation, m is the slope of the equation, or the rate of change. It is written as the rise, the change in y , over run, the change in x . The other constant, b , is the y intercept. That is because when x equals 0, $y = b$. This can be proven by substituting 0 for x in the equation $y = mx + b$, resulting to

$$y = m(0) + b$$

Since anything times 0 is 0, no matter what the value of m is, $y = b$ when $x = 0$. Here are some examples of equations that are *not* linear:

$$y = \sqrt{x} + 10$$

$$y = x^2 - 4$$

$$y = 1/x$$

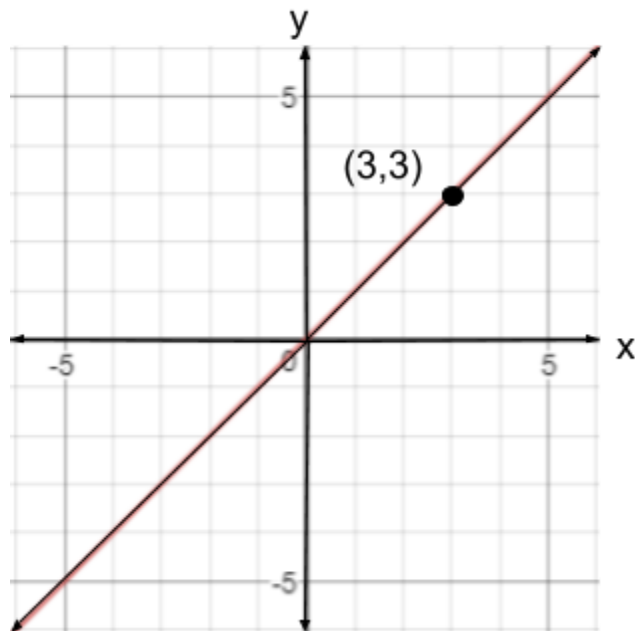
In general, in order for an equation to be linear, the degree of the equation, or the exponent with the highest value in the equation, must be 1. In addition, there can be no fractional or negative exponents being applied to the variable.

Another important thing to know is that x is the independent variable and y is the dependent variable. That means that the value of y depends entirely on the value of x . If you have learned about independent and dependent variables in science when doing an experiment, this is a similar thing. In a science experiment, the independent variable is what the person who is doing the experiment purposely changes to see how it affects the dependent variable, which is the thing that changes based on how the independent variable is changed or manipulated. That is why when we graph equations, as we will do in the next section, the x -axis will be the horizontal axis, while the y -axis is the vertical axis.

2-2 Graphing linear equations

Before we can graph a line, we must first be able to graph a point on a xy graph. A point on an xy graph is written as (x, y) . That is, the x value of the coordinate is written first, then the y value. For example, the point $(2, 5)$ means that y is 5 when x is 2.

Here is what an xy graph looks like. The graph goes on forever in all directions. As you can see, the graph passes through points $(2, 2)$, $(3, 3)$, and so on. This can be shown in a table. In a table, you pick multiple x values, calculate the y values by substituting x into the equation, then writing it down in a table.



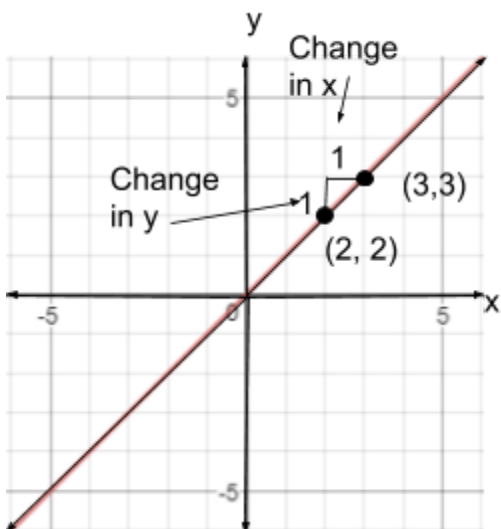
x	y
-1	-1
0	0
1	1
2	2

For example, to find the y value of $y = x$ when x is 4, replace x with 4 in the equation $y = x$. Doing this gives you the equation $y = 4$. Therefore, when $x = 4$, $y = 4$. In other words, the graph $y = x$ passes through point $(4, 4)$.

However, there is a way to graph equations without making a table. Last chapter we talked about the parts of the linear equation $y = mx + b$, including the slope, m , which is rise over run. But what does that mean? This means the $\frac{\text{Change in } y}{\text{Change in } x}$, and in a linear equation, this is constant.

To begin finding the slope of a graph, first graph the equation. This can be done by making a table and connecting the points you find.

Next, find 2 points on the line. It doesn't matter which two points you choose, though it must be 2 different points. To make finding the slope as easy as possible, choose 2 points which the x and y values of the point are whole numbers. To find the slope of the graph above, I could pick the points $(3, 3)$ and $(2, 2)$.



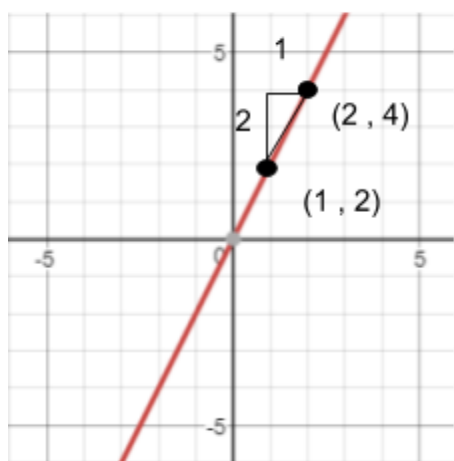
x	y
-1	-1
0	0
1	1
2	2

Next, draw a little triangle by the points, and see how long the 2 shorter sides of the triangle are. In this case, the 2 smaller sides of the triangle drawn here are both 1. The length of these sides are actually equal to the change in y and the change in x. As you can see, the y value of (3, 3) is one more than the y value of (2, 2). The same is true for the x value. This makes sense since $2 + 1 = 3$.

Finally, you can calculate the slope. Remember that the slope is just the $\frac{\text{Change in } y}{\text{Change in } x}$. Since the

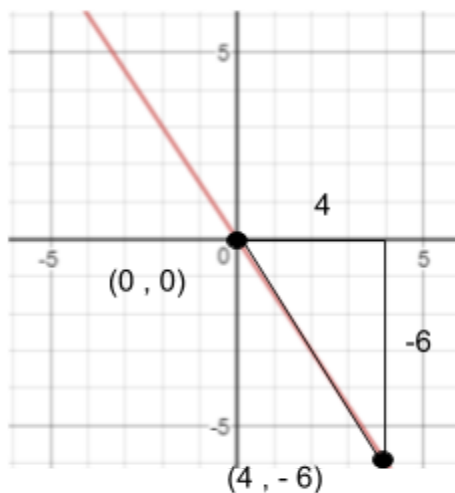
change in y is 1 and the change in x is 1, $m = \frac{1}{1} = 1$. Thus, the slope of this equation is one. Another way to figure this out is simply by looking at the equation $y = x$. When no coefficient is written on front of a variable, it is the same thing as 1 times the variable. For example, $a = 1a$, $x = 1x$, etc. Since the form of a linear equation is $y = mx + b$, and the slope is what is being multiplied by x, the slope of $y = x$ is one since x is being multiplied by 1. The slope of $y = 2x$ would be 2, since x is being multiplied by 2, and the slope of $y = -3/2x$ is $-3/2$.

$$y = 2x$$



$$m = 2/1 = 2$$

$$y = -3/2 x$$

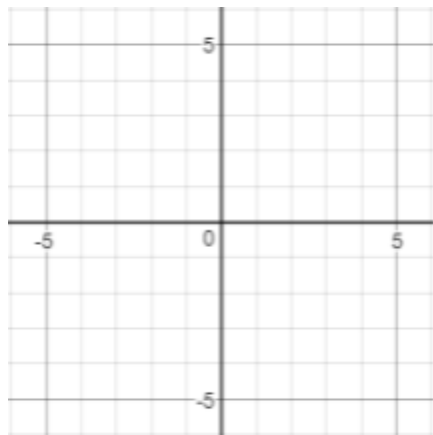


$$m = -6/4 = -3/2$$

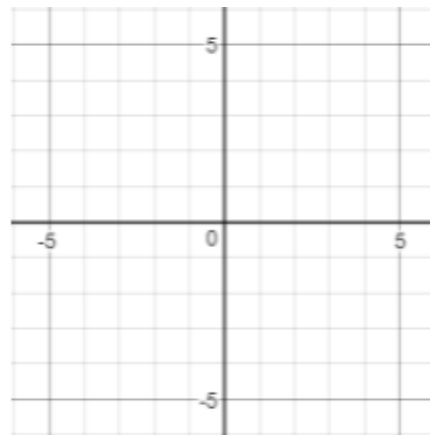
Practice

Graph the following equations.

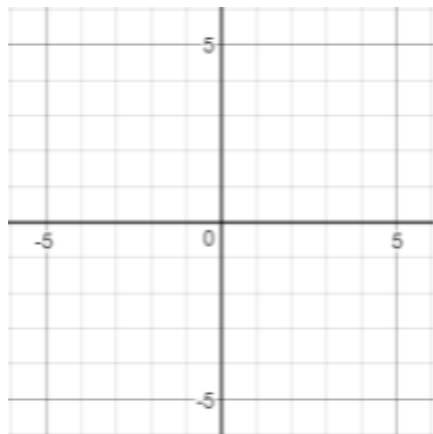
1. $y = 3x$



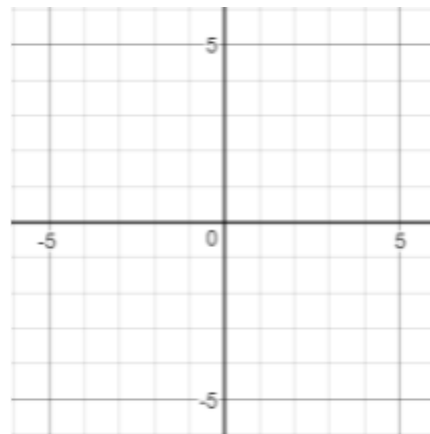
4. $y = 6x$



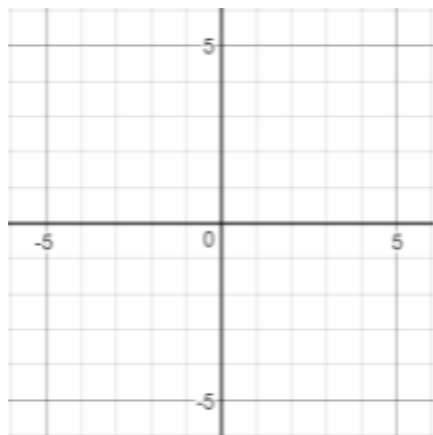
2. $y = -3x$



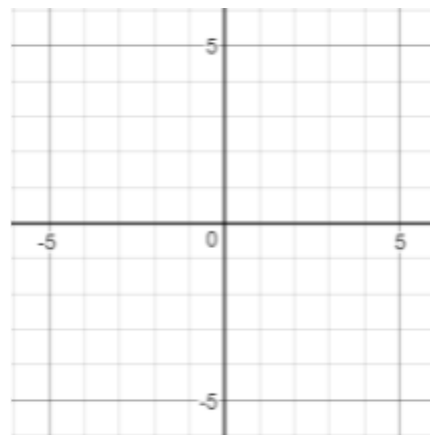
5. $y = -7/2x$



3. $y = 5/2x$

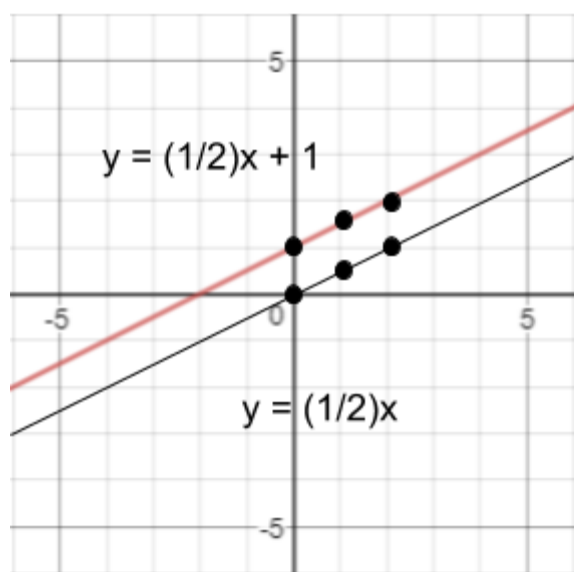


$y = -x$



2-3 Equations with nonzero y-intercepts

Recall that the form of a linear equation is $y = mx + b$. In all the previous equations we have looked at, the line passed through point $(0, 0)$, also commonly called the origin. That is because in all of the equations in 2-2 that we graphed had a b value of 0. b is called the y -intercept since the graph touches the y -axis at that point. Also said in 2-1, this is because the graph of an equation touches the y axis when $x = 0$, and since anything times 0 is 0, substituting 0 into the equation $y = mx + b$ just gives $y = b$. Therefore, any linear equation always passes through point $(0, b)$. Below is an example of an equation with a nonzero y -intercept.



$$y = (1/2)x + 1$$

x	y
0	1
1	3/2
2	2

$$y = (1/2)x$$

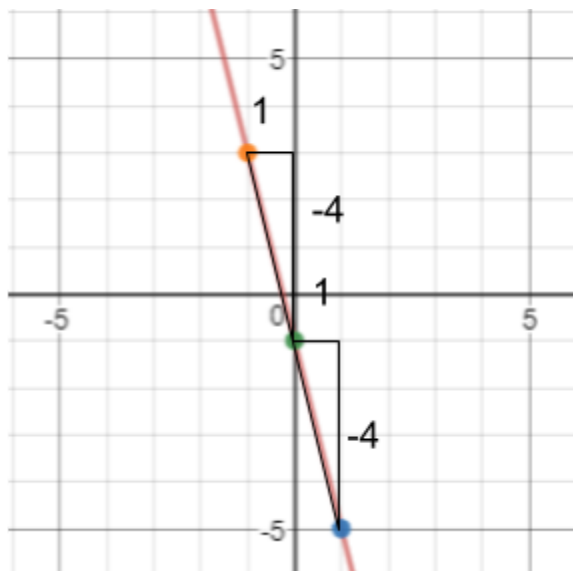
x	y
0	0
1	1/2
2	1

The easiest way to graph any linear equation is by first graphing the y -intercept, then using the slope to find which points the graph passes through. For this equation, since the slope is $1/2 = \text{rise/run}$, you can draw a point by going up one unit from the previous point and going right 2 units. The next point you can draw is up 1 unit and right 2 units of the point you just plotted. You can keep doing this forever, and as long as you go up the same numbers of units as the rise and go right the same number of units as the run from a point on the line, you can find another point on the line. It's important to understand this because it will make graphing equations much easier for you, and you will not need to write the table.

Notice that this graph does *not* pass through $(0, 0)$, but rather $(0, 1)$. Another important thing you may see is that the graph of $y = (1/2)x + 1$ is exactly as the graph of $y = (1/2)x$, except that every point of $y = (1/2)x + 1$ has y value that is 1 greater than $y = (1/2)x$.

Here is another example, this time with a negative y -intercept:

$$y = -4x - 1$$

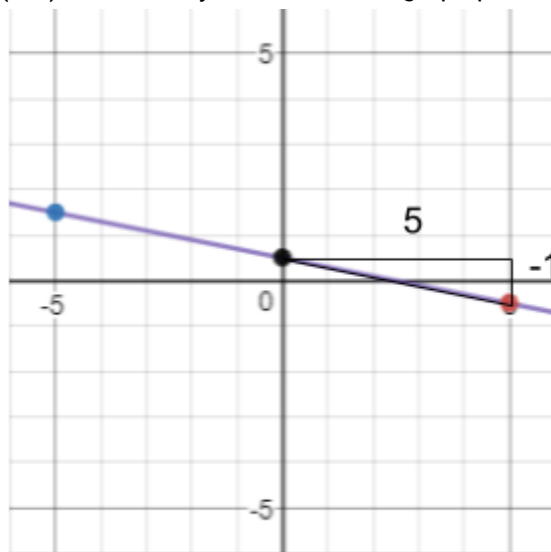


This line passes through $(0, -1)$ since the y-intercept is -1 .

The triangles were drawn to show that each point is 1 unit right and 4 units down from the previous point. This makes sense since the slope, -4 , is equal to $\frac{-4}{1}$, so the rise is -4 and the run is 1 . (Going up a negative amount is the same as going down a positive amount).

Here is one more example, $y = -(1/5)x + 1/2$. As you can see, this graph passes through $(0, 1/2)$.

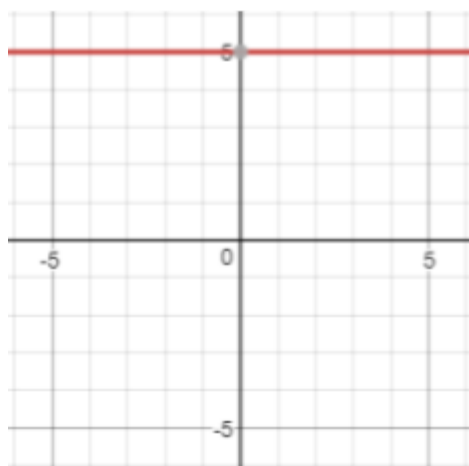
$$y = -1/5x + 1/2$$



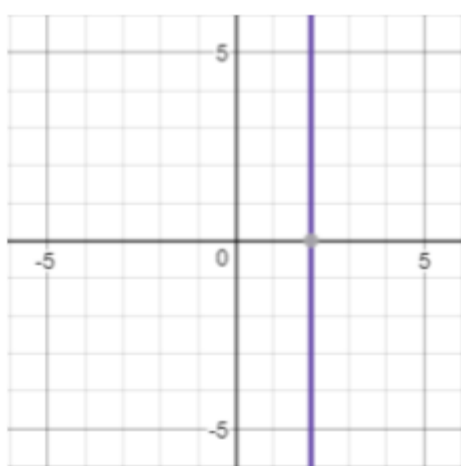
Finally, there are a couple of special equations that you should know. While they are linear equations, they are different because of their slope and that they only have one variable.

The first graph shown below, $y = 5$, has 0 slope, so it doesn't go up or down. If you think about it, $y = 5$ is the same thing as $y = 0x + 5$. Unlike the other graphs we have seen, y is not dependent on x . That just means y is always the same value, no matter what x is. The graph on the right is even stranger. It has an undefined slope, and it cannot be written in the form $y = mx + b$. The equation $x = 2$ is vertical, and whenever you have to graph an equation $x = a$, just draw a vertical line passing through $(a, 0)$. There are a couple other graphs as well for you to show as examples.

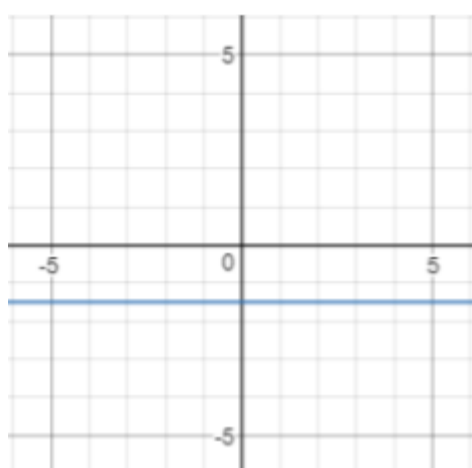
$$y = 5$$



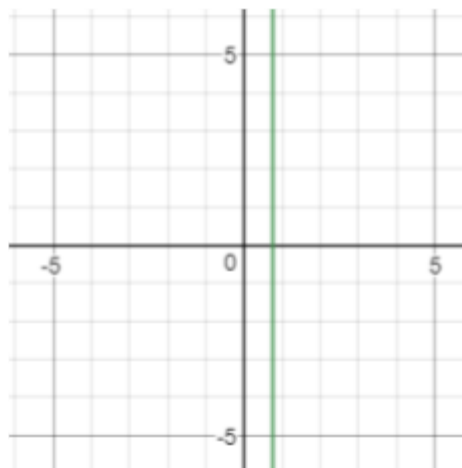
$$x = 2$$



$$y = -2$$



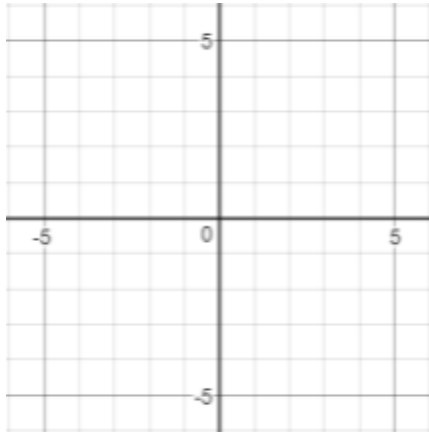
$$x = 0.75$$



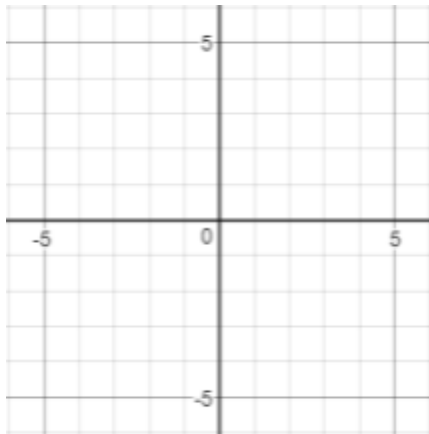
Practice

Graph the following equations.

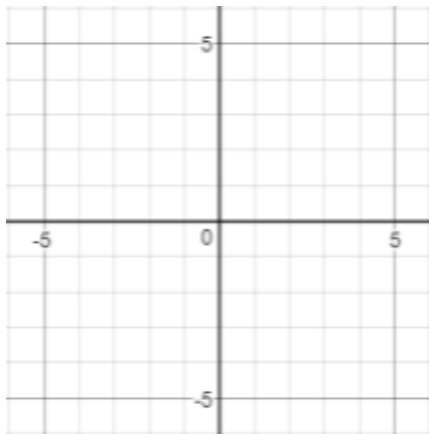
1. $y = 3x$



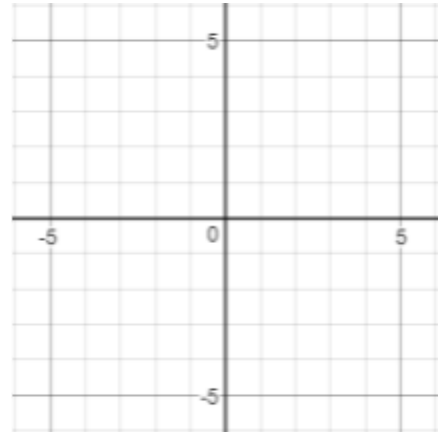
2. $y = 2x + 2$



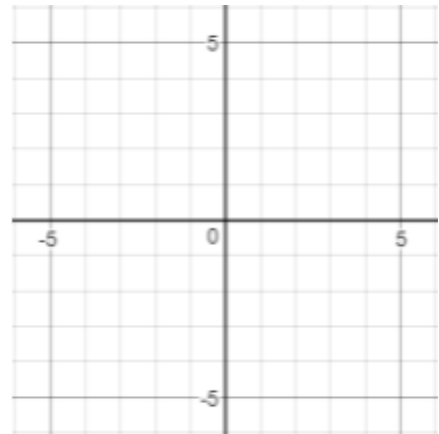
3. $y = \frac{1}{4}x - 1$



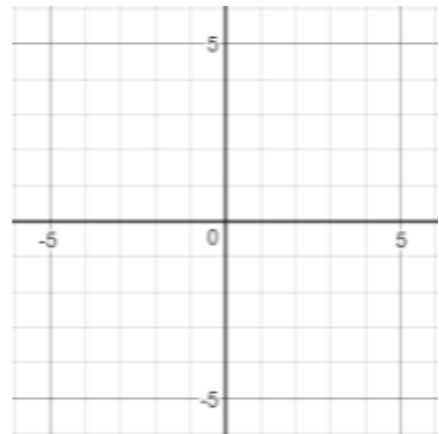
4. $y = -x - 3$



5. $y = -5x + 5$



6. $x = 4$



2-4 Transforming linear equations and other forms of linear equations

So far, we have seen many linear equations such as $y = x$, all in the form $y = mx + b$. However, this form, known as slope-intercept form, is only one way to write an equation. What if I told you that the equation

$$5x - 3y = 2$$

is also a linear equation which you can graph? The way this equation is written is called standard form, and the equation for a linear equation in standard form is $Ax + By = C$, where x and y are variables and A , B , and C are constants.

While standard form can be useful for some things we haven't learned yet, a linear equation in standard form is harder to graph than one in slope-intercept form. When we are given slope-intercept form, to graph the equation we can just plot the y-intercept, then plot more points by moving *rise* units up and *run* units right. However, just by looking at a standard form equation, we cannot tell what the slope or the y-intercept of the equation is.

If you are told to graph an equation in standard form, the best way to do it is by first changing the form of the equation into point-slope. There are only 2 steps to this process, but it's important to know!

Step 1: Subtract Ax from both sides of the equation $Ax + By = C$. You should now have $By = -Ax + C$

Step 2: Divide both sides of the equation by B . Doing this will give you $y = -\frac{A}{B}x + \frac{C}{B}$

That's all you have to do. Using this method, there are 2 ways to find the slope and y-intercept of the equation: 1) changing the equation so it is in slope-intercept form $y = mx + b$ or 2) substitute the coefficients into the formulas $m = -\frac{A}{B}$ and $b = \frac{C}{B}$.

While the 2nd way may be faster, it's more important to know the first way. The 2nd way will only work if you have $Ax + By = C$. It would not work for the equation $x + y + 2 = 0$. Let's try some examples. For the following, we need to find the slope and y-intercept of the equation.

$$x + y = 2$$

Using the 2nd method, we could find the slope and y-intercept by plugging in the coefficients and constant in the formulas. Doing this gives us $m = -1/1 = -1$, and $b = 2/1 = 2$. Using the 1st method, we would just subtract x from both sides of the equation, giving us $y = -x + 2$.

$$2x - 6y = 12$$

$$\text{2nd method: } m = (-2)/(-6) = 1/3, b = 12/(-6) = -2$$

$$\text{1st method: } -6y = -2x + 12$$

$$y = 1/3x - 2$$

$$m = 1/3, b = -2$$

$$x + 5 = y + 12$$

The 2nd method won't work for this equation, so we will just transform it into slope-intercept.

$$x + 5 - 12 = y + 12 - 12$$

$$x - 7 = y$$

$$y = x - 7$$

$$m = 1, b = -7$$

Another example:

$$2x + 3y + 5 = 10$$

First, subtract 5 from both sides.

$$2x + 3y + 5 - 5 = 10 - 5$$

$$2x + 3y = 5$$

While at first we couldn't find the slope and y-intercept using the 2nd method, now the equation is in the form $Ax + By = C$. Solving both ways,

$$\text{1st method: } 3y = -2x + 5$$

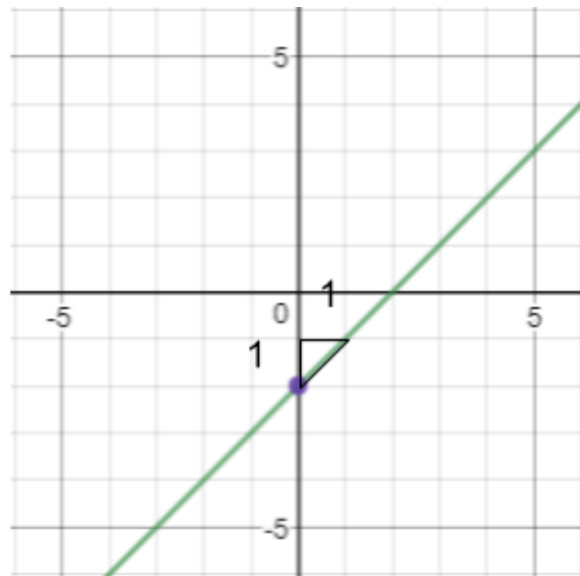
$$y = -\frac{2}{3}x + \frac{5}{3}$$

$$m = -2/3, b = 5/3$$

$$\text{2nd method: } m = -2/3, b = 5/3$$

As you can see, we applied the rules of solving equations to transform linear equations into point-intercept form, which is useful because it makes graphing the equation much easier. To show this, let's do an example. If you are told to graph the equation $3y - 3x = 6$, first do the steps shown, then graph.

$$\begin{aligned} 3x - 3y &= 6 \\ -3y &= -3x + 6 \\ y &= x - 2 \end{aligned}$$



By transforming this equation, we were able to write it in a way which we can know the slope and y-intercept easily. Overall, transforming equations can be helpful to make it easier to graph.

2-5 Graphing linear equations given information about the function

Sometimes you may need to graph a linear equation without being given an equation. Instead, you may be given one coordinate the line passes through and the line's slope, or 2 coordinates the line passes through. If you are given two points, (x_1, y_1) , and (x_2, y_2) , which a linear equation passes through, you can find the slope of the equation by plugging in these values into the following formula (and the 2's and 1's are not being multiplied by the variables):

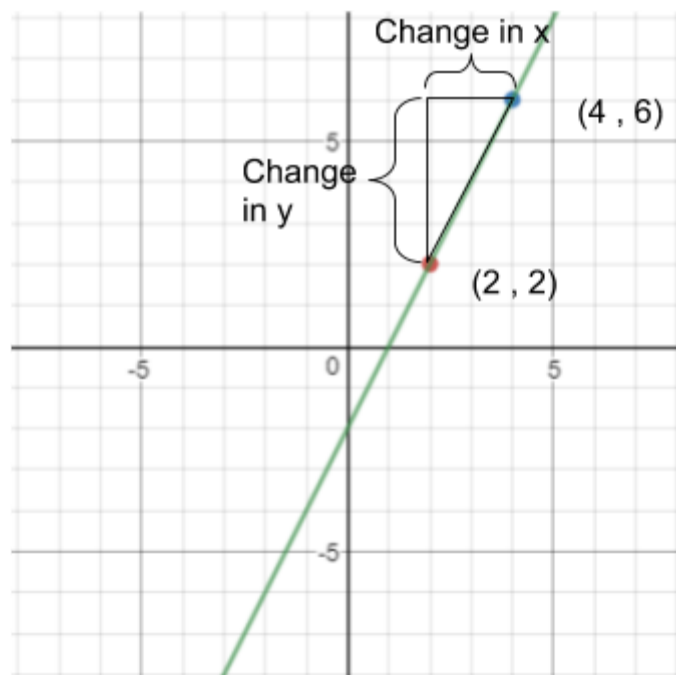
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Here is why the formula works. Remember that the slope, m , of a linear equation is $\frac{\text{rise}}{\text{run}}$, which is the same thing as the $\frac{\text{change in } y}{\text{change in } x}$. Since $y_2 - y_1$ is the change/difference in y , and $x_2 - x_1$ is the change in x , the formula works. To show this on a graph, if you were given that a linear equation passes through the points $(2, 2)$ and $(4, 6)$, then the change in x and the change in y could be shown as the following.

$$\text{Change in } x = 4 - 2 = 2$$

$$\text{Change in } y = 6 - 2 = 4$$

$$\frac{(\text{Change in } y)}{(\text{Change in } x)} = \frac{4}{2} = 2$$



After finding the change in x , change in y , and the slope, you have almost all the information you need to write the equation. However, we do not know the y -intercept. Currently, we have the equation $y = 2x + b$. In order to figure out the value of b , you can plug in x and y of one of the coordinates and solve for b . It doesn't matter which one you pick, because either way you will get the same answer, but picking the coordinate with smaller x and y values makes solving the equation for b easier.

To show that this is the case, I will plug both coordinates into the equation. These will be substituted into the equation $y = 2x + b$. Since a coordinate is written (x, y) , $x = 2$ and $y = 2$. Plugging this in to $y = 2x + b$ gives us $2 = 2(2) + b$. Likewise, plugging in $x = 4$ and $y = 6$ results to $6 = 2(4) + b$.

Solving the equations,

$$\begin{aligned}2 &= 2(2) + b \\2 &= 4 + b \\b &= 2 - 4 = -2\end{aligned}$$

$$\begin{aligned}6 &= 2(4) + b \\6 &= 8 + b \\b &= 6 - 8 = -2\end{aligned}$$

If you look at the graph, you will see that this is correct, since the graph passes through the point (0, -2). No matter which coordinate you choose to substitute, as long as the line passes through it, you will get the same answer for b.

For another example, what if you are told to find the equation of a linear function with a slope of 1 and that the line passes through the point (5, 6)? We actually do the exact same thing as the last problem, just we don't have to do the first 2 steps because the slope was already given to us.

First, write the equation given what you know. Since it is a linear equation with the slope 1, and we do not know what the y-intercept is, we can write it as

$$y = x + b.$$

Next, we substitute (5, 6) into the equation, and solve.

$$\begin{aligned}6 &= 5 + b \\b &= 1\end{aligned}$$

Now we can write the equation by plugging $m = 1$ and $b = 1$ into the equation $y = mx + b$, to get the final answer.

$$y = x + 1$$

Now that we've shown how and why the formula $(y_2 - y_1)/(x_2 - x_1)$ works, let's try using it to figure out the equation of a line given 2 points it passes through.

A line passes through (0, 1) and (-3, 6). Find the equation of the line.

To figure this one out, let's use our formula. This gives us

$$m = \frac{6-1}{-3-0} = \frac{5}{-3} = -\frac{5}{3}$$

Now that we have the slope, we can write the equation using what we know.

$$y = -\frac{5}{3}x + b$$

Usually, to solve for b, we would have to plug in a coordinate. However, this is a special case because one of the points we were given, (0, 1), is the y-intercept. Therefore, $b = 1$, so the final answer is

$$y = -\frac{5}{3}x + 1$$

Let's do one last problem for this section.

Find the equation of a line that passes through $(-1, -4)$ and $(2, 5)$.

First, we use the formula $m = (y_2 - y_1)/(x_2 - x_1)$.

$$m = \frac{5 - (-4)}{2 - (-1)} = \frac{5 + 4}{2 + 1} = \frac{9}{3} = 3$$

Be careful not to mess up the signs when doing this. Remember to put the negative signs, and that subtracting a negative number is like adding a positive number.

Now that we have the slope, we can write this equation.

$$y = 3x + b$$

Now we can choose a coordinate to substitute in the equation. I choose $(2, 5)$ because it doesn't have any negative signs.

$$5 = 3(2) + b$$

$$5 = 6 + b$$

$$b = 5 - 6 = -1$$

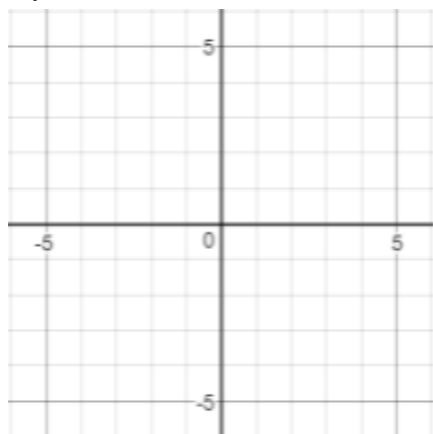
We get the final answer $y = 3x - 1$.

Overall, when given 2 points that a line passes through, first plug the x and y values of the points into the equation $m = (y_2 - y_1)/(x_2 - x_1)$ to find the slope. Then, substitute what you got for m into the equation $y = mx + b$, and finally plug in the x and y values of one of the coordinates that the line passes through to find the y-intercept.

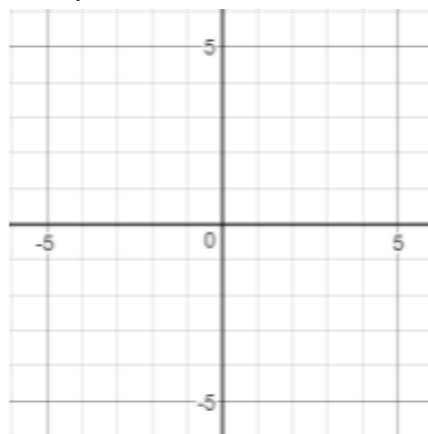
Chapter 2 Test

Graph the equation with the information given.

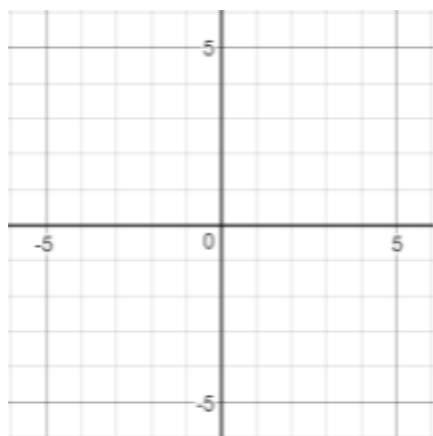
1. $y = 2x - 1$



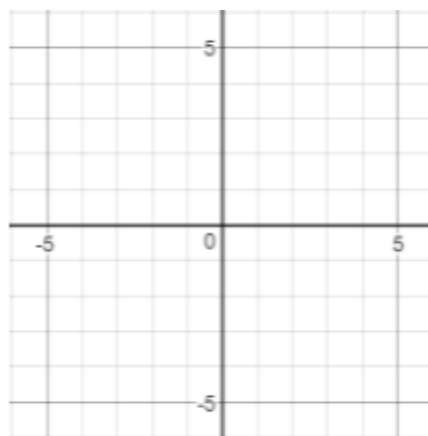
4. $y = 4$



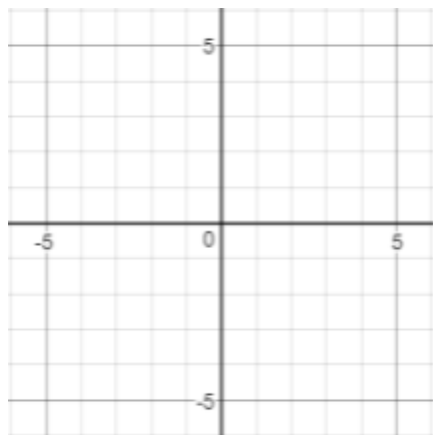
2. $y = -\frac{1}{3}x$



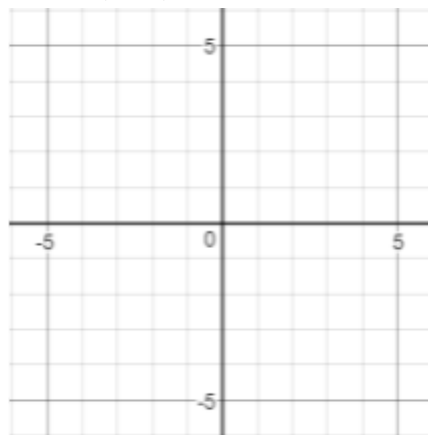
5. $\frac{3x+y}{2} - 6 = -8$



3. $-3x + 5y = 8$



6. The line passes through (1, 0) and (-4, 5)



Chapter 3

Systems of linear equations

This section is a little challenging but very important to understand. It will teach mainly how to see if 2 linear equations have no solutions, have 1 solution, or have an infinite number of solutions, as well as multiple methods of solving a system of 2 equations, including graphing, substitution, and elimination, Finally, how to solve a system of 3 equations will be taught as well.

4-1 Solving system by graphing

What is a system?

A system of equations is a set of multiple equations with multiple variables which you can solve to find where the equations intersect when graphed. There are 3 main ways to solve a systems. First, we'll look at graphing, second, we'll learn substitution, and third, elimination.

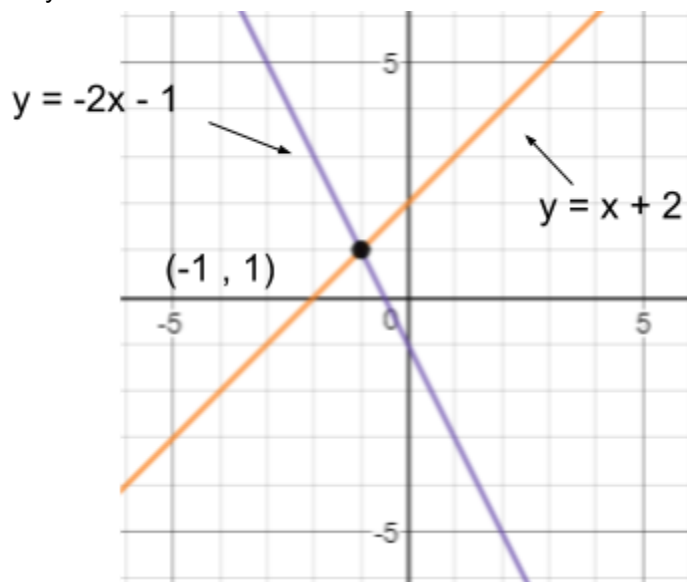
Solving a system by graphing

One way you can solve a system of 2 equations is by graphing them. However, this is only practical if there are 2 variables.

The solution to a system is where the equations given intersect. Therefore, to find the answer to a system, you could just graph the 2 equations, then see the coordinate (x , y) where they intersect. For example, if you were told to solve the system

$$\begin{aligned}y &= x + 2 \\y &= -2x - 1\end{aligned}$$

Normally, there would be an infinite number of solutions for one equation with 2 variables. Think about it: if you were told to solve the equation $x + y = 10$, then the following could be solutions (written as (x , y) coordinates): (10 , 0), (1 , 9), (2 , 8), (3 , 7), (4 , 6), (5 , 5), (6 , 4), (7 , 3), (8 , 2), (9 , 1), (10 , 0), (2.5 , 7.5), (2.001 , 7.999)... It could go on forever, and an infinite number of solutions could satisfy this equation. However, in our system above, we have 2 equations. This usually means that there is only one x value and one y value that would satisfy both equations. Remember that you can "boolean check" equations by plugging variables into the original equation after solving them to check if you solved the equation right. If the left side of the equation equals the right side of the equation, and after doing this you get $a = a$, you correctly solved the equation. You can do the same with systems by plugging in both variables into the equations. Back to our problem, all we need to do is graph the 2 equations on the same graph and see where they intersect.



As you can see, the 2 equations intersect at (-1 , 1). To prove this is true, we can substitute (-1 , 1) back into both equations. By plugging in $x = -1$ and $y = 1$ into $y = -2x - 1$ and $y = x + 2$, we get

$$1 = -2(-1) - 1$$

$$1 = 2 - 1$$

$$1 = 1 \text{ (correct)}$$

$$1 = -1 + 2$$

$$1 = 1 \text{ (correct)}$$

We just proved that $(-1, 1)$ is the correct solution to this system. If you solve a system by graphing, you won't need to do this, but you will need to substitute for other methods we will learn.

Here is another system of equations:

$$2x - 3y = -2$$

$$x = 2y - 2$$

In chapter 2 we talked about how to transform linear equations into slope-intercept form in order to graph the equation. For this system, to solve by graphing, we need to do this to both equations.

$$-3y = -2x - 2$$

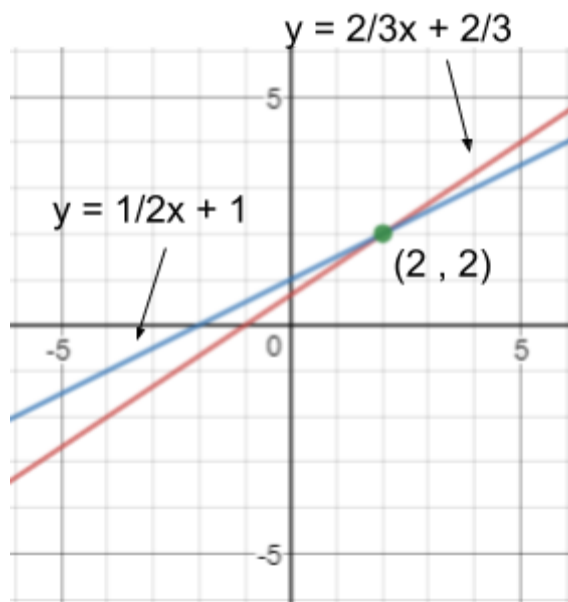
$$\underline{y = (2/3)x + 2/3}$$

$$2y - 2 = x$$

$$2y = x + 2$$

$$\underline{y = (1/2)x + 1}$$

The 2 underlined equations is what we need to graph to find the solution. Doing so gives us this:



Because the 2 lines intersect at $(2, 2)$, the solution is $(2, 2)$. Another way to write this is $x = 2$ and $y = 2$. If you want, you may check this answer.

So far, we have looked at systems with one solution, but this is not always the case.

Systems with no solutions or infinite solutions

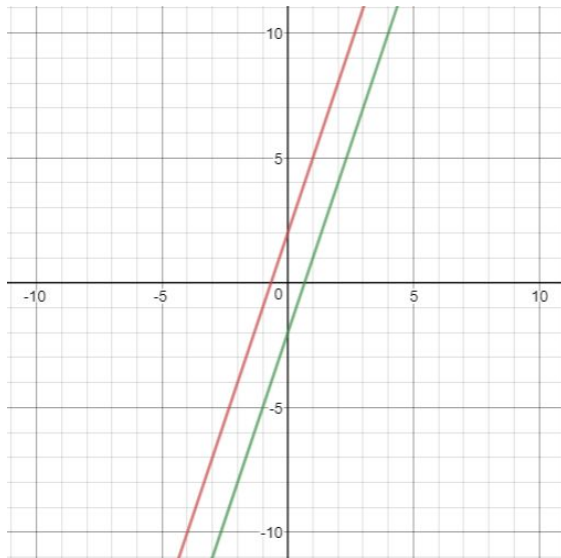
There are a couple of special situations where the solution to a system of 2 linear functions is not just one (x, y) coordinate.

Consider the following system:

$$y = 3x + 2$$

$$y = 3x - 2$$

It would seem that a system like this would have one solution. However, if you graph the functions, you will see that is not actually true.



These two linear equations will never intersect. That is because both of them have the same slope, but a different y-intercept. 2 lines with the same slope and a different y-intercept are called parallel lines. $y = 3x + 2$ is parallel to $y = 3x - 2$. Whenever you have 2 parallel equations, the 2 equations will go on forever and never touch the other function.

Here is another system of equations:

$$\frac{1}{2}x = y - 6$$

$$2y - x = -2$$

To begin solving the system by graphing, we need to transform the equations into slope-intercept form.

$$\frac{1}{2}x \cdot 2 = y - 6 \cdot 2$$

$$x = 2y - 12$$

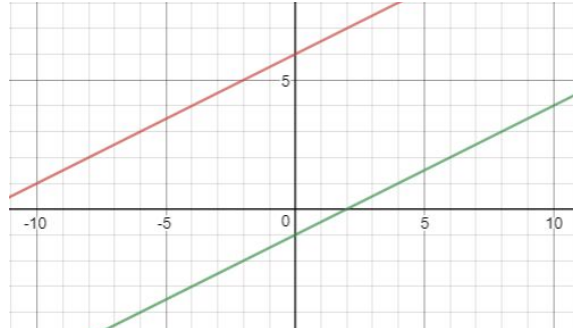
$$2y = x + 12$$

$$y = \frac{1}{2}x + 6$$

$$2y = x - 2$$

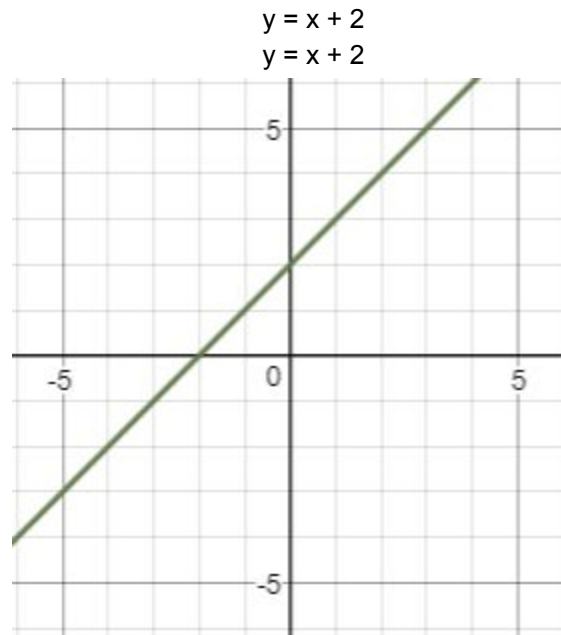
$$y = \frac{1}{2}x - 1$$

Even before we graph these equations, we can see that they have the same slope and different y-intercepts.



Since these equations have the same slope and will never intersect, there is no solution to this system.

So far, we have seen that 2 lines that are parallel have no solution. But what happens if the 2 lines given do not have just the same slope but also the same y-intercept? If the slope and y-intercept of 2 lines are the same, then it really isn't 2 different lines, but in fact the same line: the equation of both lines are exactly the same. Here is an example:

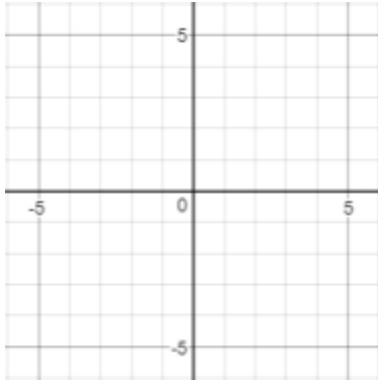


While the graph may look like just one line, it is 2 equations on top of each other. Now that we have graphed it, what is the solution? In the beginning of the section, we saw examples with one solution because the equations touched once. Then, we saw systems with no solutions because the 2 equations never touched on a graph. This is a different situation, however, because the graphs touch an infinite number of times. Therefore, a system with 2 equations that are the same have an infinite number of solutions.

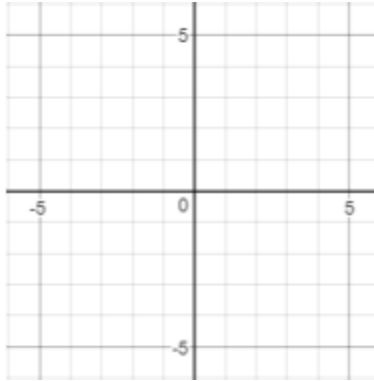
Practice

Solve the following systems by graphing. If there is no solution or an infinite number of solutions, write that as the answer.

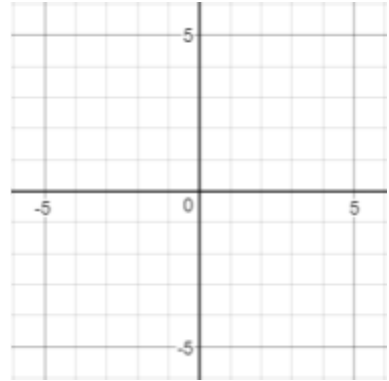
1. $y = 3x - 2$
 $x = 2y - 6$



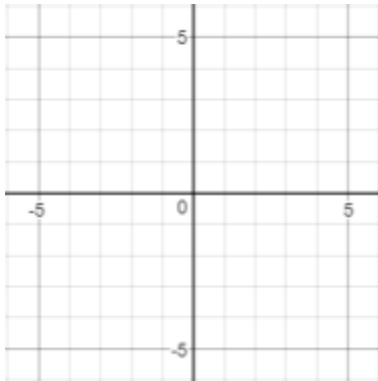
2. $x + y = 2$
 $-y - x = -4$



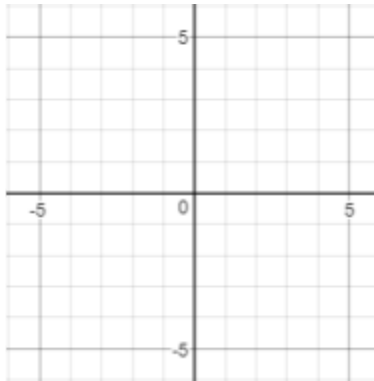
3. $x = 2y + 3$
 $3 - y = 4$



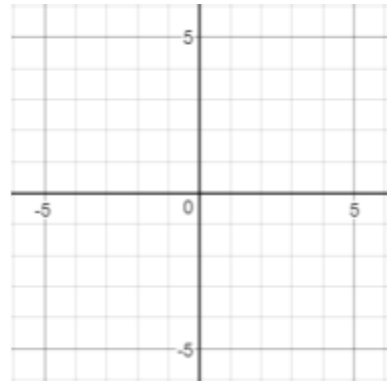
4. $y = -2x + 4$
 $2y = -3x + 4$



5. $y - 5x - 4 = 0$
 $2y = 10x + 8$



6. $y = x$
 $y = 2x$



4-2 Solving systems of 2 linear equations with substitution

As we learned in the previous section, the solution to a system of 2 equations is the point where the 2 equations touch. We could solve the system by graphing both equations. However, just like how the variable in an equation is not always x , systems are not always written in terms of x and y . While it is still possible to graph this, graphing usually is not the best way to solve a system. The 2 systems you are solving may intersect at $(100, 293)$, and if they did, it would take a very long time to graph the equations. However, there are a couple of methods you can use to solve a system without graphing it. The method we're about to use is called substitution. Take a look at this system.

$$\begin{aligned}a &= b + 2 \\a + b &= 1000\end{aligned}$$

If you tried to graph this with an a and b axis, it would take a very long time. However, there is a much easier way to solve this system. Look at the first equation, $a = b + 2$. This equation means that a is 2 more than b . In other words, it expresses a in terms of b ; it tells what the value of a is based on the value of b . Just like how the equation $y = x$ expresses y in terms of x , this equation expresses a in terms of b . Since a literally is equal to $b + 2$, you can substitute a into the second equation. If you do this, you get

$$(b + 2) + b = 1000$$

Now that there is only one variable in this system, you will be able to solve for b like any normal equation.

$$\begin{aligned}b + 2 + b &= 1000 \\2b &= 1000 - 2 \\2b &= 998 \\b &= 998/2 = 499\end{aligned}$$

Now we know that b is equal to 499, but how would you solve for a ? We will actually need to substitute again, but instead of substituting one equation into another, we will just substitute $b = 499$ into the first equation.

$$a = b + 2$$

Since $b = 499$,

$$a = 499 + 2 = 501$$

We have successfully solved for a and b for this system. Just to make sure we got it correct, we'll substitute $a = 501$ and $b = 499$ into both equations. If the left side and right side of each equation is equal, then we solved the system correctly.

$$\begin{aligned}501 &= 499 + 2 \\501 &= 501\end{aligned}$$

$$\begin{aligned}501 + 499 &= 1000 \\1000 &= 1000\end{aligned}$$

At first, this may seem like a lot of work, however with practice you will be able to solve systems very quickly. For more practice, let's solve a few more systems.

$$n + 2z = 13$$

$$3n + z = 14$$

Currently, we can't substitute anything. In order for us to use substitution, we must have one variable expressed in terms of the other variable. We can do this multiple ways. One way is to subtract $2z$ from both sides of the first equation. This gives us n by itself on one side of the equation.

$$n = 13 - 2z$$

Next, we can substitute $n = 13 - 2z$ into the second equation so that z is the only variable in the second equation.

$$3(13 - 2z) + z = 14$$

Now we have to use the distributive property since we cannot subtract $13 - 2z$. If you do not know how to do this, look at chapter 1, section 3.

$$39 - 6z + z = 14$$

We can solve this like a normal equation.

$$39 - 5z = 14$$

$$-5z = -25$$

$$z = 5$$

Next, we plug in $z = 5$ into the equation $n = 13 - 2z$ to figure out n .

$$n = 13 - 2(5) = 13 - 10 = 3$$

Our final answer is $n = 3$ and $z = 5$.

This next system should look more familiar, and it should be easy to solve.

$$y = x + 2$$

$$y = -2x - 1$$

To solve, we could plug either the first equation into the second equation, or we could plug the second equation into the first equation. It doesn't matter which one you do because you will get the same result.

$$x + 2 = -2x - 1$$

$$3x = -3$$

$$x = -1$$

Now you can substitute $x = -1$ into either equation. Again, it doesn't matter which one because you will get the same result for y either way.

$$y = -1 + 2 = 1$$

The final answer is $x = -1$ and $y = 1$. You could also write it as a coordinate, $(-1, 1)$.

Here is the last problem:

$$d = 50t$$

$$50t + d = 100$$

Substituting $d = 50t$ into the second equation,

$$50t + 50t = 100$$

$$100t = 100$$

$$t = 1$$

And finally substituting $t = 1$ into $d = 50t$,

$$d = 50(1) = 50$$

$$t = 1$$

$$d = 50$$

4-3 Solving systems with elimination

The final method of solving a system is with elimination. While this is usually the fastest method, it can be the hardest to understand. It involves subtracting equations from each other, and changing equations as necessary to put the equation in a form which makes the system easy to solve.

Before learning this method, you must first understand how to add and subtract equations. The best way to show how this works is by doing an example. If you were told to subtract $5x + 3y = 8$ from $6x + 4y = 10$, here is how you would do it.

$$\begin{array}{r} 6x + 4y = 10 \\ - (5x + 3y = 8) \\ \hline \end{array}$$

First, you would set up the equations like this, similar to how you would set up a normal subtraction equation. The important thing is to make sure that the terms multiplied by x are aligned, the terms multiplied by y are aligned, and that the terms not multiplied by a variable are aligned. The next step is to distribute the negative sign into all the terms in the second equation. We are still subtracting the equations since adding a negative number is like subtracting a positive number.

$$\begin{array}{r} 6x + 4y = 10 \\ - 5x - 3y = -8 \\ \hline \end{array}$$

Obviously, you cannot subtract $3y$ from $6x$ or subtract $5x$ from 10 . But like we learned in the first chapter, you can add “like terms”, that is, terms with all the same variables. Therefore, we can do $6x - 5x$, $4y - 3y$, and $10 - 8$. If we do this, we get $x + y = 2$, since $6x - 5x = x$, $4y - 3y = y$, and $10 - 8 = 2$.

$$\begin{array}{r} 6x + 4y = 10 \\ - 5x - 3y = -8 \\ \hline x + y = 2 \end{array}$$

It's not really difficult, just add or subtract like terms in the equation, and drop the equal sign. Just make sure that before you do this that you align like terms correctly so you don't make any mistakes and so you can do the math. For more practice let's do a few more. This time, let's add.

$$\begin{array}{r} y = 2x + 1 \\ x + y = 9 \\ \hline \end{array}$$

This time, we have to add $y = 2x + 1$ and $x + y = 9$, but we can't do it yet. The x and y terms are not aligned vertically yet, so we will have to change the top equation from slope-intercept form into standard form ($Ax + By = C$).

$$y - 2x = 2x - 2x + 1$$

$$y - 2x = 1$$

$$-2x + y = 1$$

Notice how the equation was rewritten from $y - 2x = 1$ to $-2x + y = 1$. If that last step wasn't done, we wouldn't have been able to subtract these equations. Since we changed $y = 2x + 1$ to $-2x + y = 1$, we can now solve the problem.

$$\begin{array}{r} -2x + y = 1 \\ x + y = 9 \\ \hline \end{array}$$

Now we just add like terms to get our final answer.

$$\begin{array}{r} -2x + y = 1 \\ x + y = 9 \\ \hline -x + 2y = 10 \end{array}$$

One last example:

$$\begin{array}{r} x + 2y = 1 \\ -(x + y = 9) \\ \hline \end{array}$$

We have to subtract these two equations. The like terms are already aligned, so all we need to do is distribute the negative sign.

$$\begin{array}{r} x + 2y = 1 \\ -x - y = -9 \\ \hline \end{array}$$

Finally, we can subtract the terms.

$$\begin{array}{r} x + 2y = 1 \\ -x - y = -9 \\ \hline y = -8 \end{array}$$

The x's cancel each other out, which left us only with $y = -8$. This here is exactly why adding and subtracting equations in a system is useful. When you do this, you can make one variable cancel out,

leaving just one variable and what that variable is equal to. Really, we just used elimination to solve for y in a system.

This is exactly what elimination is: adding or subtracting the 2 equations in a system to cancel one variable and get the value of the other one. Now that we can add and subtract equations, let's try solving some systems using elimination.

$$\begin{array}{l} x + y = 2 \\ 2x + y = 8 \end{array}$$

From looking at the equations, we can see that the variables are already aligned. In addition, if you look carefully, you will notice that if you subtracted the equations from each other, the y would cancel out. That is exactly what we're going to do. Subtracting the 2nd equation from the first equation gives us

$$\begin{array}{r} 2x + y = 8 \\ - (x + y = 2) \end{array}$$

$$x = 6$$

It's that easy to find x. You can subtract the first equation from the 2nd equation as well, but in some cases like this, it is easier to subtract the 2nd equation from the first so you don't get $-x = -6$, which would give you the same answer but would take an extra step. The last thing to do here is substitute $x = 6$ into one of the equations to find y.

$$\begin{array}{l} 6 + y = 2 \\ y = 2 - 6 = -4 \end{array}$$

The final answer is $x = 6$ and $y = -4$.

To get more practice with elimination so you are more comfortable using it, let's do a few more practice problems.

$$\begin{array}{ll} 2x + y + 2 = 13 & (1) \\ 3x + 3y = 21 & (2) \end{array}$$

The first thing you should notice is that the first equation is not in standard form. Subtract 2 from both sides so that it is.

$$2x + y = 11 \quad (3)$$

However, we've run into a problem. Even if we subtract equations (2) and (3) from each other, no variable will cancel. But remember you can change how an equation is written by adding, subtracting, multiplying, or dividing both sides of an equation by the same number. It is perfectly fine to multiply an equation by a constant as long as you multiply every term by the same number. Therefore, we can multiply equation (3) by 3.

$$(3) \times 3: 3(2x + y = 11) = 6x + 3y = 33 \quad (4)$$

We now have a fourth equation, $6x + 3y = 33$. While it might seem like the equation just got more complicated, we can now do equations (4) - (2) to cancel the y variable.

$$\begin{aligned}
 (4) - (2): 6x - 3x &= 33 - 21 \\
 3x &= 12 \\
 x &= 4
 \end{aligned}$$

Though it took a lot of work, we now know that $x = 4$. From here, it's easy: just substitute x into equation (1), (2), (3), or (4). Substituting $x = 4$ into equation (3) gives

$$\begin{aligned}
 2(4) + y &= 11 \\
 8 + y &= 11 \\
 y &= 3
 \end{aligned}$$

The final answer is $x = 4$ and $y = 3$.

Here is one final example. Look at the notation to see what is being done.

$$\begin{aligned}
 2x &= y - 3 & (1) \\
 2y - 7x &= 0 & (2)
 \end{aligned}$$

$$\begin{aligned}
 2x - y &= -3 & (3) \\
 -7x + 2y &= 0 & (4)
 \end{aligned}$$

$$\begin{aligned}
 (3) \times 2: 4x - 2y &= -6 & (5) \\
 (4) + (5): 3x &= 6 \\
 x &= 2
 \end{aligned}$$

Substituting $x = 2$ into (3),

$$\begin{aligned}
 2(2) - y &= -3 \\
 4 - y &= -3 \\
 -y &= -7 \\
 y &= 7
 \end{aligned}$$

As you can see, we first changed the 2 original equations (1) and (2) so that the variables aligned, making equations (3) and (4). Next, we multiplied equation (3) by 2 so that the new equation (5) could be added to equation (4) to cancel out y . Once we solved for x , we substituted $x = 2$ into equation (3) to finally solve for $y = 7$.

There are a lot of things to know about systems. Among them:

- Sometimes you may have to multiply both equations by a different number to be able to cancel out a variable
- If you are trying to solve for a system, and the variables cancel so that all you are left with is $a = b$, but a is not actually equal to b (for example, $3 = 5$), then the system has no solution.
- If you are trying to solve for a system, and the variables cancel so that all you are left with is $0 = 0$, then the system has an infinite number of solutions.
- Generally, when solving a system, you should use elimination, as it is the fastest method. However, if one of the equations defines a variable, such as $y = ax + b$, where y is defined in terms of x , and a and b are constants, then substitution may be better.

Remember to keep practicing solving systems. The more you do it, the less work it feels like, and you will become more comfortable with doing it. It will also help you be better at judging whether you should solve a system with substitution or elimination.

Practice

Solve each system for the 2 variables given. Solve using either substitution or elimination.

1. $x + y = 10$
 $6x + 4y = 48$

2. $b = 3a + 2$
 $2b + a = -3$

3. $3a - 2b = 12$
 $2a + 3b = 47$

4. $a = b + 2$
 $b = 2a - 4$

5. $x + y = 3$
 $\frac{1}{2}x + \frac{1}{4}y = \frac{5}{4}$

6. $5z + 3r = -1$
 $-2z - 4r = -8$

7. $J = A + 4$
 $J + A = 16$

8. $5x + 5 = 5y - x + 1$
 $y - x = 2y + 2y - 5$

4-4 Systems with 2 linear equations word problems

Systems can be very tedious to solve, but are also very important to know. This chapter will have word problems with systems so that you can see their real-world application/use and so you can have more practice with solving systems using elimination and substitution.

You are considering hiring a mowing company to mow your lawn. One charges \$10 plus \$6 dollars for every hour it takes the company to mow your lawn. The other company charges \$4 plus 8\$ for every hour it takes for the company to mow your lawn. At how many hours is the cost of both companies equal, and what is the cost?

The first thing to do when doing a word problem is to find out the equations. The 2 variables you can choose are c , the cost, and h , the number of hours the company takes to mow your lawn. Once you have done this, you can write your equations as

$$c = 10 + 6h \quad (1)$$

$$c = 4 + 8h \quad (2)$$

This is because the total cost is the base cost plus the number of dollars charged per hour times the number of hours. For example, the cost of hiring the first company for an hour is $c = 10 + 6(1) = 16$.

Once you have the starting equations, you can solve it just like any normal equation. Remember, we are trying to solve for the number of hours at which the cost is equal. Normally, you should use elimination, but since both equations define a variable, you can use substitution. If we substitute (1) into (2), we will be able to solve this easily.

$$10 + 6h = 4 + 8h$$

$$-2h = -6$$

$$h = 3$$

We now know that at 3 hours, the cost for both companies will be equal. However, we do not know what the cost for hiring the companies for 3 hours is. All we need to do to find that out is to plug in $h = 3$ into one of the 2 original equations.

$$c = 10 + 6(3) = 28$$

The final answer is $c = 28$ and $h = 3$. Put in words, you could write the final answer as "The 2 companies will charge the same amount, \$28, for working for 3 hours."

Here is another word problem.

In a classroom, there are 29 total students. If there are 3 more boys than girls, how many boys are there in the classroom and how many girls are there in the classroom?

This one is easier. First, you have 2 variables. They can be b for boys and g for girls. Setting this one up gives the system

$$b + g = 29 \quad (1)$$

$$b = g + 3 \quad (2)$$

Since the total number of students is 30 and there are 3 more boys than girls. If we solved this using elimination, the first thing we would do is subtract g from both sides of equation (2).

$$b - g = 3 \quad (3)$$

Now, we can add the new equation to the first equation to cancel g .

$$(1) + (3): 2b = 32$$

$$b = 16$$

Substituting $b = 16$ into (3),

$$16 - g = 3$$

$$g = 16 - 3 = 13$$

As you can see, we could have solved this system either way. It's up to you to decide which way may be easier for you to use. Here is another word problem involving money.

If 5 cookies and 3 bags of chips cost \$11, and 2 cookies and 7 bags of chips cost \$16, how much does 10 cookies and 10 bags of chips cost?

To begin solving this, we need to figure out the equations. We already know the total cost, but not how much each cookie and each bag of chips cost. Thus, our variables should be c and b , c for the cost of each cookie and b for the cost of each bag of chips (note that you cannot have 2 different things use the same letter as a variable).

If we arrange our equations like this, we can get our system.

$$5c + 3b = 11 \quad (1)$$

$$2c + 7b = 16 \quad (2)$$

Since no variable is defined, elimination is the best option to use here. Even better, the variables are already aligned vertically. However, currently we cannot cancel a variable since the coefficient of the variables are different in the equations. Remember from the last section, you can multiply a whole equation by a constant, because you can multiply both sides of an equation by the same number.

If we multiply (1) by 2 and multiply (2) by 5, the coefficient of c will be 10 in both equations since 5×2 is 10 and 2×5 is 10. We want to have the same coefficients for one variable so that we can subtract the equations and cancel a variable.

$$(1) \times 2: 10c + 6b = 22 \quad (3)$$

$$(2) \times 5: 10c + 35b = 80 \quad (4)$$

$$(4) - (3): 29b = 58$$

$$b = 58/29 = 2$$

Now we know that each bag of chips cost \$2. To find out how much each cookie costs, we substitute $b = 2$ into any of the equations.

Substituting $b = 2$ into (1),

$$5c + 6 = 11$$

$$5c = 5$$

$$c = 1$$

Now we know that each cookie costs \$1 and each bag of chips costs \$2. We have solved the system, but we are not done. The question didn't ask how much each cookie and each bag of chips cost, but how much 10 cookies and 10 bags of chips cost. We solved for how much each cookie and chips cost because otherwise we would not have been able to figure out how much 10 cookies and 10 bags of chips cost. We write the equation

$$\text{Cost} = 10c + 10b$$

And since we know that $c = 1$ and $b = 2$,

$$\text{Cost} = 10(1) + 10(2) = 10 + 20 = 30$$

Our final answer is that 10 cookies and 10 bags of chips cost \$30. Hopefully, you are becoming more comfortable solving systems. The best way to get good at solving systems is to practice them. Here is the final problem:

Nick has Quarters and dimes. Total, he has 12 coins and \$1.95. How many quarters and how many dimes does he have?

First, we need to figure out what variables to use. We can use q for the number of quarters and d for the number of dimes Nick has. Since a quarter is \$0.25 and a dime is \$0.10, we can write the system

$$0.25q + 0.1d = 1.95 \quad (1)$$

$$q + d = 12 \quad (2)$$

Elimination, again, is the best method to use. Since none of the coefficients of one variable in the equations is equal, we will have to multiply one or both of the equations by a constant. While you could divide equation (2) by 4 or 10, it will be easier to multiply equation (1) by 4 or 10. Usually, multiplying by a smaller number is the better option, but in this specific case multiplying by 10 is better, because when you multiply a number with a decimal by 10^a , you can just move the decimal place over a places left.

$$(1) \times 10: 2.5q + d = 19.5 \quad (3)$$

As you can see, multiplying this equation by 10 was easy; we just had to shift the decimal one place to the left for each number.

Next, we need to cancel d :

$$(3) - (2): 1.5q = 7.5$$

To solve for the number of quarters, just divide both sides by 1.5.

$$\frac{1.5}{1.5} q = \frac{7.5}{1.5}$$
$$q = 5$$

Substituting $q = 5$ into (2),

$$5 + d = 12$$
$$d = 7$$

Therefore, Nick has 5 quarters and 7 dimes.

The way to solve word problem systems is to first find out what variables to use, then write the system, and finally solving it. In order to do this, you must understand what the question is asking. Remember, the only way to get better at solving these problems is to keep practicing them.

4-5 Solving systems with 3 variables

Solving systems with 3 variables is probably the hardest thing you will have to do in Algebra 1. However, don't be too worried, because it's really just solving systems with 2 equations with a couple of extra steps. Remember how you cannot solve an equation with 2 variables. You must get rid of one of the variables to solve for one. This is the same idea here, except you will have to get rid of 2 variables in an equation with 3 variables. Graphing these equations will result in planes (basically a flat surface that goes on forever in all directions) because there is 3 variables, so it is impractical to solve these by graphing. You will find that elimination and substitution are almost always the best way to solve a system. Also, before you start trying to solve a system with 3 variables, make sure you have mastered solving systems with 2 variables. That way, you will be able to keep up with what is being done.

For our first problem, we'll use x , y , and z , as our variables, though of course it can be anything.

$$x + y + z = 6 \quad (1)$$

$$2x + y - z = 1 \quad (2)$$

$$x - y - 2z = -7 \quad (3)$$

You are probably wondering how you can solve this. Think back to solving a system with 2 equations. In order to solve it, you got rid of one variable. For a start, we can try getting rid of one variable in this equation. If you look carefully, you can see that if you subtract equations (1) and (2), you can get rid of y .

$$(1) - (2): -x + 2z = 5 \quad (4)$$

We have a start. However, this is still unsolvable because there is 2 variables. That is why we need to make a second equation without y . If you add equations (2) and (3), you can make another equation with only x and z .

$$(2) + (3): 3x - 3z = -6 \quad (5)$$

As you can see, in only 2 steps we have made a system of 2 equations! (4) and (5) don't have y , so we can solve it just like any normal system of 2 equations. To be able to get rid of x , let's multiply (4) by 3.

$$(4) \times 3: -3x + 6z = 15 \quad (6)$$

Now, we add (5) and (6) to cancel x since $-3x + 3x = 0$.

$$(5) + (6): 3z = 9$$
$$z = 3$$

Just like normal, substitute $z = 3$ into one of the equations with 2 variables. Substituting $z = 3$ into (4),

$$-x + 2(3) = 5$$

$$-x + 6 = 5$$

$$-x = -1$$

$$x = 1$$

Normally, we would be done, but we still don't know what y is. To figure that out, we have to substitute $z = 3$ and $x = 1$ into one of the 3 original equations. If we substitute these values into (1), which is $x + y + z = 6$, we get

$$1 + y + 3 = 6$$

$$y + 4 = 6$$

$$y = 2$$

Therefore, the final answer is $x = 1$, $y = 2$, and $z = 3$, or $(1, 2, 3)$.

This next problem is going to look a little different, but it still has 3 variables and 3 equations.

$$x + y = 3 \quad (1)$$

$$y + z = 5 \quad (2)$$

$$x + z = 4 \quad (3)$$

This one may make you run into a few problems. No matter which equations you subtract from each other, even if you cancel one variable, you still add another variable. Think about it: these equations could be rewritten as

$$x + y + 0z = 3 \quad (1)$$

$$0x + y + z = 5 \quad (2)$$

$$x + 0y + z = 4 \quad (3)$$

$0x$, $0y$, and $0z$, are all equal to 0, so adding them to the equation doesn't make a difference. Anyhow, if I tried to get rid of y by doing $(1) - (2)$, this is what would happen.

$$(1) - (2): x - z = -2 \quad (4)$$

It may seem like that you accomplished nothing because even though you were able to get rid of y , you just added a z that wasn't even there before. However, you actually have helped yourself by doing this. Each of the 3 original equations had 2 variables, and none of the equations had the same 2 variables. You may notice, though, that both (3) and (4) have only x and z , meaning that you could solve for one of the variables by solving this system with 2 variables.

If you add (3) and (4), you will cancel out z since $z - z = 0$.

$$(3) + (4): 2x = 2$$

$$x = 1$$

Next, substituting $x = 1$ into (4),

$$1 - z = -2$$

$$-z = -3$$

$$z = 3$$

You can substitute either $x = 1$ into (1) or $z = 3$ into (2). It doesn't matter which way you choose to do because both will give you the same answer.

$$x + y = 3 \quad (1)$$

$$1 + y = 3$$

$$y = 2$$

While this system may have looked stranger, we found that $x = 1$, $y = 2$, and $z = 3$ more easily than the first problem.

And finally, just for a little more practice to make sure you have got the hang of doing this, we'll solve another system without explanation. See if you can follow along.

$$x = -y + z \quad (1)$$

$$2y = -x + z - 2 \quad (2)$$

$$x + 2y - 4z = -8 \quad (3)$$

$$(2)/1: y = -(1/2)x + (1/2)z - 1 \quad (4)$$

Substituting (4) into (1),

$$x = -[-(1/2)x + (1/2)z - 1] + z$$

$$x = (1/2)x - (1/2)z + 1 + z$$

$$(1/2)x = (1/2)z + 1$$

$$x = z + 2 \quad (5)$$

Substituting (5) and (4) into (3),

$$(z + 2) + 2[-(1/2)x + (1/2)z - 1] - 4z = -8$$

$$z + 2 - x + z - 2 - 4z = -8$$

$$2z - x - 4z = -8$$

$$-2z - x = -8$$

$$-2z = x - 8$$

$$z = -(1/2)x + 4 \quad (6)$$

Substituting (6) into (5),

$$x = -(1/2)x + 4 + 2$$

$$(3/2)x = 6$$

$$3x = 12$$

$$x = 4$$

Substituting $x = 4$ into (6),

$$z = -(1/2)(4) + 4$$

$$z = -2 + 4$$

$$z = 2$$

Substituting $x = 4$ and $z = 2$ into (2),

$$2y = -4 + 2 - 2$$

$$2y = -4$$

$$y = -2$$

If you were able to follow along, you'll see that I solved for the variables only by substituting. However, as you may have been able to figure out, substitution is not a very fast method to solve systems with 3 variables, because you have to substitute more times. This time I'm solving the same system, but only with elimination.

$$x = -y + z \quad (1)$$

$$2y = -x + z - 2 \quad (2)$$

$$x + 2y - 4z = -8 \quad (3)$$

$$x + y - z = 0 \quad (4)$$

$$x + 2y - z = -2 \quad (5)$$

$$x + 2y - 4z = -8 \quad (3)$$

$$(4) - (5): -y = 2$$

$$y = -2$$

$$(5) - (3): 3z = 6$$

$$z = 2$$

Substituting $y = -2$ and $z = 2$ into (1),

$$x = -(-2) + 2 = 4$$

I just solved the exact same system, but much, much faster. Elimination may have seemed like a worse method when it was first taught because of all the work, but in reality elimination can actually be a lot easier than substitution, especially for 3 variables but for 2 variables as well.

In this section, you learned how to solve for systems of 3 variables. Though this book will not cover it, it is also possible to solve for 4 variables, 5 variables, and more. You could solve systems with more variables the same way you would solve for 3 variables. In Algebra 2, you may be taught to solve systems with 3 or more variables using matrices, but it is important to know how to solve it using these methods because this will not only be faster to solve but also you could solve for even more variables. If you understand well how to solve systems with 3 equations, you should be able to understand everything in Algebra. This is one of the hardest parts of Algebra, so it is fine if you do not understand it right away.

Chapter 3 test

Solve each system of equations for all variables in the equation.

1. $x + y = 7$
 $y = x + 3$

2. $2a + 2b = 2$
 $4a + 2b = 4$

3. $2a + 3b = 28$
 $3a + 5b = 45$

4. $18 - a - b = 0$
 $2b = a$

5. $x + y - z = 0$
 $x - y + z = 2$
 $-x + y + z = 5$

6. $a = b + c - 5$
 $b = a - c + 5$
 $c = a + b + 1$

7. $3x - 4y = -7$
 $2y + 3z = 23$
 $x - 5z = -22$

8. $2a - b = c$
 $a + b + c = 0$
 $2c - a = 2$

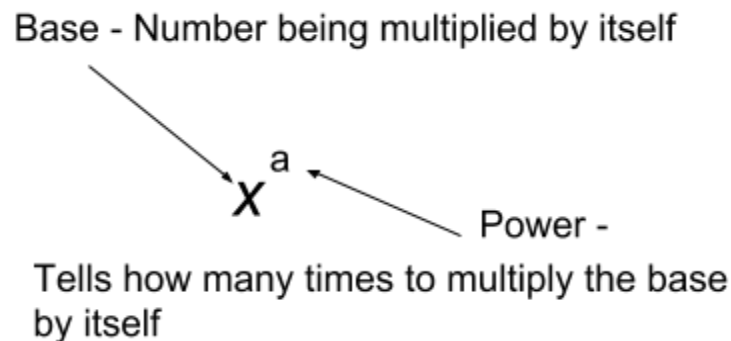
Chapter 4

Properties of exponents and roots

While you have learned some basic things about exponents and roots, this section will get more in depth with some properties of n th roots and exponents, as well as real, irrational, and imaginary numbers.

4-1 Properties of exponents

You should already know how to solve some simple exponents, such as $2^3 = 2 \times 2 \times 2 = 8$ and $7^3 = 7 \times 7 \times 7 = 49 \times 7 = 343$. Basically, you just multiply a number by itself however many times the exponent tells you to. While it may take a while to do these by hand, you understand how to do them. Before moving forward, make sure you know that the 2 parts to an exponent are the power and the base. Also know that something “squared” is a number raised to the 2nd power and something “cubed” is a number raised to the 3rd power.



However, there are some properties that can make simplifying exponents more easily. There are several properties. One property will be listed at a time, and then a couple examples of each one. It is a good idea to memorize these properties because you will use exponents a lot in math.

$$x^a \cdot x^b = x^{a+b}$$

This property basically means that if you have 2 numbers being multiplied together that have the same base, you can just add the exponents of the number. Examples:

$$3^2 \times 3^3 = 3^{2+3} = 3^5$$

$$5^7 \cdot 5^{22} = 5^{7+22} = 5^{29}$$

Another property:



$$(xy)^a = x^a y^a$$

If you have 2 numbers being multiplied together in parenthesis, and the parenthesis is raised to a power, then you can “distribute” the power to both numbers. This may not seem like a helpful property, but it is if you have a variable or if you have an exponent you don’t know off the top of your head.

$$(4 \cdot 9)^4 = 4^4 \cdot 9^4$$

$$(2x)^2 = 2^2 x^2 = 4x^2$$

$$6^3 = (2 \times 3)^3 = 2^3 \times 3^3 = 8 \times 27 = 216$$

The last example is important. Many people know that 2 cubed is 8 and that 3 cubed is 27, but most people don't know what 6 cubed is. With this property, 6 was "broken down" into 2 x 3, making it possible to simplify 2 already-known exponents.

Another property:

$$x^0 = 1$$

It is not impossible to take the 0th power of a number, and a number to the 0th power is *not* 0. Any positive number, whether it's (-100) or 4.92, raised to the 0th power is 1, and any negative number raised to the 0th power is -1.

$$1000000^0 = 1$$

$$3^0 = 1$$

$$(-49)^0 = 1$$

Another important property:

$$x^{-a} = \frac{1}{x^a}$$

$$\frac{1}{x^{-a}} = x^a$$

It may seem impossible to multiply a number by itself a negative number of times, but doing so is really the same as doing 1 divided by the exponent with a positive power. Examples:

$$x^{-3} = \frac{1}{x^3}$$

$$4^{-5} = \frac{1}{4^5}$$

$$\frac{1}{3^{-3}} = 3^3$$

This property is important to know because whenever you simplify exponents, you never want to leave negative exponents in your final answer. .

$$\left(\frac{x^a}{x^b}\right) = x^{a-b}$$

This is the opposite of the multiplication property.

$$\left(\frac{9^9}{9^8}\right) = 9^{9-8} = 9$$

$$\left(\frac{2^2}{2^4}\right) = 2^{2-4} = 2^{-2} = \frac{1}{2^2} = \frac{1}{4}$$

This next property is important to know as well for when learning factoring.

$$(x^a)^b = x^{ab}$$

If you have a power raised to a power, then you can multiply the powers.

$$(6^2)^6 = 6^{2 \times 6} = 6^{12}$$

$$(((2)^2)^2)^2 = 2^{2 \cdot 2 \cdot 2} = 2^8$$

In this next property, you distribute a power again, to both the numerator and the denominator.

$$\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$$

With this property, you can take the exponent of fractions.

$$\left(\frac{1}{2}\right)^4 = \left(\frac{1^4}{2^4}\right) = \frac{1}{16}$$

$$\left(\frac{3}{2}\right)^{-2} = \frac{3^{-2}}{2^{-2}} = \frac{2^2}{3^2} = \frac{4}{9}$$

That is the last property for this section. This may seem like a lot to memorize, but you really should memorize these, and it shouldn't be too hard because these properties somewhat follow a pattern. For a final recap, all the properties will be listed again here. This time, I'll include the names of the properties because they will be referred to later.

Properties of exponents

$$x^a \cdot x^b = x^{a+b} \text{ Product of powers property}$$

$$\text{Power of a product property } (xy)^a = x^a y^a$$

$$x^0 = 1 \text{ The zero exponent property}$$

$$\text{The negative exponent property } x^{-a} = \frac{1}{x^a}$$

$$\frac{1}{x^a} = x^{-a} \text{ The negative exponent property}$$

$$\text{The quotient of powers property } \left(\frac{x^a}{x^b}\right) = x^{a-b}$$

$$(x^a)^b = x^{ab} \text{ The power of a power property}$$

$$\text{The power of a quotient property } \left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$$

Now that you know the properties of exponents, it's time to use them to solve and simplify some problems.

$$\frac{x^5}{x^3} = x^{5-3} = x^2$$

For this problem, the power of a quotient property was used to simplify this equation. When simplifying exponents, you need to put these exponents in a form where a) there are no negative exponents, such as x^{-5} , and b) each variable appears only once in the final answer. For example, $\frac{xy}{x}$ is not fully simplified because x appears twice in the fraction. In addition, any constants raised to a power must be fully expanded. If you had 2^4 , you would have to multiply it out. Therefore, 2^4 is not completely simplified, but 16 is. Simplify literally means to put something in its simplest form. Now that that is out of the way, here is our first problem.

$$(7^0)^4 = 7^0 = 1$$

The power of a power was taken, which was $0 \times 4 = 0$, and because any number to the power of 0 is 1, our final answer is 1.

$$2^{-2} \times 2^1 = 2^{-2+1} = 2^{-1} = \frac{1}{2^1} = \frac{1}{2}$$

Since 2 exponents with the same base are being multiplied, the powers can be added, and the fraction can be rewritten as the fraction 1/2 using the negative exponent property.

$$(2xy)^3 = 2^3 \cdot x^3 y^3 = 8x^3 y^3$$

Using the power of a product property, the exponent was distributed to all terms in the parenthesis.

$$\left(\frac{xy}{y^2}\right)^{-2} = \frac{(xy)^{-2}}{(y^2)^{-2}} = \frac{x^{-2}y^{-2}}{y^{-4}} = \frac{y^4}{x^2 y^2} = \frac{y^2}{x^2}$$

This problem used a lot of the properties. First, the exponent -2 was distributed to the numerator and the denominator. Next, the exponent was distributed to xy and y^2 by using the power of a product property for the numerator and the power of a power property for the denominator. Next, because the

exponents were negative, the numerator was moved to the denominator and the denominator was moved to the numerator. Finally, y was in the top and bottom of the fraction, so the exponents were subtracted to get the final answer.

$$2^{\frac{1}{2}} \cdot 2^{\frac{1}{2}} = 2^{\frac{1}{2} + \frac{1}{2}} = 2$$

You've probably never seen a fractional exponent before, but the properties of exponents still apply even to fractional exponents. Even if you have no idea what $2^{\frac{1}{2}}$ is, you could still be able to figure out that $2^{\frac{1}{2}} \cdot 2^{\frac{1}{2}}$, which is the same thing as $(2^{\frac{1}{2}})^2$, is 2.

Practice

Simplify the following expressions completely.

1. $y^2 \cdot y^3 =$

2. $\frac{x}{x^4} =$

3. $(a^3)^2 =$

4. $(a^{-2})^{-18} =$

5. $\frac{a^{\frac{1}{2}}b}{a^{\frac{1}{2}}} =$

6. $20^{0x} =$

7. $(\frac{x^{-2}y}{y^{-3}})^5 =$

8. $abc \cdot abc =$

9. $(abc)^2 =$

10. $\frac{a}{b} \cdot \frac{c}{a} \cdot \frac{b}{c} =$

11. $y^2 \cdot y^{-2} =$

12. $(\frac{z^{-1}}{c^{-1}})^{-5} =$

13. $3^{-3} =$

14. $\frac{2^{-3}}{2} =$

15. $7^{7^0} =$

$$16. (a^{\frac{1}{n}})^n =$$

$$17. 4^{\frac{1}{2}} =$$

$$18. (-5)^2 =$$

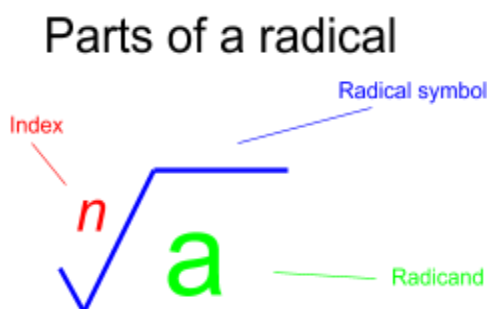
$$19. (-2)^5 =$$

$$20. a^c \cdot a =$$

$$21. (\frac{a}{b})(\frac{a}{b})^c =$$

4-2 radicals with different indexes

In chapter 0, square roots were discussed. Remember, the square root of a number is the number that, when you multiply it by itself, is equal to the radicand, which is the number under a radical ($\sqrt{}$).



However, square roots don't have an index. If a radical does not have an index, it is automatically assumed that the index is 2. Just like how $3 = 3^1$, $\sqrt{2} = \sqrt[2]{2}$.

What does the index actually mean? It actually tells you how many times the answer must be multiplied by itself to get the radicand. In other words, if $\sqrt[n]{a} = b$, then $b^n = a$. Since every radical that has been dealt with so far has an index of 2, the answer to the radical was the number that when multiplied by itself equaled the radicand. Following this rule, if $\sqrt{a} = b$, then b^2 , or $b \times b$, is equal to a . But if the index was anything different, for example $\sqrt[3]{a} = b$, then $b^3 = a$. Something important to know is that most of the time, the solution to a radical is an irrational number- that means it's a number with a decimal that goes on forever and ever without having a repeating pattern. As a result, they cannot be written as fractions. Conversely, a rational number may have an infinite decimal but has a repeating pattern. $2.3333\dots$ is a rational number because it has a pattern of repeating 3 forever. In addition, it can be written as the fraction $\frac{7}{3}$. On the other hand, the square root of 2 is an irrational number. Therefore, you should use a calculator to solve for radicals. It would be too hard to keep guessing what the answer is.

It may be hard to understand this using a bunch of variables, so we'll do some practice problems with numbers instead.

$$\sqrt{4} = ?$$

Because the index is 2, the number that when it's squared equals 4 needs to be found. In other words, if a is the solution to this problem, then for what value of a does $a \cdot a = 4$? If you can remember back to chapter 0, 4 is a square number, just like 9, 16, 25, 36, 49, 64, etc., so the answer to this problem is a rational number, 2.

$$\sqrt{4} = \sqrt{2 \cdot 2} = 2 \quad \text{Because there are two 2's.}$$

Here is a property of radicals:

$$\sqrt{bc} = \sqrt{b}\sqrt{c} \quad (\text{when } b \text{ and } c \text{ aren't both negative})$$

Basically, if you can rewrite the number under a radical as 2 numbers being multiplied, you can split those numbers into 2 different radicals. $\sqrt{b}\sqrt{c}$ is the same thing as $\sqrt{b} \cdot \sqrt{c}$. One thing you must remember is this property cannot be used if both b and c are negative. We'll solve those later.

You probably already knew how to take the square root of a square number such as 4, 9, 16, 25, etc. However, what is

$$\sqrt{5} = ?$$

The square root of 5 will be an irrational number, a number that has an decimal that goes on forever without end and does not have a pattern, so just use a calculator to find the answer to this one.

$$\sqrt{5} = 2.236...$$

The three dots (...) indicate that the number keeps going on. It would literally take forever to write the entire answer to the square root of 5. Numbers like this, which has an infinite number of decimal places with no repeating pattern and cannot be written as a fraction is called an irrational number. When you need to write an irrational number as the answer, list as many decimal places as your teacher wants.

So far, only problems with an index of 2 have been done (in other words, only square roots). Radicals with indexes other than 2 can also have whole number or irrational solutions.

$$\sqrt[3]{8} = ?$$

Since the index is 3, the answer to the problem is the number that, when raised to the 3rd power, is equal to 8.

Therefore, if "?" represent the answer to this problem,

$$?^3 = 8$$

This equation can be found by cubing both sides of the original equation. Solving this gives us

$$? = 2$$

Therefore the cube root of 8 is 2 since $2^3 = 8$. Another way to solve this is to rewrite the radical as the following:

$$\sqrt[3]{8} = \sqrt[3]{2 \cdot 2 \cdot 2} = 2 \quad \text{Because there are three 2's; 2 is being multiplied by itself 3 times}$$

For future reference, a cube root is a radical with an index of 3, similar to how a radical with an index of 2 is a square root. If the index is an integer n , the radicand is a (getting $\sqrt[n]{a}$), and $n > 3$, then it is problem is read aloud as the n th root of a . $\sqrt[5]{28}$ is read as the fifth root of 28.

The index of a radical is almost always greater than or equal to 2. In the above problem, because the index was 3, the problem was asking what number raised to the third power (or cubed) is equal to the radicand. If the index is 5, the problem would be asking what number raised to the fifth power is equal to the radicand.

Following this pattern, if you know that the following is true:

$$2^2 = 4,$$

$$2^3 = 8,$$

$$2^4 = 16,$$

$$2^5 = 32,$$

$$a^c = b,$$

$$\text{Then } \sqrt{4} = 2$$

$$\text{Then } \sqrt[3]{8} = 2$$

$$\text{Then } \sqrt[4]{16} = 2$$

$$\text{Then } \sqrt[5]{32} = 2$$

$$\text{Then } \sqrt[c]{b} = a$$

4-3 n th roots of fractions and fractional exponents

If you can remember from 4-1, the problem $(2^{1/2})^2$ was done. Using the properties of exponents, it can be figured out that this is equal to $2^{1/2 \times 2} = 2$. What is the value of $2^{1/2}$? A problem like 2^2 is easy because you just multiply 2 by itself 2 times. It doesn't really make sense to multiply a number by itself 1/2 times. Consider this: the square root of a number (the radicand) is the number that, when you multiply it by itself, will equal the radicand. In the previous section, it was said that if you know $a^n = b$, then $\sqrt[n]{b} = a$.

Therefore, because it is known that $(2^{1/2})^2 = 2$, then $\sqrt{2} = 2^{1/2}$. Another way to look at this, because

$$(2^{1/2})^2 = 2,$$

By taking the square root both sides of the equation, the result is

$$2^{1/2} = \sqrt{2}.$$

As a general rule:

$$a^{1/n} = \sqrt[n]{a}$$

However, this rule only applies to fraction exponents with a numerator of 1. What is the value of $3^{2/3}$? If we raised this number to the 3rd power, the denominator of the fraction would cancel out, leaving us with

$$(3^{2/3})^3 = 3^2 = 9$$

Since $(3^{2/3})^3 = 9$, then by taking the cube root of both sides, we get that $3^{2/3} = \sqrt[3]{9}$, or $\sqrt[3]{3^2}$ since $3^2 = 9$. If you do this same thing with other numbers raised to a fractional power, you can actually convert these fractional exponents into radicals.

$$5^{4/5} = ?$$

Because $(5^{4/5})^5 = 5^4$,

$$5^{4/5} = \sqrt[5]{5^4}$$

By taking the fifth root of both sides of the equation. If you kept doing this, you will notice this pattern:

$$a^{\frac{b}{c}} = \sqrt[c]{a^b}$$

Using this method, you can convert any fractional exponent into a radical or vice versa. Why is this useful to know? If you were told to simplify the following problem:

$$\sqrt[3]{3^2} \times \sqrt[7]{27}$$

You could first convert these into fractional exponents,

$$3^{\frac{2}{3}} \times 3^{\frac{3}{7}} \quad (\text{note that } 27 \text{ is } 3 \text{ cubed, so } \sqrt[7]{3^3} \text{ is } 3^{\frac{3}{7}})$$

Then use the properties of exponents. When you have 2 numbers with the same base but different exponents being multiplied, you can add the powers (Product of a power property).

$$3^{\frac{2}{3} + \frac{3}{7}} = 3^{\frac{15}{21}} = 3^{\frac{5}{7}} = \sqrt[7]{3^5} = \sqrt[7]{243}$$

You could put $3^{\frac{5}{7}}$ as your final answer, but in most cases your math teacher will want you to give your answer in the same form as the problem originally had, meaning that if the problem originally had a radical, then if you cannot write the final answer without a radical or exponent, then write the final answer as a radical as well. That means that when simplifying, don't write your answer as $\sqrt{9}$ or $(3^{\frac{1}{2}})^2$ if the original problem was $\sqrt[3]{27}$, but just 3. In this case, because the original problem was a radical, you should write out the answer as $\sqrt[7]{243}$.

4-4 Simplifying square roots

In the first section of this chapter, you learned how to simplify exponents. This chapter shows how to simplify radicals. First, how would you simplify the following:

$$\sqrt{8} =$$

There is no whole number that can solve this; 1 squared is 1, 2 squared is 4, and 3 squared is 9. However, you can rewrite this as the following:

$$\sqrt{8} = \sqrt{4 \cdot 2}$$

How does this help? If you can remember from the previous chapter, you can split the 2 numbers up (using $\sqrt{bc} = \sqrt{b}\sqrt{c}$):

$$\sqrt{4 \cdot 2} = \sqrt{4}\sqrt{2}$$

But remember, the square root of 4 is the same thing as 2, since 2 squared is 4 ($\sqrt{4} = 2$), so we can rewrite this as

$$\sqrt{4}\sqrt{2} = 2\sqrt{2}$$

Also, since $2\sqrt{2} = 8$, you can figure out that generally $a\sqrt{b} = \sqrt{b \cdot a^2}$

From $\sqrt{8} = 2\sqrt{2}$, which is read as "2 times the square root of 2." In general, if the number under a square root radical, the radicand, can be divided by a square number (such as 4, 9, 16, 25, etc.), then you can simplify the square root. Following this rule, if you could simplify these radicals.

$$\sqrt{27} = \sqrt{9 \cdot 3} = \sqrt{9}\sqrt{3} = 3\sqrt{3}$$

$$\sqrt{50} = \sqrt{25 \cdot 2} = \sqrt{25}\sqrt{2} = 5\sqrt{2}$$

$$\sqrt{40} = \sqrt{4 \cdot 10} = \sqrt{4}\sqrt{10} = 2\sqrt{10}$$

However, there is a problem if you try to take the square root of a negative number. What is $\sqrt{-4}$? This problem is asking "what number multiplied by itself twice is -4?" But only way to multiply 2 numbers and get a negative number back is if you multiply a positive number and a negative number. Recall from chapter 0 that 2 positive numbers multiplied by each other or 2 negative numbers multiplied by each other is always positive. You will reach the conclusion that there is no real number that can be multiplied by itself will equal any negative number, including -4. To be able to solve this kind of problem, a new number

was invented called an imaginary number, written as i . $i = \sqrt{-1}$, and is called an imaginary number because there is no real number that can solve $\sqrt{-1}$. Likewise, any number with i or a negative radical with an even index is called an imaginary number. A real number is any number that is not an imaginary number. Because i is the square root of negative 1, $i^2 = -1$. Here are a couple examples:

$$\sqrt{-4} = \sqrt{-1}\sqrt{4} = i \cdot 2 = 2i$$

$$\sqrt{-25} = \sqrt{-1}\sqrt{25} = i \cdot 5 = 5i$$

Interestingly enough, 2 imaginary numbers multiplied together can give you a real number answer.

$$\sqrt{-4}\sqrt{-25} = i\sqrt{4} \cdot i\sqrt{25} = 2i \cdot 5i = 10i^2 = -10, \text{ since } i^2 = -1.$$

As mentioned earlier in the chapter, you cannot do $\sqrt{b}\sqrt{c} = \sqrt{bc}$ if b or c is negative. Here's why:

$$\sqrt{-4}\sqrt{-25} = \sqrt{(-4)(-25)} = \sqrt{100} = 10 \dots \text{ but this is wrong.}$$

But this is actually not equal to 10. Before, we solved this exact same problem and got -10, not positive 10. Make sure you do not multiply the radicals like this the 2nd way, because you will get the answer wrong.

So far, we learned that because no real number can be multiplied by itself to get a negative result, we have to use the imaginary number i . A negative number squared is positive, and a positive number squared is also positive. That being said, earlier in the chapter we said that

$$\sqrt{4} = 2$$

But to make that a little more accurate, you'd actually want to write this (which is important later):

$$\sqrt{4} = \pm 2$$

The answer is read as "plus or minus 2", and it means that there is actually 2 answers to this problem: positive 2 and negative 2. If you think about it, -2^2 and 2^2 are both 4. That would also mean that $\sqrt{36} = \pm 6$ because 6^2 and $(-6)^2$ are both 36.

There is also a way to simplify the square root of fractions. For example, what is the square root of $\frac{1}{2}$?

$$\sqrt{\frac{1}{2}} =$$

First, you can rewrite this as

$$\left(\frac{1}{2}\right)^{\frac{1}{2}}$$

Then, using one of the properties of exponents, you can distribute the exponent to the top and the bottom of the fraction.

$$\frac{1^{\frac{1}{2}}}{2^{\frac{1}{2}}}$$

Now you can rewrite this as

$$\frac{\sqrt{1}}{\sqrt{2}}$$

In general,

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Back to simplifying our problem, $\sqrt{\frac{1}{2}}$, we are not done yet. $\sqrt{2}$ is not a whole number, but $\sqrt{1}$ is 1. Therefore, we can rewrite this as

$$\frac{\sqrt{1}}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

However, we are not done. Many math teachers will not allow you to write an answer with a radical in the denominator of the fraction. You will have to “rationalize” the denominator, or somehow take out the square root in the denominator of this fraction.

There is an easy way to do this. Since $(\sqrt[n]{a})^n = a$, $(\sqrt{a})^2 = a$. Also remember from chapter 1, if you multiply or divide the numerator and denominator (top and bottom) of a fraction by the same number, then the value of the fraction is the same $\left(\frac{a}{b} = \frac{ac}{bc}\right)$.

Applying these rules to this problem, we know that $(\sqrt{2})^2 = 2$. If we multiply the denominator of this fraction by $\sqrt{2}$, we will have 2 in the denominator, so there will be no radical in the denominator. However, $\frac{1}{\sqrt{2}} \neq \frac{1}{2}$ ($\neq$ means “is not equal to”). We must multiply the top and bottom of the fraction by the same number so that we don’t change the value- multiplying a number by $\frac{c}{c}$, where c is any number, doesn’t change the value since $\frac{c}{c} = 1$. Doing so gives us this:

$$\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

This is our final answer. Because we are multiplying the fraction by $\frac{\sqrt{2}}{\sqrt{2}}$, which is equal to 1, we aren't changing the value of the fraction- similar to how $\frac{1}{2} = \frac{2}{4}$. Notice that when you are asked to simplify fractions under a radical, you are really being asked to rationalize the denominator and simplify the result if needed (like $\sqrt{8} = 2\sqrt{2}$). Here are some more examples.

$$\sqrt{\frac{2}{3}} = \frac{\sqrt{2}}{\sqrt{3}} = \frac{\sqrt{2}}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{6}}{3}$$

$$\sqrt{\frac{1}{5}} = \frac{\sqrt{1}}{\sqrt{5}} = \frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

$$\sqrt{\frac{7}{4}} = \frac{\sqrt{7}}{\sqrt{4}} = \frac{\sqrt{7}}{2}$$

$$\sqrt{\frac{5}{2}} = \frac{\sqrt{5}}{\sqrt{2}} = \frac{\sqrt{5}}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{10}}{2}$$

$$\sqrt{\frac{32}{45}} = \frac{\sqrt{32}}{\sqrt{45}} = \frac{4\sqrt{2}}{3\sqrt{5}} = \frac{4\sqrt{2}}{3\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{4\sqrt{10}}{3 \cdot 5} = \frac{4\sqrt{10}}{15}$$

There is one last thing you need to know: how to solve the square root of a square root. How do you solve $\sqrt{\sqrt{32}}$? If we use our exponent rules again, we can rewrite this as

$$\sqrt{\sqrt{32}} = \sqrt{32^{\frac{1}{2}}}$$

But we can take this a step further.

$$\sqrt{32^{\frac{1}{2}}} = (32^{\frac{1}{2}})^{\frac{1}{2}}$$

And since this is a power to a power, we can just multiply the powers to get

$$(32^{\frac{1}{2}})^{\frac{1}{2}} = 32^{\frac{1}{4}} = \sqrt[4]{32}$$

Therefore, $\sqrt{\sqrt{32}} = \sqrt[4]{32}$. In general,

$$\sqrt[a]{\sqrt[b]{c}} = \sqrt[ab]{c} \text{ (when a and b are positive integers.)}$$

Knowing this, you could figure out that the following values are equivalent:

$$\sqrt[3]{\sqrt{3}} = \sqrt[6]{3}$$

$$\sqrt[3]{\sqrt[4]{2}} = \sqrt[12]{2}$$

$$\sqrt{\sqrt{100}} = \sqrt[4]{100} \text{ or } \sqrt{10}$$

$$\sqrt[3]{\sqrt{3}} = \sqrt[6]{3} = 3$$

$$\sqrt{\sqrt[3]{98}} = \sqrt[6]{98} = \sqrt[12]{98}$$

In Recap:

$$\text{if } \sqrt[n]{b} = a, \text{ then } a^n = b$$

$$a^{\frac{b}{c}} = \sqrt[c]{a^b}$$

$$\sqrt{b}\sqrt{c} = \sqrt{bc} \text{ when } b \text{ and } c \text{ aren't both negative}$$

$$\sqrt[n]{a^n} = a$$

$$\sqrt[n]{a^b} = (\sqrt[n]{a})^b$$

$$\sqrt{b \cdot a^2} = a\sqrt{b} \text{ and } \sqrt[n]{b \cdot a^n} = a\sqrt[n]{b}$$

$$\sqrt{-a} = i\sqrt{a}$$

$$i^2 = -1$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$\sqrt[a]{\sqrt[b]{c}} = \sqrt[ab]{c}$$

$$\frac{\sqrt{a}}{\sqrt{b}} = \frac{\sqrt{a}\sqrt{b}}{b}$$

Practice

Write all of the following expressions in simplest form. Only the positive or 0 answer is needed. Assume $i = \sqrt{-1}$.

$$1. \sqrt{27} =$$

$$2. i^3 =$$

$$3. \sqrt[5]{32} =$$

$$4. \frac{\sqrt{27}}{\sqrt{3}} =$$

$$5. \sqrt[9]{9^9} =$$

$$6. \sqrt{\sqrt{\sqrt{19}}} =$$

$$7. \sqrt{-\frac{4}{7}} \sqrt{-\frac{7}{4}} =$$

$$8. \sqrt{\frac{25}{6}} =$$

$$9. \frac{1}{2}\sqrt{2}$$

$$10. \frac{7i}{10i} =$$

$$11. \sqrt{\left(-\frac{4}{7}\right)\left(-\frac{7}{4}\right)} =$$

$$12. \sqrt{\sqrt{8}} \cdot \sqrt{4} =$$

$$13. \sqrt{a^3} =$$

$$14. \sqrt[3]{3^2} \times \sqrt[2]{27} =$$

$$15. \sqrt{\frac{1}{\sqrt{16}}} =$$

$$16. 2(\sqrt[3]{x})^3 =$$

$$17. 2\sqrt{2} \cdot \frac{1}{2}\sqrt{16} =$$

Chapter 5

Quadratic Equations

So far, you have learned what linear equations are and how to graph them, as well as see where 2 intersect. This chapter teaches how to graph a new type of equation, the quadratic equation.

5-1 What are functions?

So far, you have graphed equations in the form $y = mx + b$. You choose a value of x , plug it into the equation, and then determine what the y value of the line is at that x value. However, you can also write y as $f(x)$.

$$f(x) = mx + b$$

$f(x)$ is the exact same thing as y , and they are usually interchangeable. $f(x)$ is read as “ f of x ”, and it stands for function of x . There is a reason for this notation. If you are given the equation $y = -2x + 10$, and you need to know what the y value of this function is at $x = 5$, then you would do the following:

$$y = -2(5) + 10 = 0$$

Since $x = 5$, 5 was substituted for x into the equation. However, another way to write that 5 is being plugged in for x without writing it as a sentence is first writing the equation with $f(x)$ instead of y .

$$f(x) = -2x + 10$$

Remember, this is the same thing as $y = -2x + 10$, but we are writing $f(x)$ instead. With equations written in function notation, you can show substitution easily by replacing x with a constant (a number that is not multiplied by a variable). In this case, If I wanted to show I was substituting 5 for x in this equation, since $x = 5$, we can just put a 5 instead of x at any place x was in the original equation.

$$f(5) = -2(5) + 10 = 0.$$

$f(5)$ shows that you plugged in 5 into the function. While you are doing the same thing, writing the equation like this makes it clear you are plugging in 5 for the variable of this equation. If you know that $f(5) = 0$, then you know that the equation passes through $(5, 0)$. Think about this: $y = f(x)$, and if a function passes through (x, y) , then it passes through $(x, f(x))$. You could choose any value for x and this would still be true. That being said, how would you solve this?

For a linear function $f(x)$, $f(2) = 3$ and $f(7) = 10$. What is the equation of the function?

Looking at this problem, we know that if $f(2) = 3$, then when $x = 2$, $f(2)$, which is the same thing as y , is 3, since 2 was substituted for x in the unknown equation, and the result was 3. This also means that $f(x)$ passes through $(2, 3)$. Since $f(7) = 10$, it also passes through $(7, 10)$. In other words, $f(a) = b$ is just another way of saying that the function passes through (a, b) . We can use the equation

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

To find the slope of the function. Since the points are $(2, 3)$ and $(7, 10)$,

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{10 - 3}{7 - 2} = \frac{7}{5}$$

To find the y- intercept, since the equation of this function is

$$y = \frac{7}{5}x + b$$

, plugging in (2 , 3):

$$3 = \frac{7}{5}(2) + b$$

$$b = \frac{1}{5}$$

$$y = \frac{7}{5}x + \frac{1}{5} \text{ (plugging in } b \text{ back into the first equation)}$$

In conclusion, if you have a function $f(x)$, and a number a is substituted into the equation for x , the result is $f(a)$ since $x = a$.

In addition, if you have multiple functions, you can name them differently. One function could be called $f(x)$, another could be written as $g(x)$, and another as $h(x)$ or $t(x)$ just so it is easy to tell the difference between multiple functions.

To make sure that you really understand how functions work, if you have a function

$$f(x) = x + a$$

Then the following would be true:

$$f(1) = 1 + a$$

$$f(2) = 2 + a$$

$$f(3) = 3 + a$$

$$f(-5) = -5 + a$$

$$f(a) = a + a = 2a$$

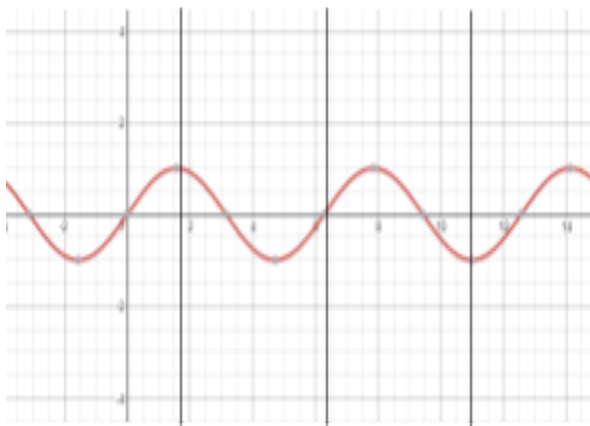
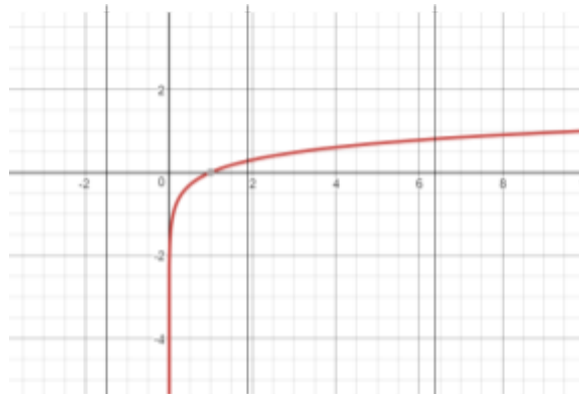
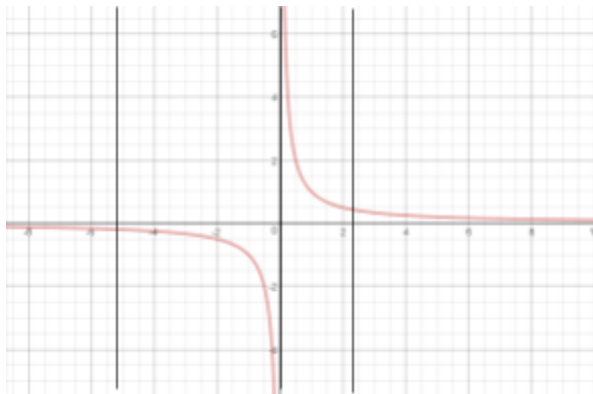
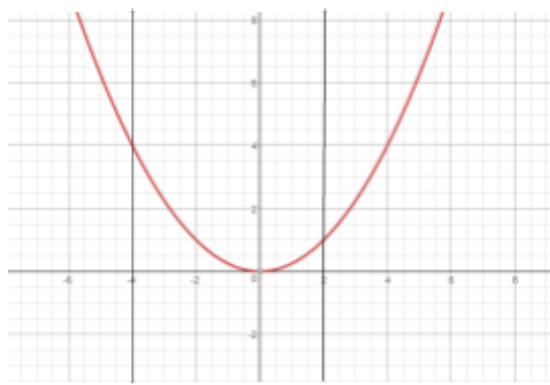
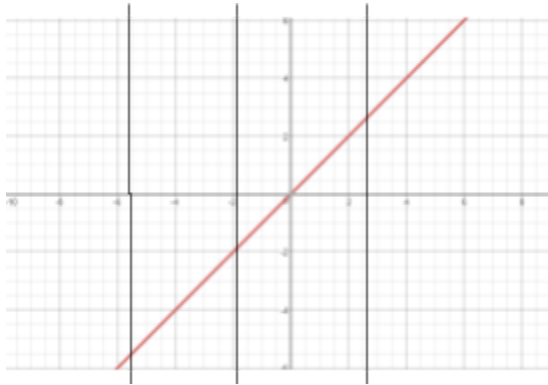
$$f(b) = b + a$$

$$f(2^2) = 2^2 + a = 4 + a$$

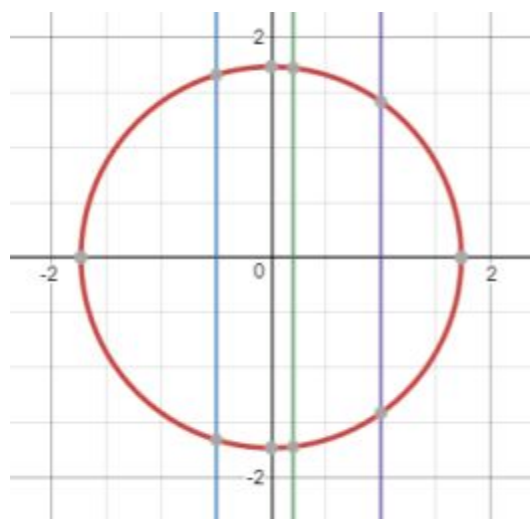
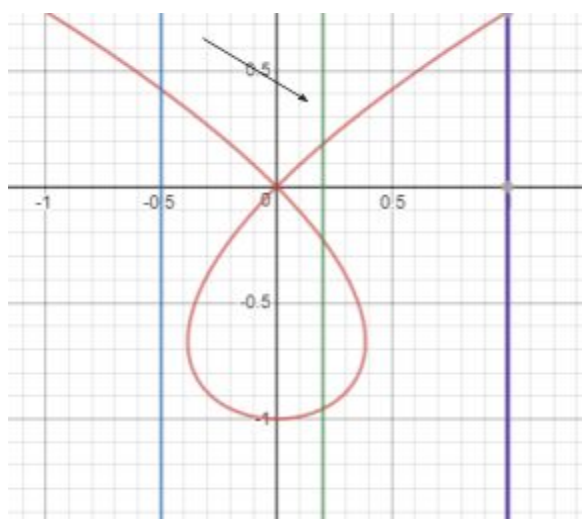
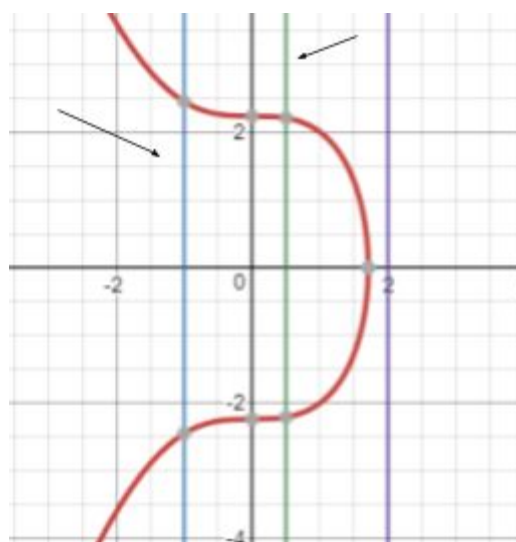
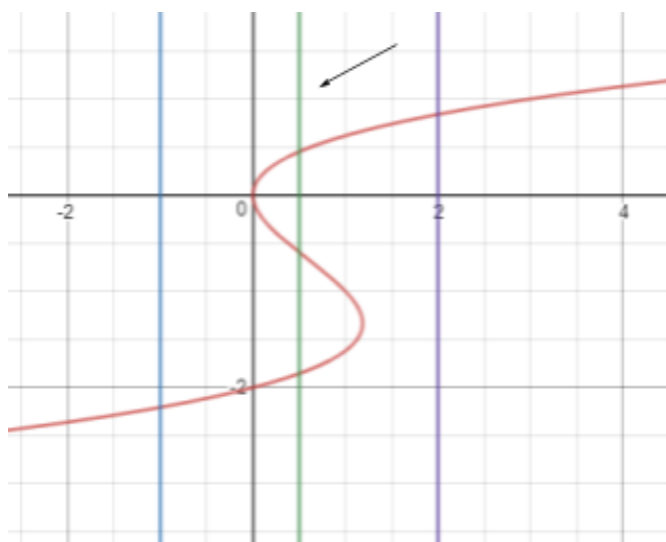
$$f(a^2) = a^2 + a$$

In each case, x is replaced by a number, and that number is plugged into the equation in the place of x .

It is important to know that not any equation is a function, and that you can only use function notation for functions. But what is a function? In short, a function is an equation which for every single x value, there is only one y value. Any equation that is not a function is called a relation. The easiest way to tell if the graph of an equation is a function is to draw a vertical line and see if the vertical line passes through it more than once. If it does, then the equation has more than one y value for one x value, and is not a function. However, if at every point a vertical line can be drawn the lines only pass through the equation at maximum of 1 time, then it is a function. Here are some examples of functions:



This is called the vertical line test. If a curve passes the vertical line test, meaning that a vertical line drawn at any point crosses the curve at a maximum of 1 time, then it is a function. Here are some examples of equations that are not functions:



In the previous chapters, we only graphed linear functions, or functions that can be written as $y = mx + b$. In this chapter, you will learn how to graph quadratic functions.

5-2 The general form of a quadratic equation

Before we start graphing quadratic equations, you need to know what makes a function quadratic. A linear function is linear if $y = mx + b$, where m and b are any real numbers. This is called the general form of a linear equation. The general form of a quadratic equation is

$$f(x) = ax^2 + bx + c \quad (a \neq 0)$$

It is the same as the general form of a linear equation except ax^2 is added. a cannot be 0 because if it were, $f(x) = bx + c$, since anything multiplied by 0 is 0, and this would just be a linear equation. b and c can be 0.

Some examples of quadratic equations are:

$$y = x^2$$

$$y = -5x^2 + 3x$$

$$f(x) = \frac{1}{2}x^2 + 4x$$

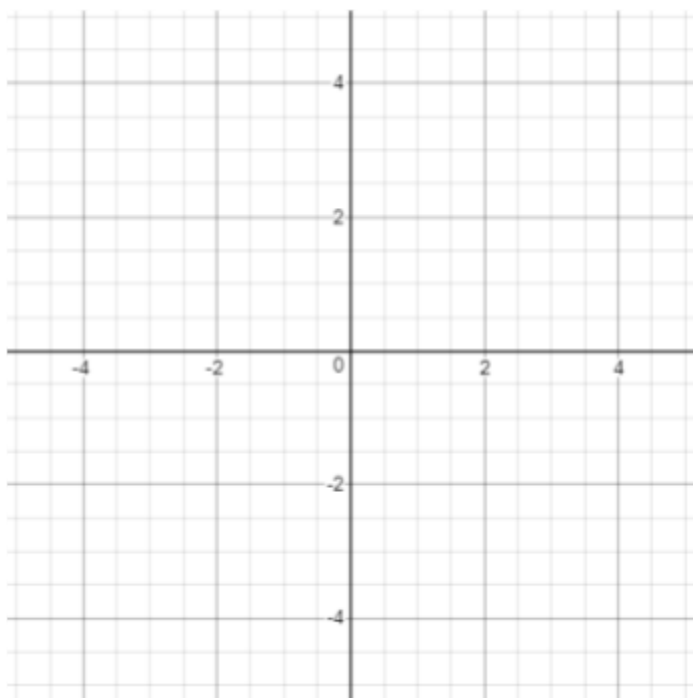
$$f(x) = x^2 - x - 23$$

The pattern is that if the highest power/exponent of x in the equation is 2, and that the variable is always raised to a positive whole number or 0, then it is quadratic.

5-3 Graphing Quadratic Equations that $a=1$

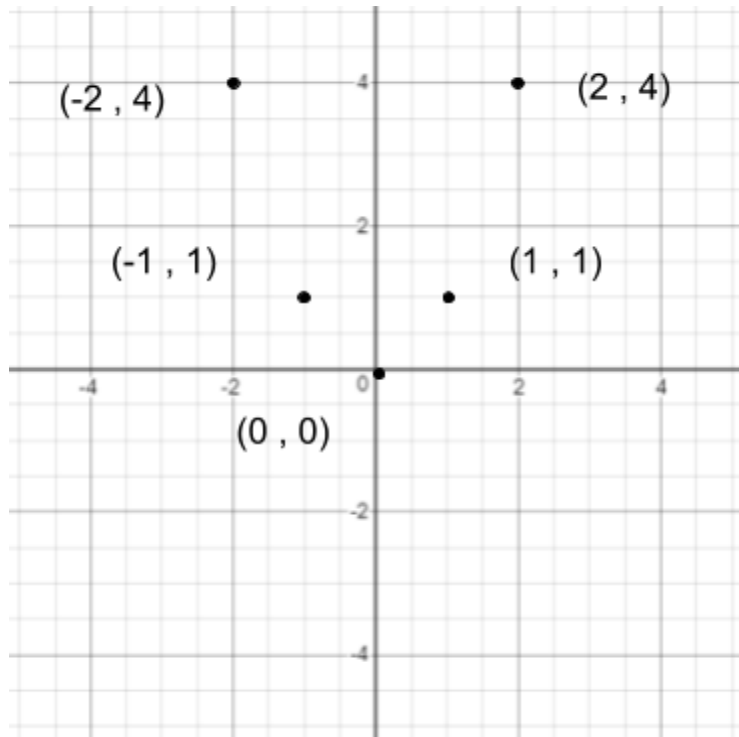
What do quadratic equations look like when they are graphed? To graph any equation, such as a linear equation, like always, we just plug in numbers for x and get a y value. Now we just need to do that for a quadratic function. This is what the most basic quadratic function, $f(x) = x^2$, looks like when graphed.

x	$f(x)$
-3	$f(-3) = (-3)^2 = 9$
-2	$f(-2) = (-2)^2 = 4$
-1	$f(-1) = (-1)^2 = 1$
0	$f(0) = 0^2 = 0$
1	$f(1) = 1^2 = 1$
2	$f(2) = 2^2 = 4$
3	$f(3) = 3^2 = 9$

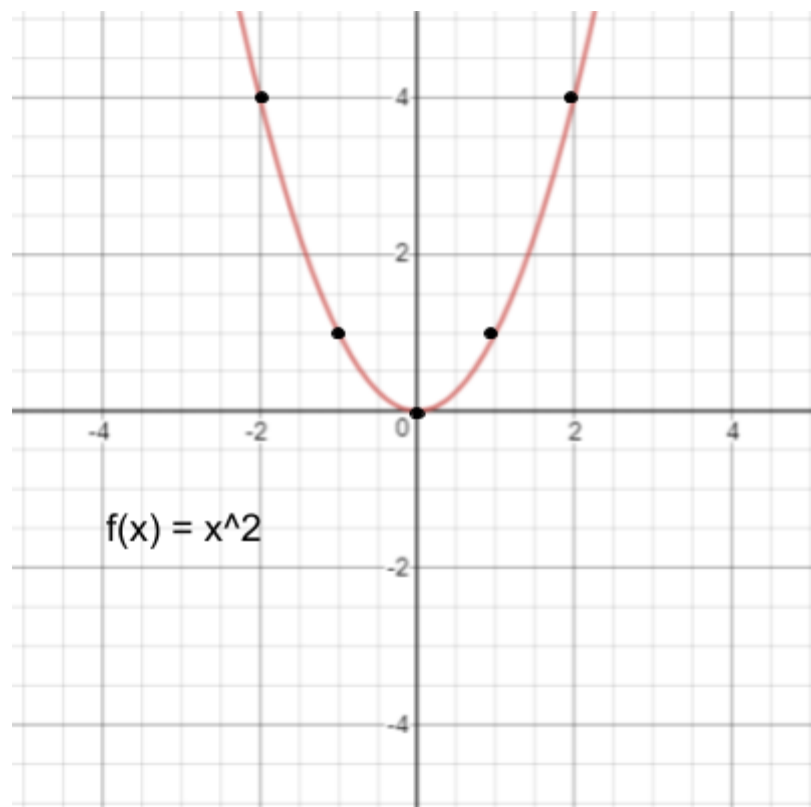


Note: “^” means “raised to the power of”, and indicates an exponent.

This is because $f(-3) = (-3)^2 = 9$, $f(-2) = (-2)^2 = 4$, $f(-1) = (-1)^2 = 1$, $f(0) = 0$, and so on. This means the function passes through $(-3, 9)$, $(-2, 4)$, $(-1, 1)$, $(0, 0)$, and so on. Once you find the points the graph goes through, just draw a curved line through the points. Here are the coordinates drawn:



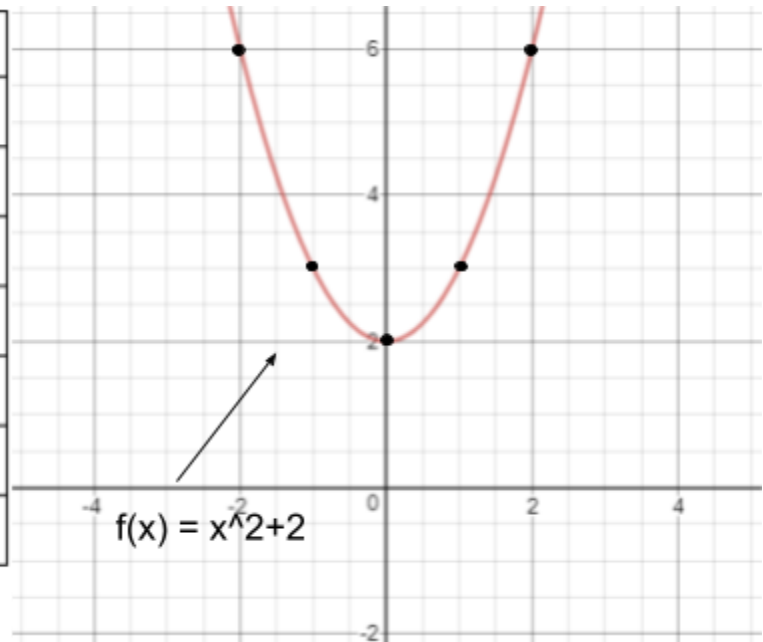
Once you know the coordinates the function passes through and plot the points, all you have to do is connect the dots. When you do that, it will look like this:



The general shape of a quadratic function, unlike a linear function, is not a straight line, but rather curves. This shape is called a parabola.

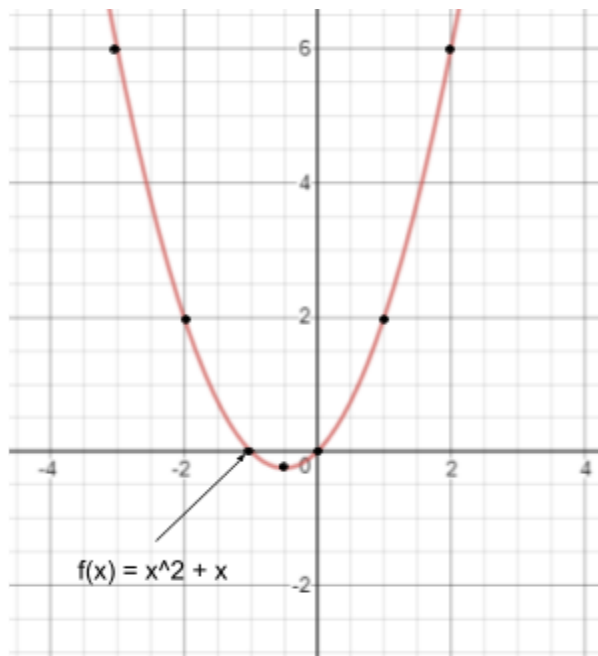
Here is the graph of $f(x) = x^2 + 2$.

x	f(x)
-3	$f(-3) = (-3)^2 + 2 = 9 + 2 = 11$
-2	$f(-2) = (-2)^2 + 2 = 4 + 2 = 6$
-1	$f(-1) = (-1)^2 + 2 = 1 + 2 = 3$
0	$f(0) = 0^2 + 2 = 2$
1	$f(1) = 1^2 + 2 = 3$
2	$f(2) = 2^2 + 2 = 6$
3	$f(3) = 3^2 + 2 = 11$



Here is another example of a quadratic equation, $f(x) = x^2 + x$.

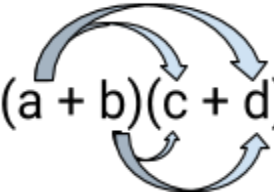
x	y
-3	6
-2	2
-1	0
-1/2	-1/4
0	0
1	2
2	6



What if you were told to graph $y = (x + 3)(x - 2)$? While it may not look like a quadratic equation, you can distribute these terms to make it into the form $y = ax^2 + bx + c$. In Chapter 1, it was shown that


$$a(b + c) = ab + ac$$

And also that


$$(a + b)(c + d) = ac + ad + bc + bd$$

If we expand the 2 factors like this (don't forget that the 2 is negative!),

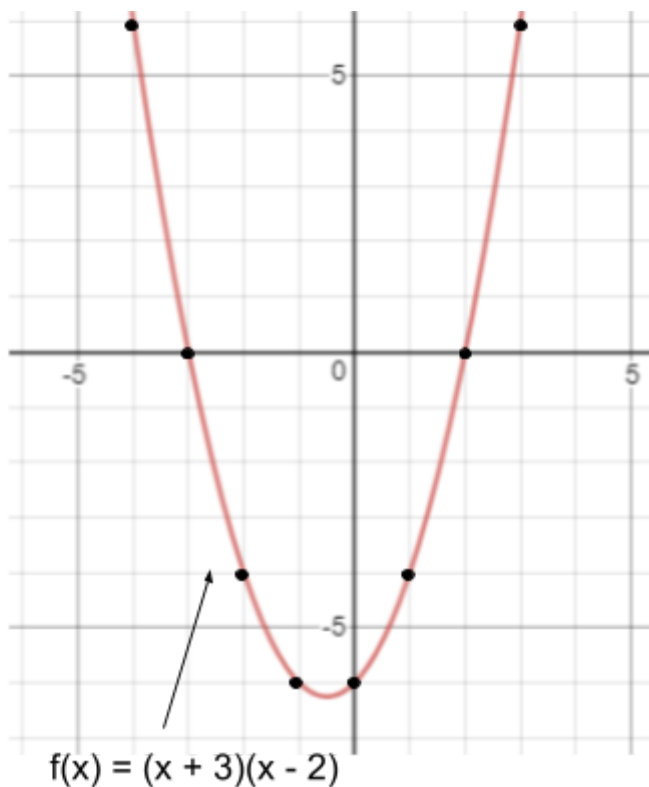
$$(x + 3)(x - 2) = x(x) + x(-2) + 3(x) + 3(-2) = x^2 - 2x + 3x - 6$$

Since $-2x$ and $3x$ are like terms, we can combine them to get

$$x^2 - 2x + 3x - 6 = x^2 + x - 6$$

Now the equation is written in Quadratic form. $a = 1$, $b = 1$, and $c = -6$. However, when you're actually graphing, doing this is not necessary. The terms were distributed to show you that the equation is quadratic. Because the function is a quadratic function, the graph will be a parabola.

x	f(x)
-3	$f(-3) = ((-3) + 3)((-3) - 2) = 0(-5) = 0$
-2	$f(-2) = ((-2) + 3)((-2) - 2) = 1(-4) = -4$
-1	$f(-1) = ((-1) + 3)((-1) - 2) = 2(-3) = -6$
0	$f(0) = (0 + 3)(0 - 2) = 3(-2) = -6$
1	$f(1) = (1 + 3)(1 - 2) = 4(-1) = -4$
2	$f(2) = (2 + 3)(2 - 2) = 5(0) = 0$
3	$f(3) = (3 + 3)(3 - 2) = 6(1) = 6$

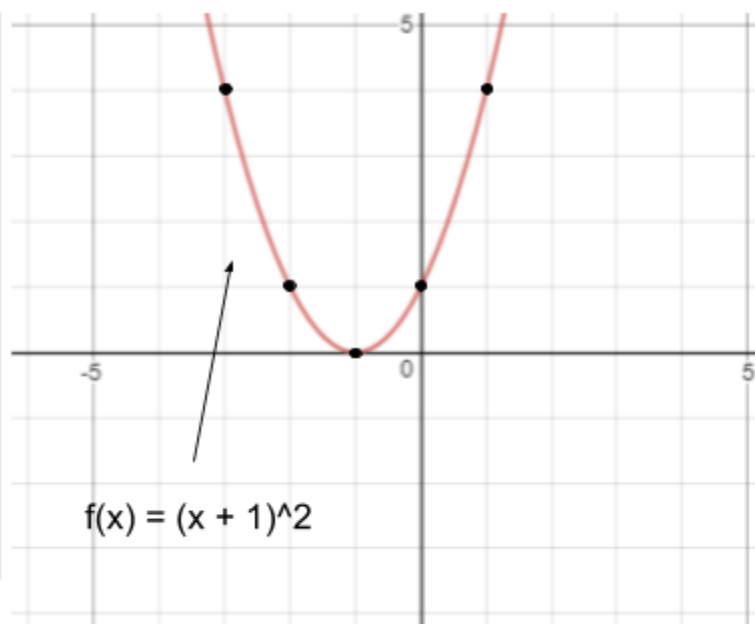


You can do something similar with the function $f(x) = (x + 1)^2$. To prove it's a function, you could expand it as well.

$$(x + 1)^2 = (x + 1)(x + 1) = x^2 + x + x + 1 = x^2 + 2x + 1$$

Again, this is just to show it's a quadratic equation. When you graph it, you could just use the original equation. It doesn't matter which one you use since you will get the same y values.

x	f(x)
-2	$f(-2) = ((-2) + 1)^2 = (-1)^2 = 1$
-1	$f(-1) = ((-1) + 1)^2 = 0^2 = 0$
0	$f(0) = (0 + 1)^2 = 1$
1	$f(1) = (1 + 1)^2 = 2^2 = 4$
2	$f(2) = (2 + 1)^2 = 3^2 = 9$

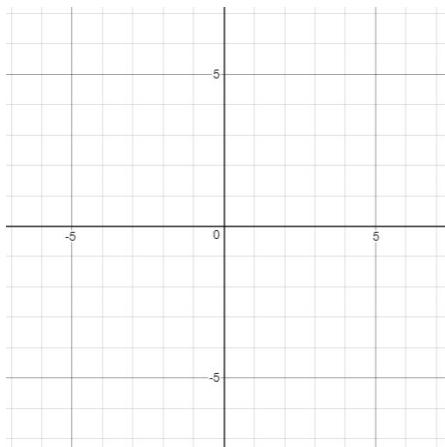


Just remember that whenever you graph a function, to find out what coordinates the function passes through, just plug in a value for x, get back a value for y, and you know a point that the function passes through.

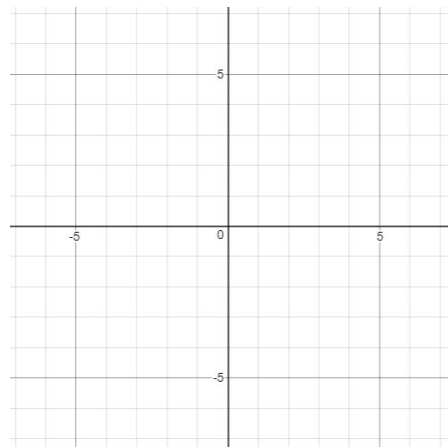
Practice

Graph the following quadratic equations.

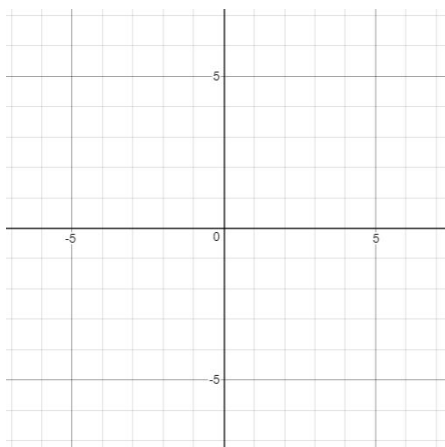
1. $f(x) = x^2 - 3$



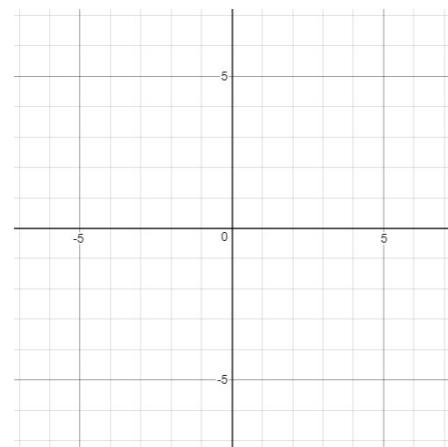
2. $y = x^2 - x$



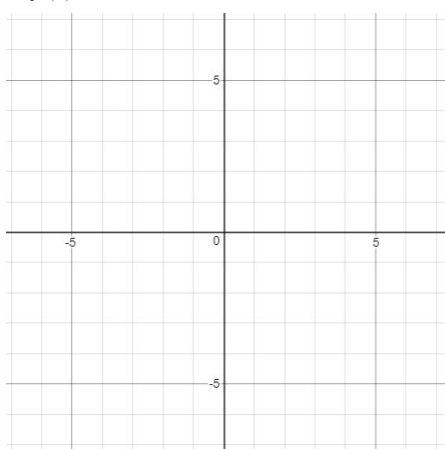
3. $f(x) = x^2 - 3x$



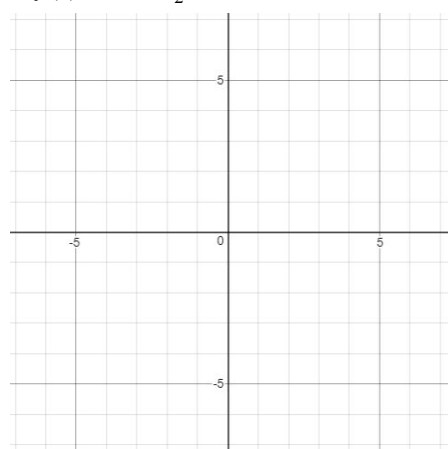
4. $f(x) = x^2 - 3x - 3$



4. $f(x) = x^2 + 2x + 1$



5. $f(x) = x^2 + \frac{1}{2}x - 2$

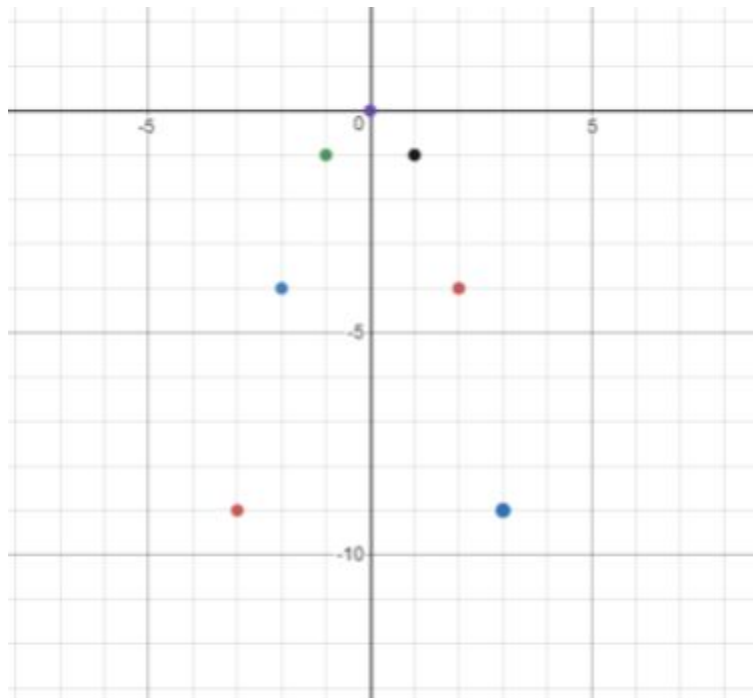


5-4 Graphing quadratics that $a \neq 1$

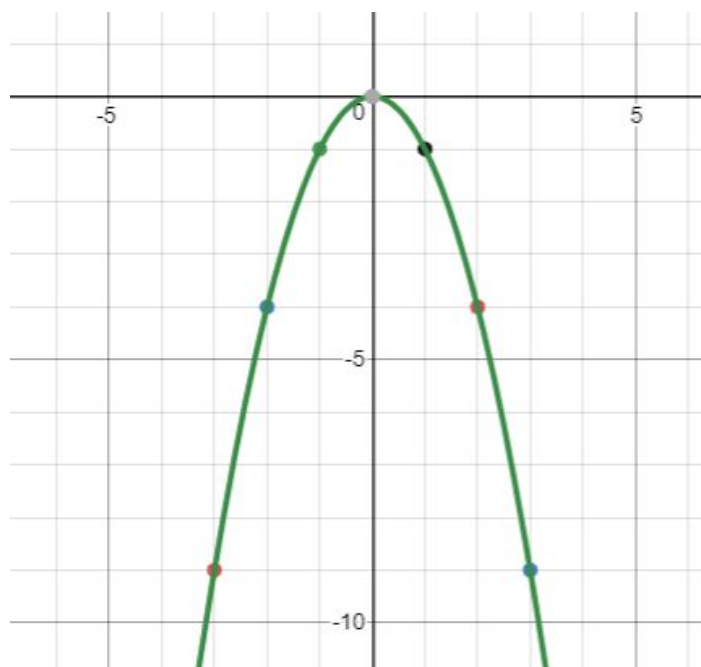
So far, we have graphed some equations in the form $f(x) = ax^2 + bx + c$. All the functions we have graphed either had $a = 0$ or $a = 1$. How would you graph $y = -x^2$? First, changing y to $f(x)$, we get the function $f(x) = -x^2$. Plugging in values of x , we get the following table:

x	$f(x)$
-4	-16
-3	-9
-2	-4
-1	-1
0	0
1	-1
2	-4
3	-9
4	-16

This means the function passes through $(-4, -16)$, $(-3, -9)$, $(-2, -4)$, and so on. Plotting the points on a graph looks like this:

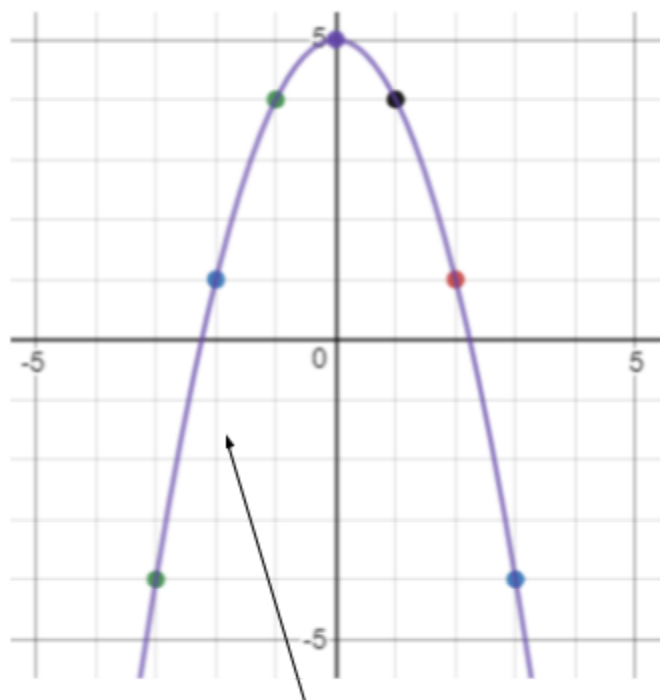


Connect the dots, and you will have the graph of the function.



For $f(x) = -x^2 + 5$, you'd do the same thing.

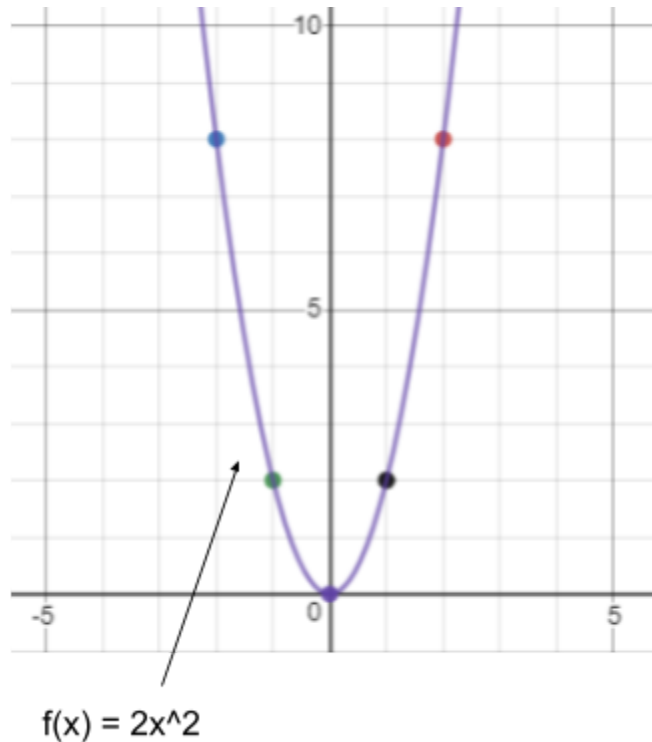
x	y
-3	-4
-2	1
-1	4
0	5
1	4
2	1
3	-4



$$f(x) = -x^2 + 5$$

How would one go about graphing $f(x) = 2x^2$? Like always, we plug in numbers for x , get a number for y , plot the coordinates, and connect the dots to get a parabola shape.

x	y
-3	18
-2	8
-1	2
0	0
1	2
2	8
3	18

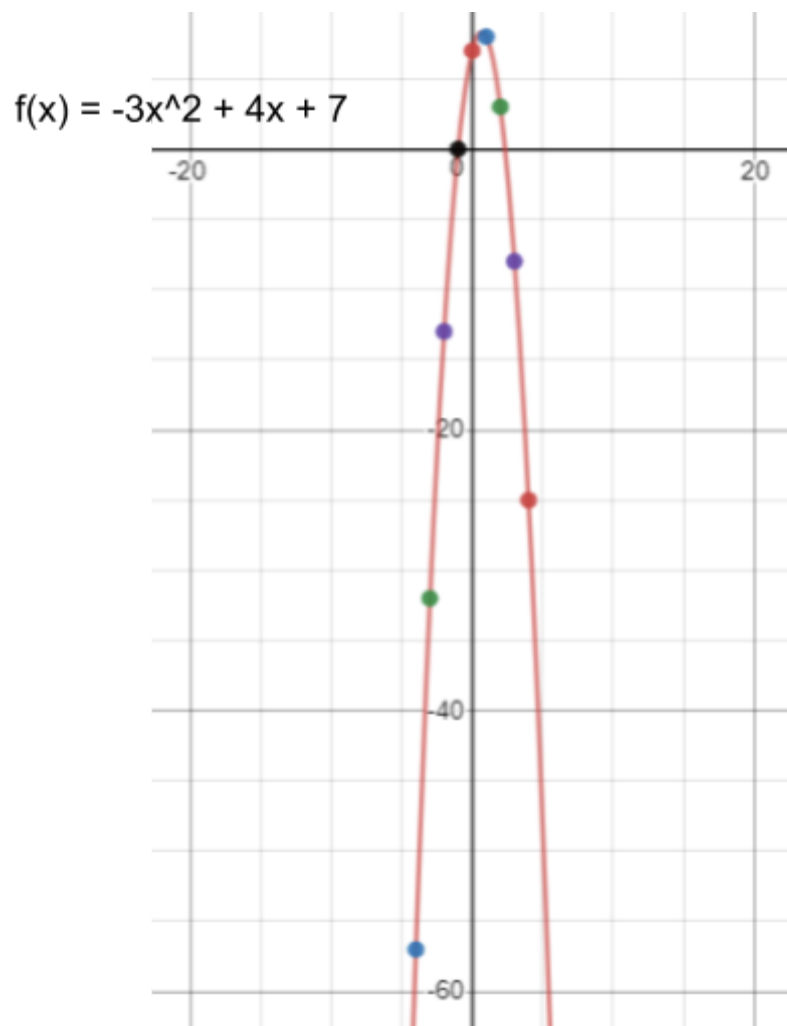


A quadratic that would take a little bit more work to graph would be $f(x) = -3x^2 + 4x + 7$. First, make a table.

x	f(x)
-4	$-3(-4)^2 + 4(-4) + 7 =$ $-3(16) - 16 + 7 =$ $-48 - 9 = -57$
-3	$-3(-3)^2 + 4(-3) + 7 =$ $-3(9) - 12 + 7 =$ $-27 - 5 = -32$
-2	$-3(-2)^2 + 4(-2) + 7 =$ $-3(4) - 8 + 7 =$ $-12 - 1 = -13$
-1	$-3(-1)^2 + 4(-1) + 7 =$ $-3(1) - 4 + 7 =$ $-3 + 3 = 0$
0	$-3(0)^2 + 4(0) + 7 =$ 7
1	$-3(1)^2 + 4(1) + 7 =$ $-3 + 4 + 7 = 8$

2	$-3(2)^2 + 4(2) + 7 =$ $-3(4) + 8 + 7 =$ $-12 + 15 = 3$
3	$-3(3)^2 + 4(3) + 7 =$ $-3(9) + 12 + 7 =$ $-27 + 19 = -8$
4	$-3(4)^2 + 4(4) + 7 =$ $-3(16) + 16 + 7 =$ $-48 + 23 = -25$

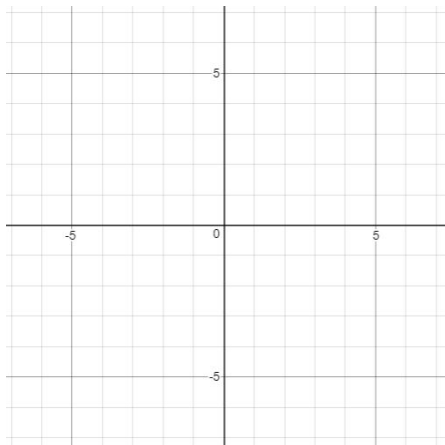
On a graph, the points look like this:



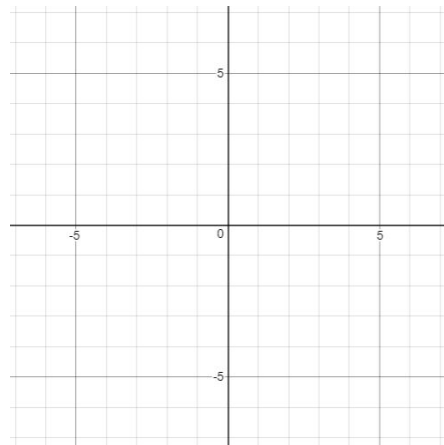
Chapter 5 Test

Graph the following Quadratic functions.

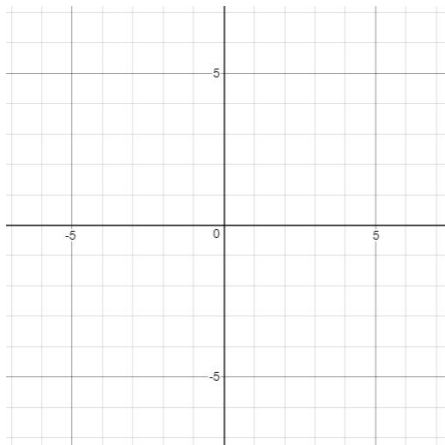
1. $f(x) = -x^2$



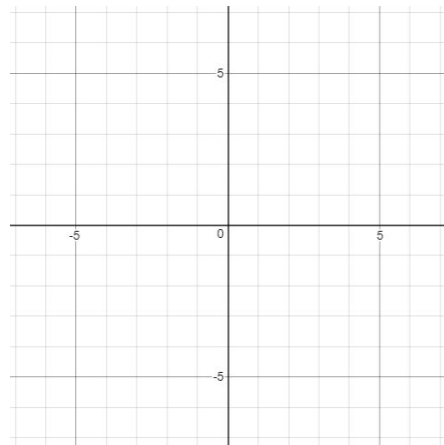
2. $f(x) = x^2 + 2x + 1$



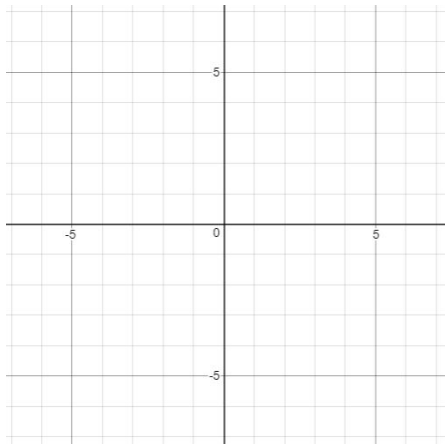
3. $f(x) = 4x^2$



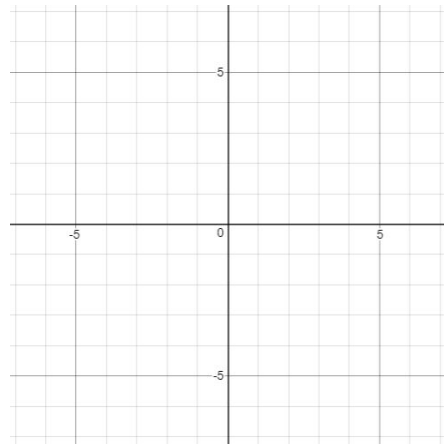
4. $f(x) = -x^2 + 3$



5. $f(x) = 4x^2 + 4x - 1$



6. $f(x) = -\frac{1}{2}x^2 + 4x + 6$



Chapter 6

The parts of a Quadratic Function and Factoring

You know what a Quadratic function is and how to graph it. This chapter teaches about the x-intercept, y-intercept, vertex, and line of symmetry of a quadratic function, as well as how to find them.

6-1 The y-intercept of a Quadratic Equation

The y-intercept of a quadratic equation is the same as the y-intercept of a linear function: it is simply the coordinate where it crosses the y-axis. When any function has an y-intercept, it is when $x = 0$. Therefore, to find the y -intercept of any function, you can plug in 0 for x. Knowing that the general form for a quadratic function is

$$f(x) = ax^2 + bx + c ,$$

Plugging in $x = 0$ gives

$$f(0) = a(0)^2 + b(0) + c = c .$$

Therefore, the y-intercept is just $(0, c)$, since the x-coordinate of the y-intercept is 0, and the y-value when you plug in $x=0$ is c , of a quadratic equation. For example, if you were asked to find the y-intercept of this function:

$$f(x) = 2x^2 - x + 4$$

It is just $(0, 4)$. A few more examples:

The y-intercept of $y = x^2 + 2$ is $(0, 2)$.

The y-intercept of $y = 10x^2$ is $(0, 0)$.

The y-intercept of $f(x) = -10x^2 - 2$ is $(0, -2)$.

The y-intercept of $f(x) = 3x^2 - 7x + 6$ is $(0, 6)$.

However, if the quadratic function is written in the form $f(x) = (ax + b)(cx + d)$, then you will first have to expand it. Doing this:

$$f(x) = (ax + b)(cx + d) = ax(cx) + ax(d) + b(cx) + bd = (ac)x^2 + (bc)x + bd$$

To find the y-intercept of $f(x) = (ax + b)(cx + d)$, we must plug in $x = 0$. Doing this gives us

$$f(x) = (ac)0^2 + bc(0) + bd = bd .$$

Therefore, the y-intercept when the equation is in factored form, as this is called, $(0, bd)$, where the y-value is the 2 constant terms multiplied together. Some examples:

The y-intercept of $y = (x + 3)(x + 2)$ is $(0, 2 \times 3)$ or $(0, 6)$.

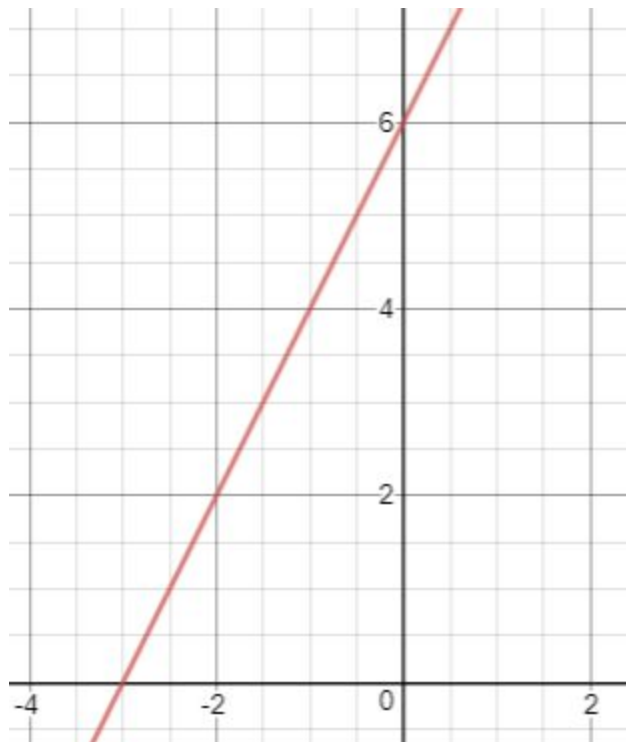
The y-intercept of $y = (2x + 2)(x - 5)$ is $(0, -10)$.

The y-intercept of $f(x) = (-4x + 3)(7x - 5)$ is $(0, -15)$.

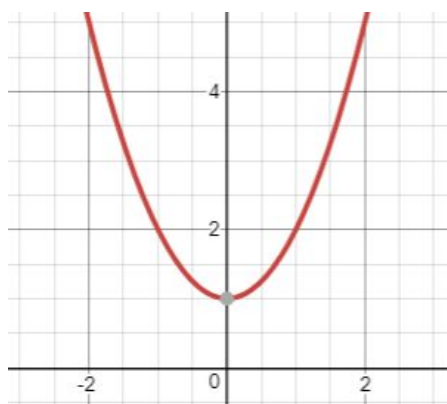
The y-intercept of $f(x) = (x - 1)^2$ is $(0, 1)$ since $(x - 1)^2 = (x - 1)(x - 1)$.

6-2 Finding the x-intercept of a Quadratic by Graphing

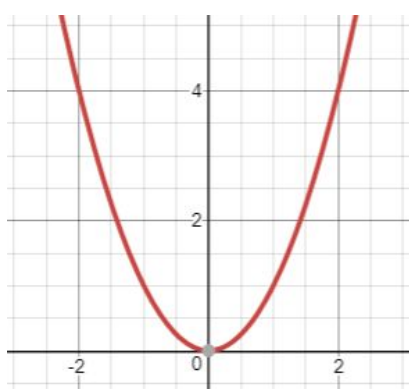
You have learned that the y-intercept of a function is the coordinate at which a function passes through the y-axis, when $x = 0$. The x-intercept of a function is/are the coordinate(s) at which it passes through the x-axis, when $y = 0$. Remember, a function cannot have more than one y-value for one x-value. That is why a function can only have at most one y-intercept. However, many functions can have more than one x-intercept. If we look at this linear function, $y = 2x + 6$,



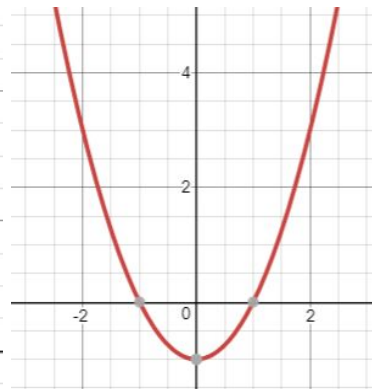
We can see that the y-intercept is (0, 6) and the x-intercept is (-3, 0). All linear functions will have exactly one x-intercept, but, quadratic equations can have 0, 1, or 2.



0 x-intercepts



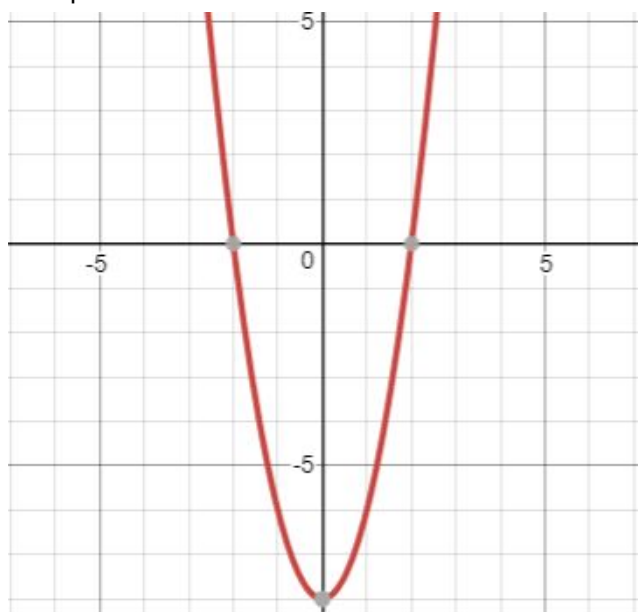
1 x-intercept



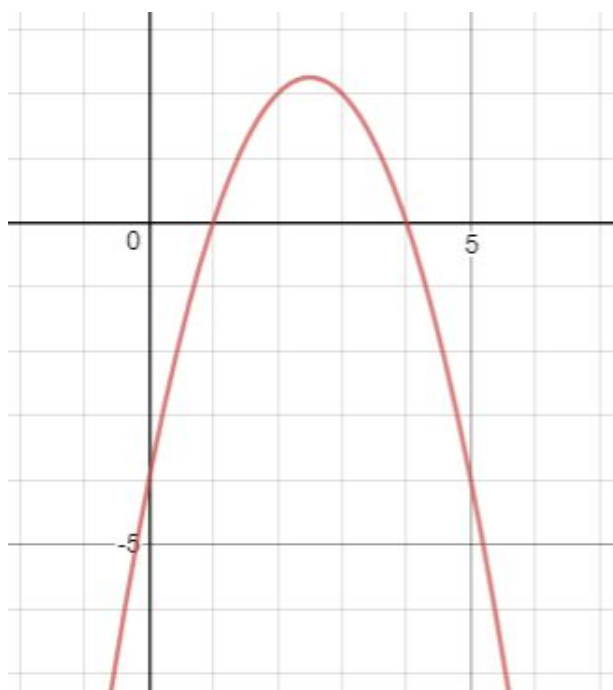
2 x-intercepts

If you are given a graph of a quadratic function and are told to find the x-intercepts, all you have to do is estimate where it crosses the x-axis. If it doesn't cross the x-intercepts, write that there are no x-intercepts.

Examples:

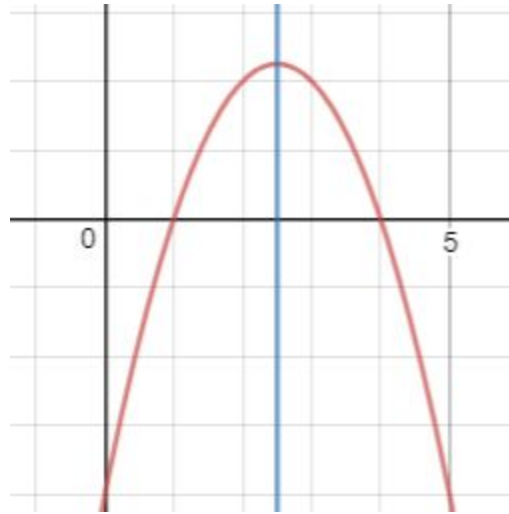


The x-intercepts are $(-2, 0)$ and $(2, 0)$.



The x-intercepts are $(1, 0)$ and $(4, 0)$.

If you look at any quadratic equation, you will see that it is symmetrical. All quadratics can be divided into 2 symmetrical parts with a line called the line of symmetry.



The blue line above, $x = \frac{5}{2}$, is the Line of symmetry for this quadratic equation. An important thing to notice is that in a quadratic, the x-intercepts are always the same distance away from the line of symmetry. This is useful because if you knew that the line of symmetry is $x = -2$ and that one of the roots is $(-3, 0)$, then you could figure out that the other root is $(-1, 0)$, since -3 and -1 are both 1 unit away from -2.

6-3 How to factor Quadratic functions when $a=1$

You know how to change a quadratic equation from factored form to standard form, and that process is called expanding. Doing the process in reverse, or changing a quadratic function such as $x^2 + 5x + 6$ into $(x + 2)(x + 3)$, is called factoring. In order to change

$$x^2 + bx + c \text{ (notice that } a=1\text{)}$$

to

$$(x + g)(x + h),$$

All you need to do is find 2 numbers that first multiply together to equal c and can also be added together to equal b . You then plug those 2 numbers, g and h , into $(x + g)(x + h)$. While this may look hard when shown with variables, with some practice this can be done fairly easily. For example, consider the quadratic

$$x^2 + 6x + 8.$$

To factor this, you need to find 2 numbers that add up to 6 and multiply to equal 8. The only 2 whole numbers that multiply to equal 8 is 2 and 4. Check that the 2 numbers also add up to 6, and all you have to do now is write 2 and 4 for g and h .

$$x^2 + 6x + 8 = (x + 2)(x + 4)$$

Given the quadratic $x^2 + 4x + 4$, you can figure out that 2 and 2 add up to 4 and multiply to 4, and rewrite it as $(x + 2)(x + 2)$.

The 2 numbers you are looking for can also be negative. To factor $x^2 - 7x + 10$, the 2 numbers need to add up to -7 and multiply to 10. Because the numbers add up to -7, at least one of the numbers must be negative. However, since the third term, 10, is positive, knowing that two negative numbers cannot be multiplied together to get a positive number, you can conclude that both numbers must be negative. The only 2 negative whole numbers that multiply to 10 are -2 and -5. Therefore, $x^2 - 7x + 10 = (x - 2)(x - 5)$.

A lot of the time, the third number can have many factors. For example, to factor $x^2 + 10x - 24$, you first need to find 2 numbers that multiply to -24, meaning one number is positive and the other is negative. 24 and -1, -24 and 1, 12 and -2, -12 and 2, -8 and 3, 8 and -3, -6 and 4, and 6 and -4 all equal -24 when multiplied together. To narrow down these options, look at b , 10. Since 10 is positive, you know that the term with the greater magnitude must be positive. That rules out half of the options. The next thing you do is guess and check. You can quickly figure out that $24 - 1$, $8 - 3$, and $6 - 4$ are all not 10. You will find that $12 - 2$ is 10, so 12 and -2 are your numbers, so $x^2 + 10x - 24 = (x + 12)(x - 2)$.

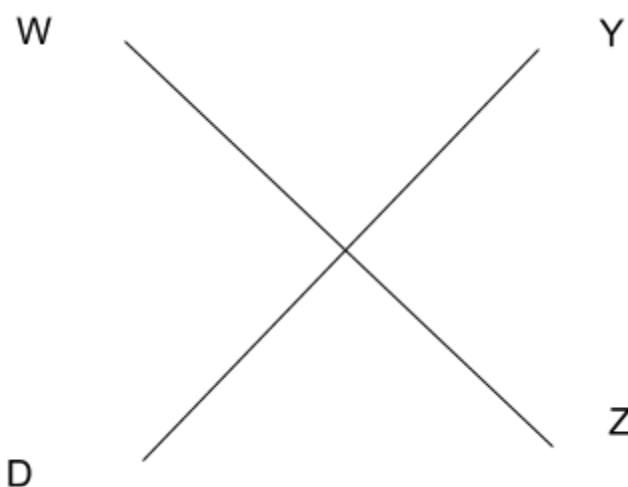
Finally, know that not all Quadratic functions can be factored with 2 whole numbers. You won't be able to factor $x^2 + 7x - 11$, since there aren't 2 whole numbers that multiply to -11 and add to 7.

6-4 How to factor Quadratic equations when $a \neq 1$

While factoring equations when $a = 1$ is relatively easy after some practice, factoring equations that $a \neq 1$ is much more difficult. If you are given a quadratic function

$$y = ax^2 + bx + c,$$

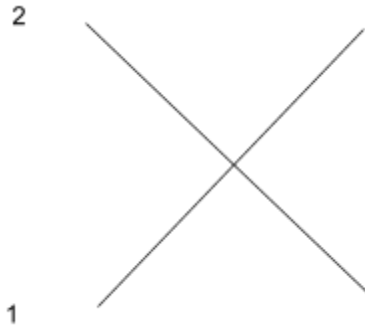
to factor it, you must find 4 numbers and arrange them like this.



$WD = a$, $YZ = c$, and $WZ + YD = b$ (that is why W and Z are connected and why D and Y are connected), and once you find all four numbers you write the factored form $(Wx + Y)(Dx + Z)$. To factor like this, you will have to guess and check frequently. To factor any quadratic as quickly as possible, first figure out if either a or c are prime numbers, meaning they have no factors except the number itself and 1. Some examples of prime numbers are 3, 5, 7, 11, and more. If you know that either one of these values are prime, then you can figure out 2 of the numbers above right away. For example, here is how you would factor the following.

$$2x^2 + 9x + 7$$

First, since you know that $a = 2$, $WD = 2$.



Next, since you know that $c = 7$, $YX = 7$. However, the following diagram would be INCORRECT:



Because $2(7) + 1(1) = 15$, which is NOT equal to b , which is actually 9. Instead, you can switch the 7 and the 1.



This is the correct diagram, because $2(1) = 2 = a$, $7(1) = 7 = c$, and $2(1) + 7(1) = 9 = b$. Now that you have the correct diagram, you just write the factored form as $(Wx + Y)(Dx + Z)$, which in this case would just be $(2x + 7)(x + 1)$. If you aren't sure you factored a quadratic correctly, just expand it (the opposite of factoring), and if you get the original equation you started with from doing this, you factored the quadratic correctly.

To make sure you understand how the diagram works, look at the following.

$$3x^2 - 23x - 8 = (3x + 1)(x - 8)$$


Because $3(1) = 3$, $1(-8) = -8$ and $3(-8) + 1(1) = -23$.

$$4x^2 + 4x + 1 = (2x + 1)(2x + 1)$$


Because $2(2) = 4$, $1(1) = 1$, and $2(1) + 2(1) = 4$.

$$6x^2 + 7x + 1 = (6x + 1)(x + 1)$$


Because $6(1) = 6$, $1(1) = 1$, and $6(1) + 1(1) = 7$.

Additionally, you can factor out a “common term” from a quadratic. To be able to factor out a common term, all terms in the quadratic must be divisible by the common term, meaning they can be divided by the common term and get a whole number. Here are some examples of factoring out a common term:

$$\begin{aligned} 2x^2 + 2x + 2 &= 2(x^2 + x + 1) \\ 3x^2 + 6x + 9 &= 3(x^2 + 2x + 3) \\ \frac{1}{2}x^2 - 4x + 8 &= \frac{1}{2}(x^2 - 8x + 16) = \frac{1}{2}(x - 4)^2 \\ -7x^2 - 7 &= -7(x^2 + 1) \\ -x^2 + 1 &= -(x^2 - 1) = -(x + 1)(x - 1) \\ x^2 + 5x &= x(x + 5) \\ 5x^2 + 5x &= 5x(x + 1) \\ 2x^2 + 3x &= x(2x + 3) \end{aligned}$$

If you distribute the answer, you will get what you started with. If you look at the 3rd or 5th example, the quadratic function was changed into a number multiplied by a quadratic in the form $x^2 + bx + c$, which is much easier to factor than a quadratic in the form $ax^2 + bx + c$.

6-5 How to find x-intercepts using factoring

You've probably wondered why you even need to learn how to factor. In short, factoring is useful because the factored version of any function can easily tell you that functions x-intercepts.

If a function is written in the form $(x - a)(x - b) = 0$, then its roots are a and b.

If a function is written in the form $(Ax - B)(Cx - D) = 0$, then its roots are $\frac{B}{A}$ and $\frac{D}{C}$.

Here is why. You may recall that for any x-intercept, $y = 0$. You can set $y = (x - a)(x - b)$ equal to 0 and get $(x - a)(x - b) = 0$. When any 2 things are multiplied together to equal 0, at least one of the of the 2 products must be 0. $(x - a)$ equals 0 when $x=a$, and $(x - b) = 0$ when $x=b$. Likewise, $(Ax - B)(Cx - D) = 0$ when $x = \frac{B}{A}$ or $x = \frac{D}{C}$. When you solve an equation for x so that it equals 0, the answers you get are often called roots or solutions of the quadratic.

Following these formulas, you can figure out:

The roots of $(x + 2)(x - 4) = 0$ are $x = -2$ and 4.

The root of $(x - 1)(x - 1) = 0$ is $x = 1$

The roots of $(x + 3)(x + 5) = 0$ are $x = -3$ and -5.

The roots of $(2x - 1)(x + 1) = 0$ are $x = 1/2$ and -1.

The roots of $(x + \frac{3}{2})(3x - 3) = 0$ are $x = -3/2$ and 1.

It's easy to find the roots of those quadratics because they are in factored form. However, if it's not in factored form and you need to find out what the x-intercepts are, then you can either graph the function or factor it and then find the roots. For example, to find out what the roots of $y = x^2 - 5x + 4$ are, first factor it. The 2 numbers that multiply to 4 and add up to -5 are -4 and -1, so the factored form is

$$y = (x - 4)(x - 1).$$

To find the x-intercepts, you must set y equal to 0 in order to find out the x-values that the y-value of the equation is equal to 0.

$$0 = (x - 4)(x - 1)$$

And finally, setting $x - 4 = 0$ and $x - 1 = 0$, you obtain $x = 1, 4$. Similarly, if you were given the equation $y = 2x^2 - 9x + 9$ and factored it to $(2x - 3)(x - 3)$ and set it equal to 0, then you could figure out that $x = 3/2, 3$.

6-6 The Quadratic Formula

Factoring is useful to find the x-intercepts of quadratic equations. But how can you find the x-intercepts of quadratic equations that can't be factored, such as $x^2 + 3x - 8$ or $6x^2 + 7$? There is a famous formula, called the Quadratic Formula, that can be used to find the roots of any Quadratic equation. It is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (\pm \text{ means plus or minus})$$

Where a, b, and c are the coefficients in $ax^2 + bx + c$. You must remember this formula because it is so important since it is used so often in algebra 1, algebra 2, and more difficult math. While you can use this formula to find the roots of any quadratic, you would generally use factoring if possible since it is faster. Let's try using it so you can understand how to use it.

Example 1: Find the roots of $x^2 + 5x - 7$.

Step 1: figure out that a=1, b=5, and c=7

Step 2: Plug in a=1, b=5, and c=7 into the Quadratic formula and simplify.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-5 \pm \sqrt{5^2 - 4(1)(-7)}}{2(1)} = \frac{-5 \pm \sqrt{25 + 28}}{2} = \frac{-5 \pm \sqrt{53}}{2}$$

And that's it! Notice that if you plug in something factorable such as $x^2 + 5x + 6$ into the quadratic formula, you will still get the same answer.

Example 2: Find the x-intercepts of $x^2 + 5x + 6$.

a=1, b=5, c=6:

$$x = \frac{-5 \pm \sqrt{5^2 - 4(1)(6)}}{2(1)} = \frac{-5 \pm \sqrt{1}}{2} = \frac{-5 \pm 1}{2} = -3, -2$$

(-3, 0) and (-2, 0)

Remember, solving for the roots tells you the x-intercepts, and quadratics can have 0, 1, or 2 x-intercepts. What happens if you put a quadratic into the quadratic formula that doesn't have x-intercepts, like $x^2 + 1$?

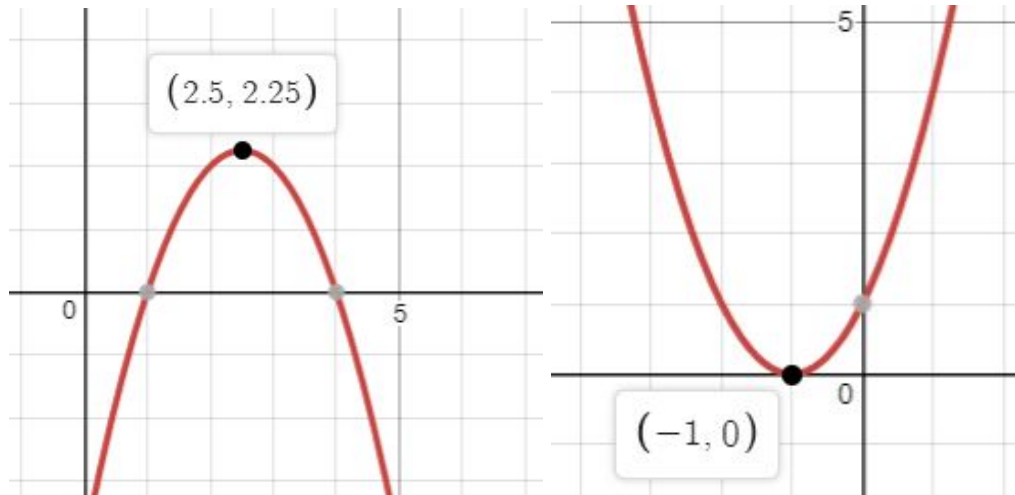
$$a=1, b=0, c=1$$

$$x = \frac{0 \pm \sqrt{0^2 - 4(1)(1)}}{2(1)} = \frac{\pm \sqrt{-4}}{2} = \frac{\pm \sqrt{-1} \sqrt{4}}{2} = \frac{\pm 2i}{2} = i, -i$$

If you ever get an imaginary or complex number (complex number being the sum of a real number and an imaginary number) after plugging the coefficients of a quadratic equation into the quadratic formula, then there are no x-intercepts and no real roots. DO NOT write the roots are -i and i. If you ever get imaginary roots, just write "No real roots" as your answer.

6-7 Finding the vertex of a quadratic equation

So far, you've learned about what the y-intercept and x-intercept(s) of a quadratic function are and how to find them. Another part of a quadratic function is the vertex. The vertex is the minimum value or the maximum value of a quadratic function.



So how do you find the vertex of a quadratic function? There are 2 ways. With the first method, you have to do something called “completing the square”. Basically, you have to change the quadratic equation to the form $y = d(x - h)^2 + k$, also known as vertex form. Once you do this, you can find out the vertex, which, no matter what d is, is always (h, k) . Here are the steps you have to follow, using the function $2x^2 + 4x + 1$, to find the vertex.

Step 1: Factor a out of the first 2 terms.

Work:

$$2(x^2 + 2x) + 1$$

Step 2: Once you have the equation in the form $A(x^2 + BX) + C$, take one half of the coefficient of the 2nd term in the parenthesis, B , square it (raise it to the 2nd power), add that number to BX inside the parenthesis and subtract AB from C .

Work:

Since $A=2$, $B=2$, and $C=1$:

One half of 2 is $\frac{2}{2} = 1$.

1^2 is 1.

Adding 1 to $2x$ inside the parenthesis and subtracting 2 times 1 from 1,

$$2(x^2 + 2x + 1) + 1 - 2(1) = 2(x^2 + 2x + 1) - 1$$

Step 3: Factor the terms inside the parenthesis, and write it as a square term

Work:

1 and 1 multiply to 1 and add up to 2. Therefore, $x^2 + 2x + 1 = (x + 1)(x + 1) = (x + 1)^2$

Since $x^2 + 2x + 1 = (x + 1)^2$, $2(x^2 + 2x + 1) - 1 = 2(x + 1)^2 - 1$

Step 4: Now that the function is in the form $y = d(x - h)^2 + k$, identify (h , k) as the vertex

Work:

Since h = -1 and k = -1 in $y = 2(x - 1)^2 - 1$, the vertex is (-1 , -1).

Here is another example of finding the vertex by completing the square, this time for $y = x^2 + 4x + 1$.

$x^2 + 4x + 1 = 1(x^2 + 4x) + 1 = 1(x^2 + 4x + (2^2)) + 1 - 2(1) = (x^2 + 4x + 4) + 1 = (x + 2)^2 - 3$, vertex is (-2 , -3).

If you are comfortable with completing the square, this method can be faster than the second method. However, if completing the square is too hard to remember, then use this second method. You may want a calculator.

In $ax^2 + bx + c$,

The vertex is $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$

For example, for the equation $x^2 + 6x + 8$, since a = 1 and b = 6, the vertex is $\left(-\frac{6}{2(1)}, f\left(-\frac{6}{2(1)}\right)\right)$, and since $-\frac{6}{2(1)}$ is -3, then the vertex is $(-3, f(-3))$. $f(-3)$ is $(-3)^2 + 6(-3) + 8 = -1$, so the vertex is $(-3, -1)$.

The highlighted formula is one you should definitely remember, especially if you forget how to complete the square or don't understand how to complete the square.

One last thing you should know is that the Line of Symmetry is the same as the x-coordinate of the vertex. The x-coordinate of the vertex, as shown above, is $-\frac{b}{2a}$, so the line of symmetry is $-\frac{b}{2a}$. Consider this problem:

The Line of Symmetry for the function $y = 2x^2 + 4x + 6$ is $x = -1$. What is the coordinate of the vertex?

Since the line of symmetry is the x-coordinate of the vertex, you can figure out that the vertex is $(-1, f(-1))$. $f(-1)$ is 4, so the vertex is $(-1, 4)$.

Finally, think about this question:

The vertex of a parabola is (2 , 5) and one x-intercept is (0 , 0). Find the other root.

The x-coordinate of the vertex is the line of symmetry, so the line of symmetry is $x=2$. Since the roots of a quadratic function are the same distance away from the line of symmetry, and the one root you're given is 2 units to the left of the line of symmetry, the other root must be the same distance, 2 units, but to the right of the line of symmetry. Therefore, the other x-intercept is (4 , 0).

Chapter 6 Test

1. What is the y-intercept of $f(x) = (x + 3)(x - 2)$?
2. How many x-intercepts does $y = 3x^2 + 2x + 10$ have?
3. a. What is the line of symmetry of $f(x) = -2x^2 + 4x + 2$?
b. What is the vertex of this function?
c. How many x-intercepts does this function have?
4. Factor $9x^2 + 3x$.
5. Factor $x^2 - 1$.
6. Factor $4x^2 + 14x + 10$.
7. Expand/Distribute $-(x + 2)(2x - 5)$ and thus find the y-intercept.
8. Solve for x. $(x - 4)(x - 4) = 0$.
9. Given that the roots for a quadratic equation are $(-1, 0)$ and $(3, 0)$, what is the line of symmetry?
10. What is the vertex of $(x - \frac{1}{2})^2 + \frac{5}{2} - 3$?
11. a. Find the vertex of $y = -6x^2 + x$.
b. Is the vertex of this function its maximum value or minimum value?
12. What is the y-intercept of $(ax + 1)(x - b)$?
13. What are the roots of $x^2 + 6x + 12 = 0$?
14. What are the roots of $(x + 3)(2x - 10) = 0$?
15. What are the roots of $(x - 3)^2 - 1$?
16. Rewrite $f(x) = 2(x^2 + 4x)$ in vertex form.
17. Factor $4x^2 - 1$.
18. Factor $-2x^2 - 5x - 2$.
19. Solve for x. $2x^2 - 2x - 13 = 0$.
20. Find the y-intercept, x-intercepts, line of symmetry, and vertex of $y = x^2 - 11x + 18$.

Chapter 7

Other Types of Functions and Transformations

This chapter shows how to graph exponential functions, square root functions, and absolute value functions.

7-1 Exponential Functions

Exponential functions are written in this form:

$$y = ab^{cx+d} + e$$

($a \neq 0$, $b \neq 0$ or 1 , $c \neq 0$, where a, b, c, d , and e are real numbers)

That means the following equations would be exponential functions:

$$y = 2^x$$

$$y = 3^{x-1}$$

$$y = 4^{2x}$$

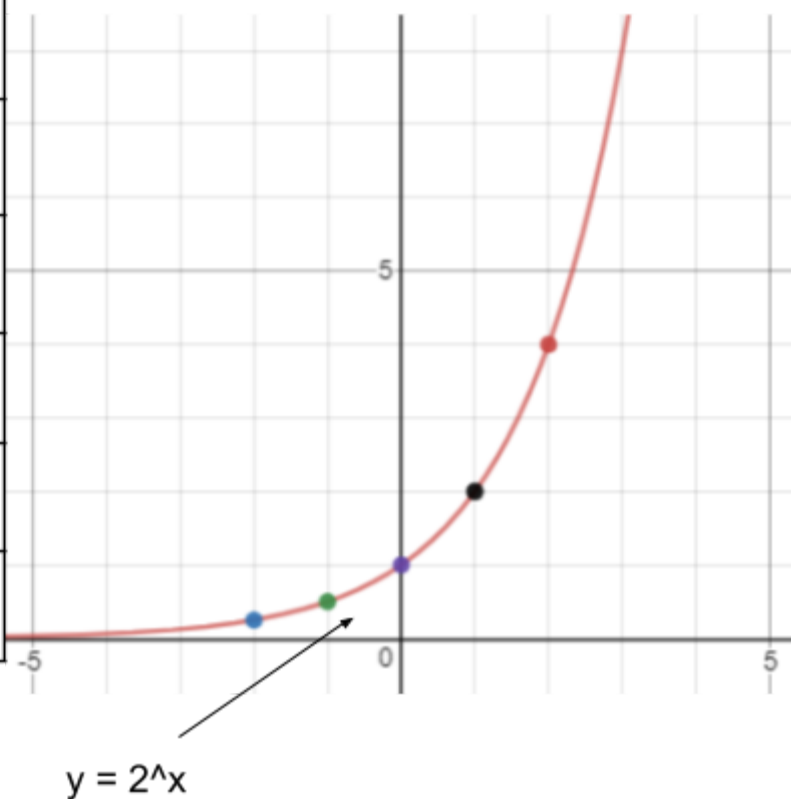
$$y = 5^x + 3$$

$$y = 6(6^x)$$

$$y = 2(3^{2x+4}) + 7$$

So what does an exponential function look like when graphed? To do that, we need to plug in values for x and get values for y . Here is the table and graph for $y = 2^x$.

x	y
-2	$2^{-2} = 1/(2^2) = 1/4$
-1	$2^{-1} = 1/(2^1) = 1/2$
0	$2^0 = 1$
1	$2^1 = 2$
2	$2^2 = 4$



As you can see, the line gets steeper and steeper. Here is the graph of $y = -3^x + 1$.

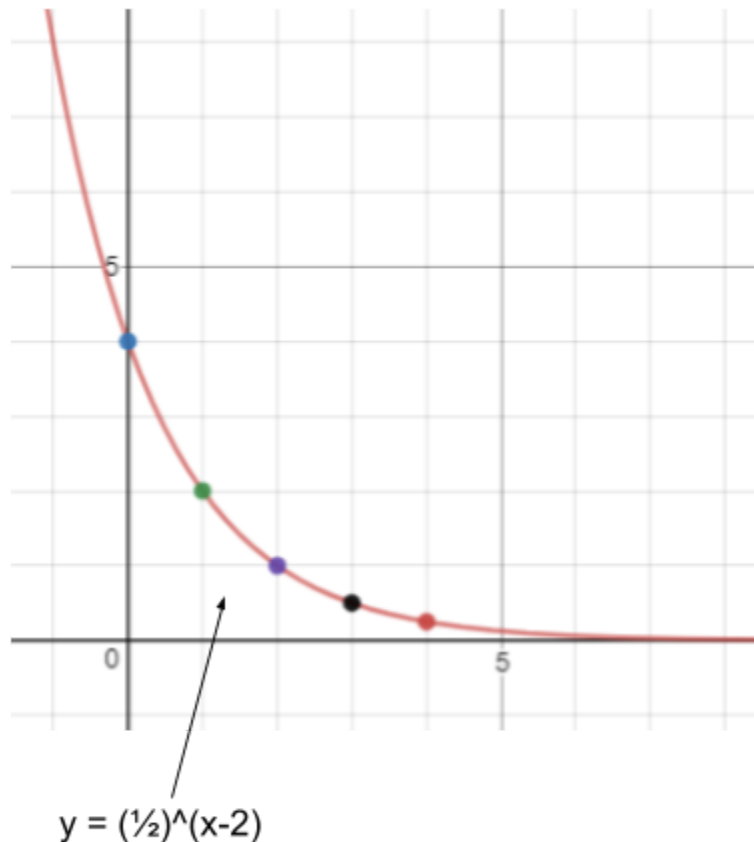
x	y
-2	$-3^{-2}+1 = -1/9 + 1 = 8/9$
-1	$-3^{-1}+1 = -1/3 + 1 = 2/3$
0	$-3^0+1 = -1+1=0$
1	$-3^1+1 = -3+1 = -2$
2	$-3^2+1 = -9+1 = -8$



$$y = -3^x + 1$$

This graph also gets steeper, but it goes down instead of going up. Here is an example of an exponential function that gets less steep, $y = (\frac{1}{2})^{x-2}$.

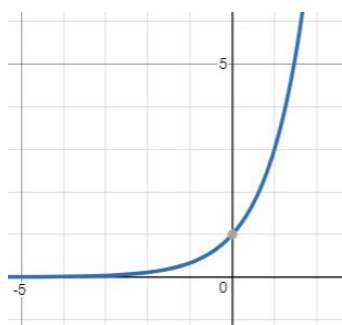
x	y
0	$(\frac{1}{2})^{(0-2)} =$ $(\frac{1}{2})^{(-2)} =$ $2^2 = 4$
1	$(\frac{1}{2})^{(0-1)} =$ $(\frac{1}{2})^{(-1)} =$ $2^1 = 2$
2	$(\frac{1}{2})^{(2-2)} =$ $\frac{1}{2}^0 = 1$
3	$(\frac{1}{2})^{(3-2)} =$ $(\frac{1}{2})^1 = \frac{1}{2}$
4	$(\frac{1}{2})^{(4-2)} =$ $(\frac{1}{2})^2 =$ $(\frac{1^2}{2^2}) =$ $\frac{1}{4}$



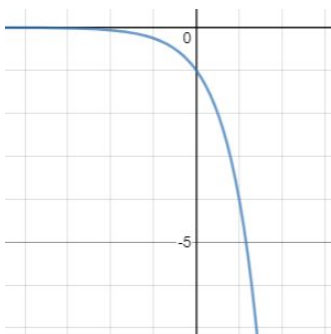
There are 2 types of exponential equations: growth and decay. If you have a function $y = ab^{cx+d} + e$:

- If $a > 1$ or $a < -1$, it is a growth function.
- If $0 < a < 1$ or $-1 < a < 0$, it is a decay function.

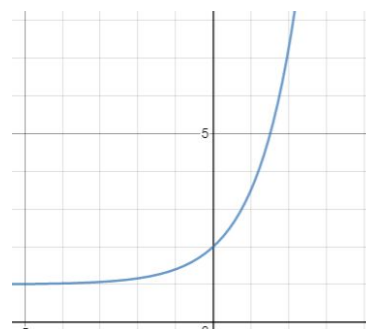
Just by looking at a graph of an exponential function, you can tell if it is a growth or decay function if it constantly gets steeper or if it constantly gets less steep. If it constantly gets steeper, then it is a growth function, while if the graph constantly gets less steep, it is a decay function. Here are some examples of growth functions.



$$y = 3^x$$

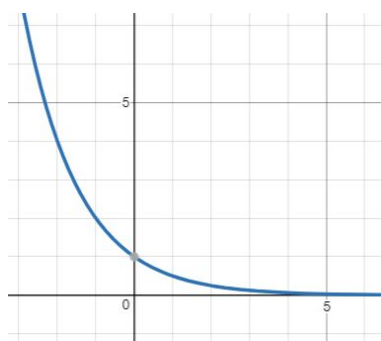


$$y = -4^x$$

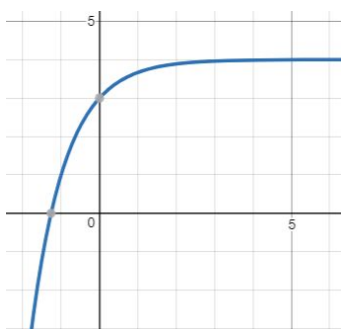


$$y = \left(\frac{5}{2}\right)^x + 1$$

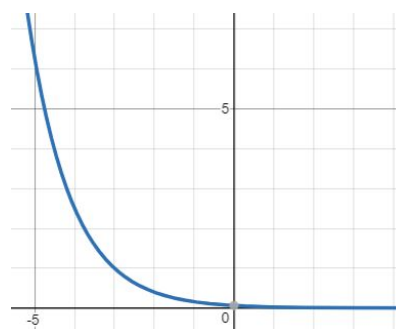
Here are some examples of decay functions.



$$y = \left(\frac{1}{2}\right)^x$$



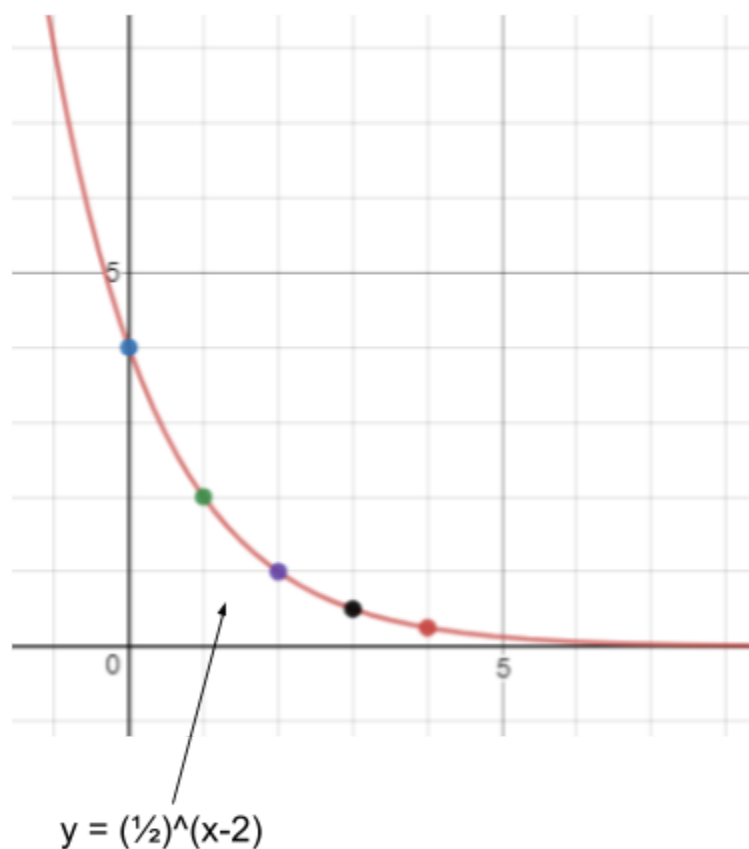
$$y = -\left(\frac{1}{3}\right)^x + 3$$



$$y = \left(\frac{2}{5}\right)^{x+3}$$

Graphing these functions are not hard, as long as you can remember the rules for exponents. Plug in a value for x , get a value back for y , and connect the coordinates you find with a curved line.

x	y
0	$(\frac{1}{2})^{(0-2)} =$ $(\frac{1}{2})^{(-2)} =$ $2^2 = 4$
1	$(\frac{1}{2})^{(0-1)} =$ $(\frac{1}{2})^{(-1)} =$ $2^1 = 2$
2	$(\frac{1}{2})^{(2-2)} =$ $\frac{1}{2}^0 = 1$
3	$(\frac{1}{2})^{(3-2)} =$ $(\frac{1}{2})^1 = \frac{1}{2}$
4	$(\frac{1}{2})^{(4-2)} =$ $(\frac{1}{2})^2 =$ $(\frac{1^2}{2^2}) =$ $\frac{1}{4}$



7-2 Square root functions and Domain and Range

Another type of function is a square root function. They are written in the form

$$y = a\sqrt{bx + c} + d \quad (a \neq 0, b \neq 0)$$

Some examples of square root functions include:

$$y = \sqrt{x}$$

$$y = 2\sqrt{x}$$

$$y = \sqrt{-3x}$$

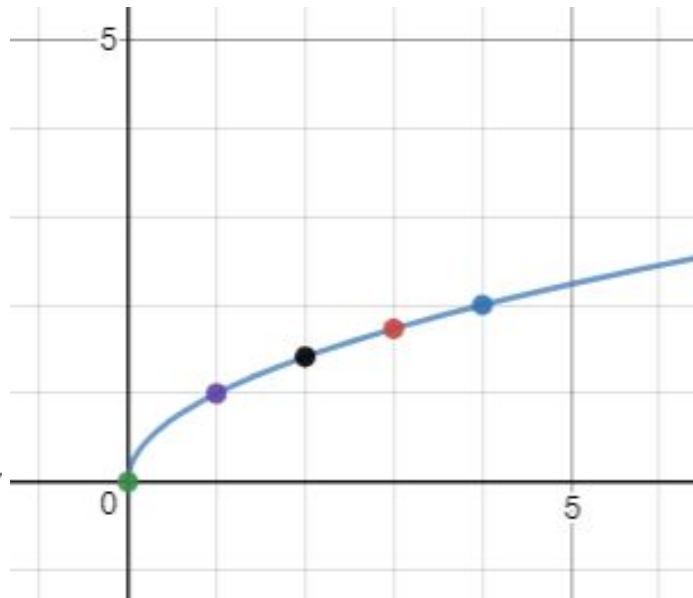
$$y = -\sqrt{2x} + 5$$

$$y = \sqrt{x-7}$$

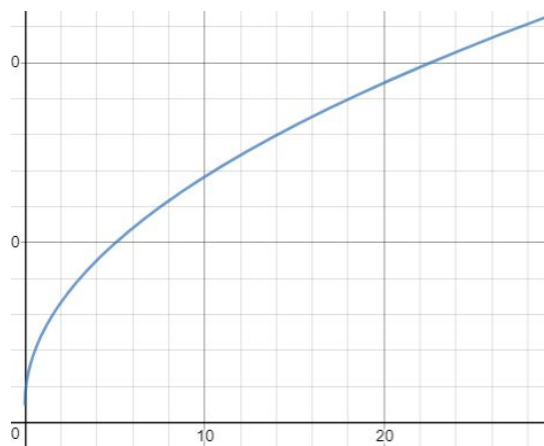
In words, when graphed, a square root function looks like half of a sideways parabola. For example, this is the graph of $y = \sqrt{x}$ (you may want to use a calculator to calculate the square roots to make this easier to plot and graph.)

x	y
0	$0^{1/2} = 0$
1	$1^{1/2} = 1$
2	$2^{1/2}$
3	$3^{1/2}$
4	$4^{1/2} = 2$

$$y = x^{1/2}$$



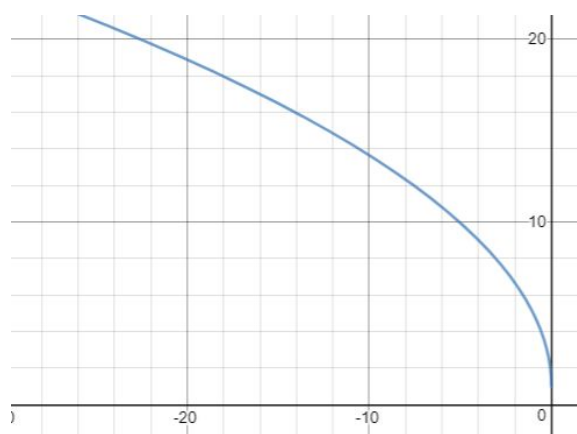
Notice that there are no values of y when $x < 0$. This is because if you plugged in a negative number for x , you'd get an imaginary number for y . Here are a few more square root functions and their graphs.



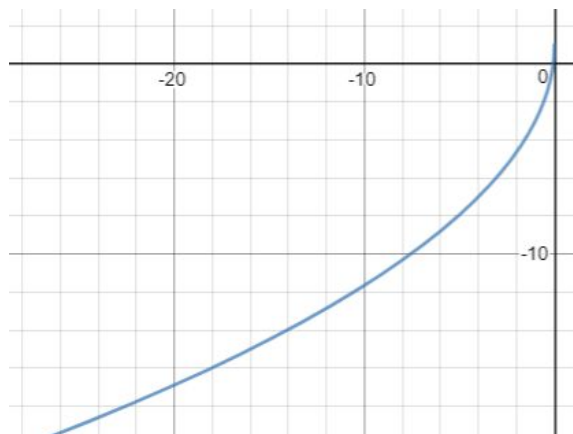
(1) $y = 2\sqrt{4x} + 1$.



(2) $y = -2\sqrt{4x} + 1$



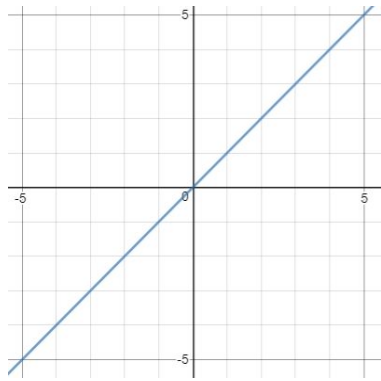
(3) $y = 2\sqrt{-4x} + 1$



(4) $y = -2\sqrt{-4x} + 1$

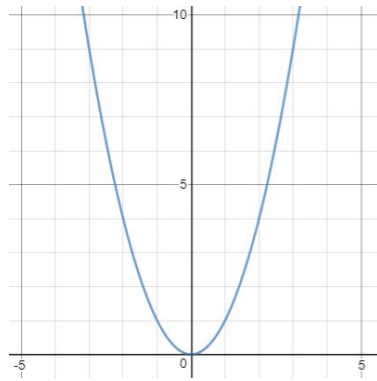
Often, when you graph square root functions, you will be asked to give the domain and range of the function. The domain of a function states the range of x-values of the function, and the range states the range of y-values of the function.

You will find that every single square root function has a restricted domain and restricted range. In graphs (1) and (2) above, the domain is $x \geq 0$. In graphs (3) and (4), the domain is $x \leq 0$. In graphs (1) and (3), the range is $y \geq 1$, and in graphs (2) and (4) above the range is $y \leq 1$. Here are some more examples of domain and range.



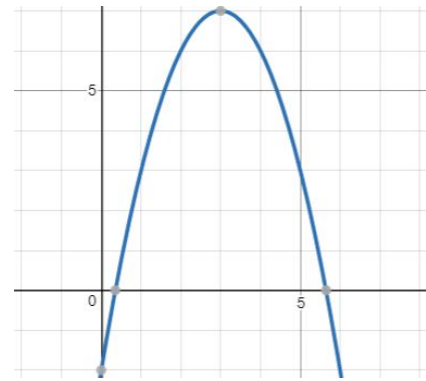
$$y = x$$

Domain: All real numbers
Range: All real numbers



$$y = x^2$$

Domain: All real numbers
Range: $y \geq 0$



$$y = (x - 3)^2 + 7$$

Domain: All real numbers
Range: $y \leq 7$

Remember, you can't graph imaginary numbers, so when graphing square root functions, plug in x-values that you think will give you real number y-values.

7-3 Cubic Functions

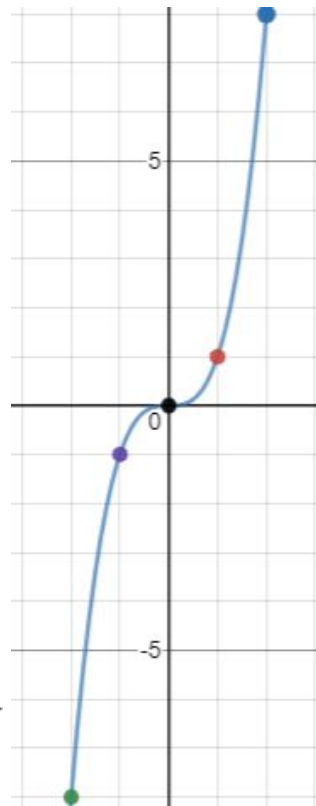
Previously, you have graphed functions such as $y = x$ and $y = x^2$. What do you think $y = x^3$ would look like if it was graphed? It is actually quite different than the other graphs we have looked at before. First, a function is a cubic function if it is written in this form:

$$f(x) = ax^3 + bx^2 + cx + d \quad (a \neq 0)$$

Here is the most basic cubic function, $f(x) = x^3$. Remember that when you raise a number a to the third power, it is the same as $a \times a \times a$.

x	y
-2	-8
-1	-1
0	0
1	1
2	8

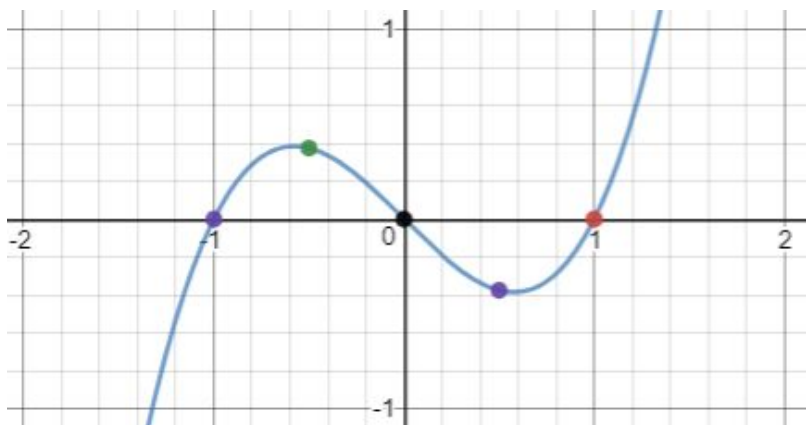
$$f(x) = x^3$$



At first, it gets less steep, then changes direction and becomes steeper. However, not all cubic functions are like this, and the only real way to know what a cubic function looks like is to graph it. This is the graph of the cubic function $f(x) = x(x + 1)(x - 1)$.

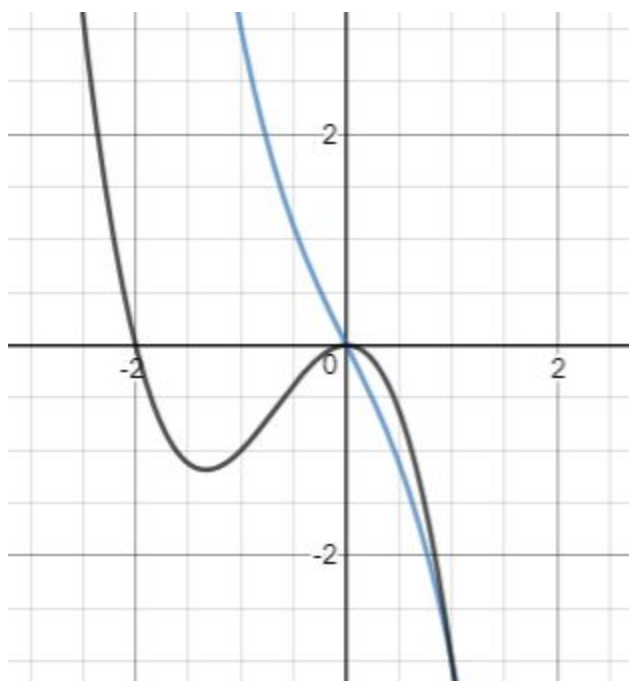
x	f(x)
-1	0
-1/2	3/8
0	0
1/2	-3/8
1	0

$$f(x) = x(x+1)(x-1)$$



These 2 functions are also cubic, even though they look a bit different, because the degree of the function, or the highest power that the variable is raised to, is 3.

— is $y = -2x^3 - 2x$
 — is $y = -2x^3 - 2x^2$



As you will quickly realize, if you haven't already, cubic functions can look very different from one another. There are few patterns though there are 2 you should know:

- If the equation is written in the form $y = ax^3 + bx^2 + cx + d$, then d is the y-intercept.
- If you are given the cubic function in the form $y = (x - a)(x - b)(x - c)$, you can determine that the roots are $x = a, b, c$. and thus know that it passes through $(a, 0)$, $(b, 0)$, $(c, 0)$.

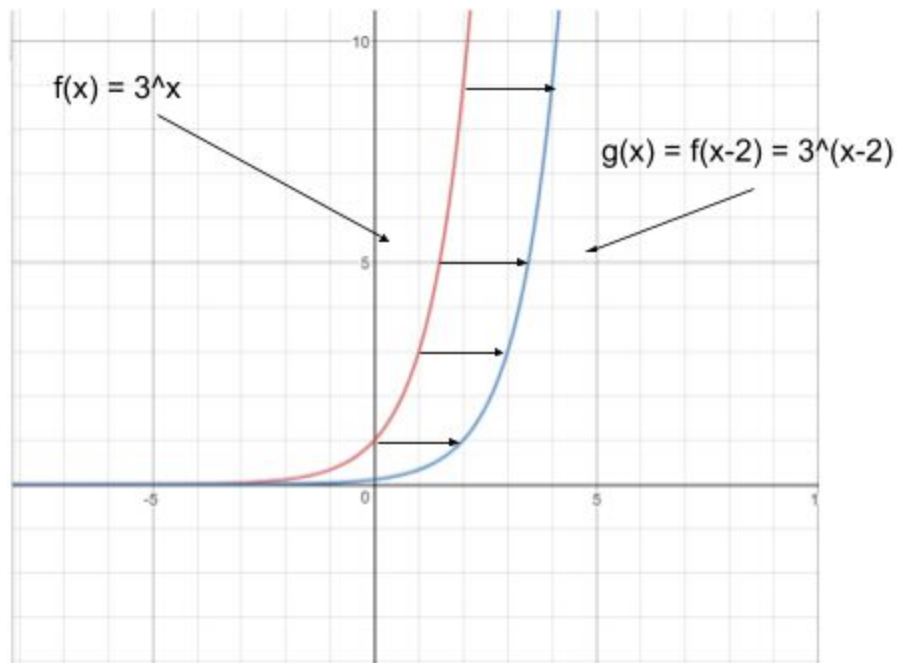
In conclusion, do what we have always done when graphing functions: plug in an x-value, get back a y value, and do that enough times until you know enough coordinates the function passes through to be able to draw a line through them.

7-4 Transformations of Functions

This section goes all the different types of transformations for functions.

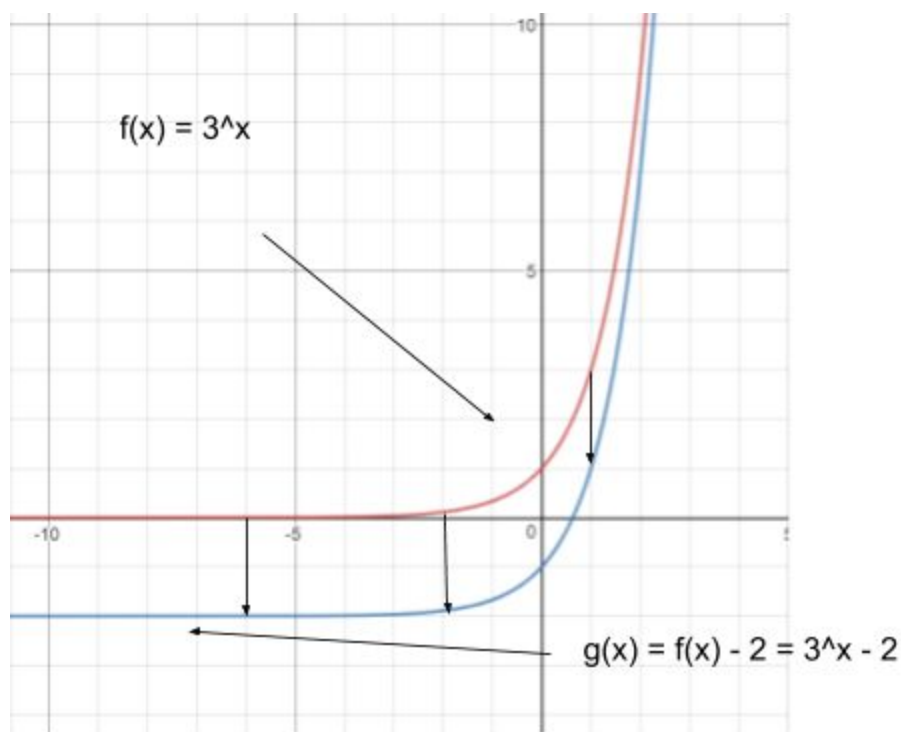
1. Translations (slides/ shifts)

For any function, which can be written as $f(x)$, The function $f(x - a) + b$ is translated a units right and b units up from the original function $f(x)$ for any function. Shifting a negative number of units right is the same as shifting the function a positive number of units left, and shifting a negative number of units up is the same as shifting a positive number of units down. This can also be written as $f(x + a) - b$ is shifted a units left and b units down from the Original function. $f(x + a)$ means you are substituting $x + a$ into the function $f(x)$ for every value of x . Here are some examples.



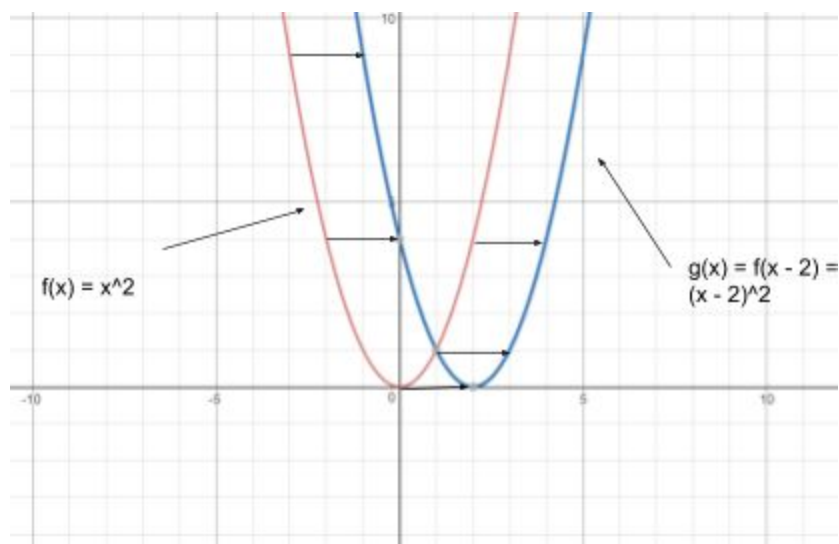
$g(x)$ is the same thing as $f(x - 2)$. Notice that 3^{x-2} is the same thing as 3^x except there is an $x - 2$ instead of an x . Also notice that when the 2 graphs have the same y value, they are exactly 2 units apart. $g(x)$ is $f(x)$ shifted right 2 units.

We can also shift $f(x)$ down 2 units:

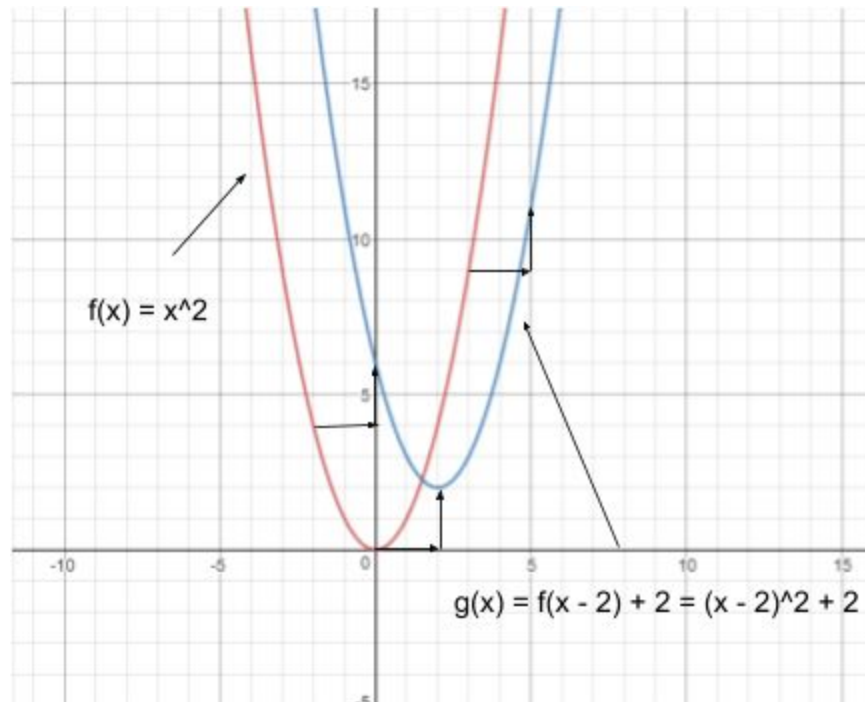


This is pretty similar to the other graph, except $g(x)$ is shifted down 2 units instead of right 2 units. Again, notice that when the x value of these 2 functions are the same, the functions are exactly 2 units apart.

Since this same thing works for any function, this would also work for a quadratic.



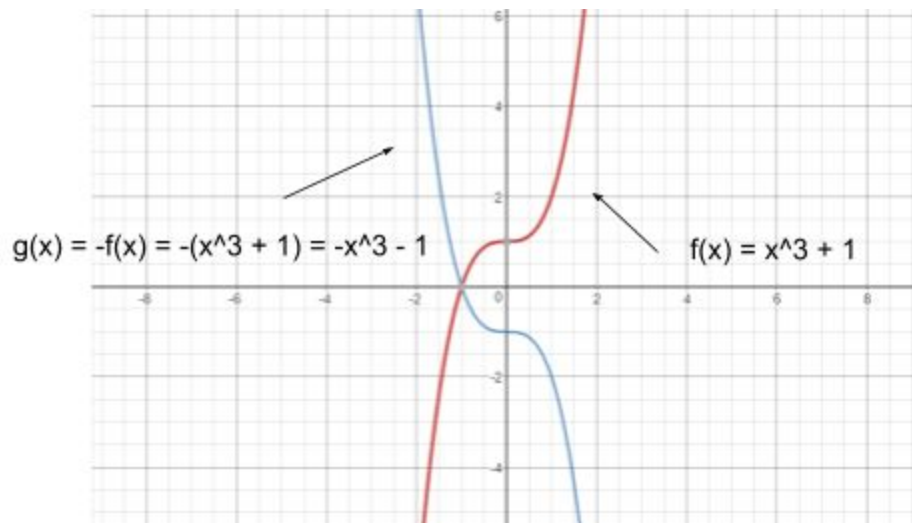
Each point of $f(x - 2)$ is shifted 2 units to the right of $f(x)$. You can shift functions in 2 directions at once as well. Look at the following:



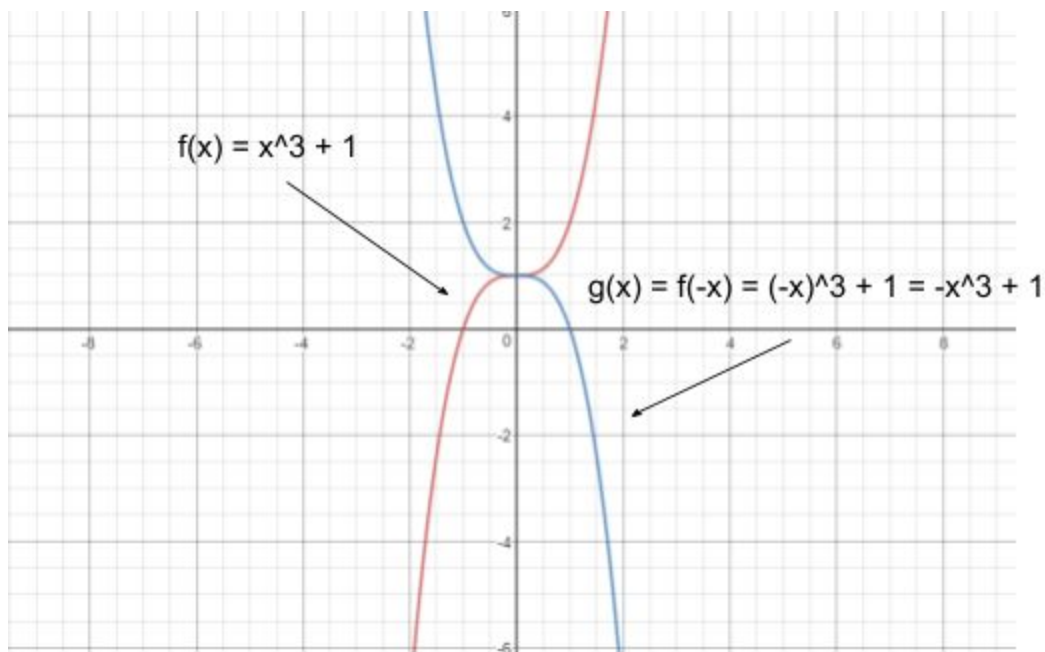
In this function, $f(x - 2) + 2$ is shifted 2 units up and 2 units right. Since $f(x)$ passes through $(1, 1)$, $g(x)$ passes through $(1 + 2, 1 + 2)$, or $(3, 3)$. You might have noticed that $g(x)$ is in vertex form. In general, if $f(x)$ passes through $(p, f(p))$, then $f(x - a) + b$ passes through $(p + a, f(p) + b)$ because each point is shifted right a units and up b units. If you would like to play around with this more, you can use this link to desmos: <https://www.desmos.com/calculator/7eu8kn70vl>. Just use the slider to shift the function. In this graph, a quadratic is used, but you could use any function and this would work.

2. Reflections

In addition to functions being shifted, they can also be reflected across lines. Firstly, $-f(x)$ is $f(x)$ reflected across the x axis. This is because $f(x) \cdot -1 = -f(x)$, which means that every value of the function will be multiplied by negative 1. By doing this, every value is reflected across the x axis.

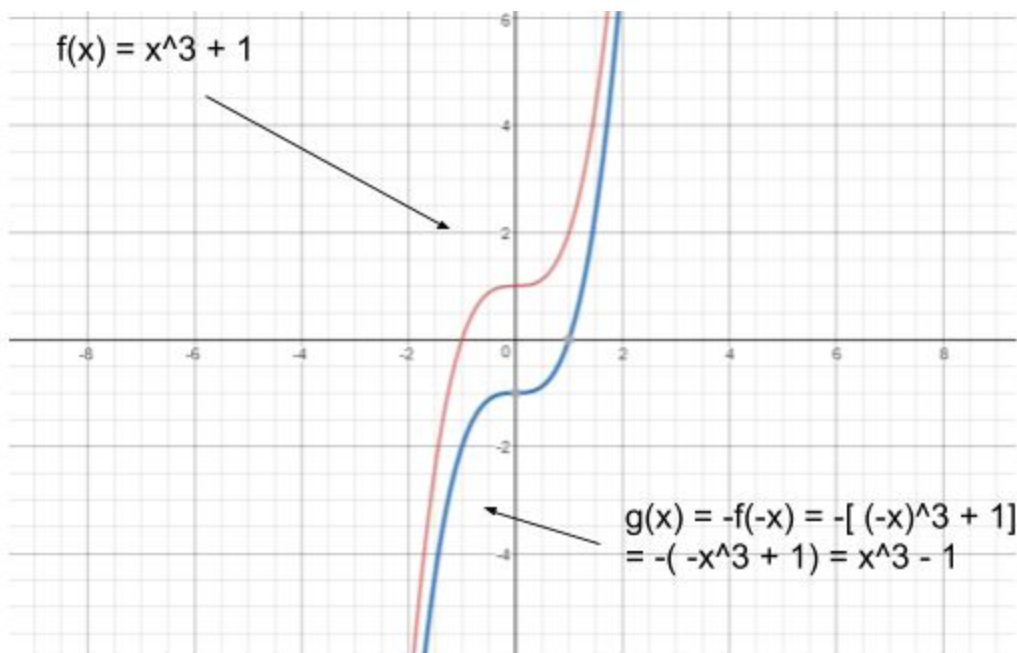


$g(x)$ is $-f(x)$. Notice that if $f(x)$ passes through point $(p, f(p))$, then $g(x)$ passes through $(p, -f(p))$. On the other hand, the function $f(-x)$ is $f(x)$ reflected across the y axis.



$-x$ was simply substituted into the function $f(x)$, similar to how $f(2)$ would substitute 2 into the function $f(x)$. $g(x)$ is this time $f(x)$ reflected across the y axis.

Similar to how functions can be translated/shifted both up or down and left or right, they can also be reflected across the x and y axes. In this case, you would write $g(x)$ equal to $-f(-x)$.



If you don't see how this is the the function reflected from both axes, you can look at the graph before this one and simply reflect it across the x axis, since you will get this exact graph.

When a function is reflected across both axes, if $f(x)$ passes through $(p, f(p))$, then $-f(-x)$ passes through $(-p, -f(p))$.

Finally, you can also reflect functions across the line $y = x$. The function you get back, $f^{-1}(x)$, is called the inverse function. To find the inverse function, follow these 4 steps:

- 1) Replace $f(x)$ with y .
- 2) Switch the x and the y . Replace any x 's in the function with a y and replace any y 's in the function with an x .
- 3) Isolate the y term by itself, so that the equation is solved for y in terms of x .
- 4) Replace y with $f^{-1}(x)$.

Following these steps, you can find the inverse of any one-to-one function. Remember, a function is an equation which only has one element of the range for every element of the domain- in other words, an equation which doesn't have 2 or more y values for one x value. A one-to-one function means that not only is there only one element of the range for every value of the range, but there is also only one element of the domain for every value of the range, which means that there can't be 2 x values with one y value. Hence, $f(x) = x^2$ is not a one-to-one function, even though it is a function, and therefore its inverse will not be a function.

An example of a function you can find the inverse of is $f(x) = x^3 + 1$. Using the steps:

- 1) Changing $f(x)$ to y : $y = x^3 + 1$
- 2) Switching the x and y : $x = y^3 + 1$
- 3) Solving the equation for y :
 - a) $y^3 = x - 1$
 - b) Taking the cube root of both sides, $y = \sqrt[3]{x - 1}$

4) Changing y to $f^{-1}(x)$: $f^{-1}(x) = \sqrt[3]{x-1}$

The graph of these functions would look like this:

The line $y = x$ is shown here so you can see that $f^{-1}(x)$ is $f(x)$ reflected across the y axis clearly. Notice that if $f(x)$ passes through (p, q) , then $f^{-1}(x)$ passes through (q, p) .

If you would like to see the reflections of different functions, use this link.

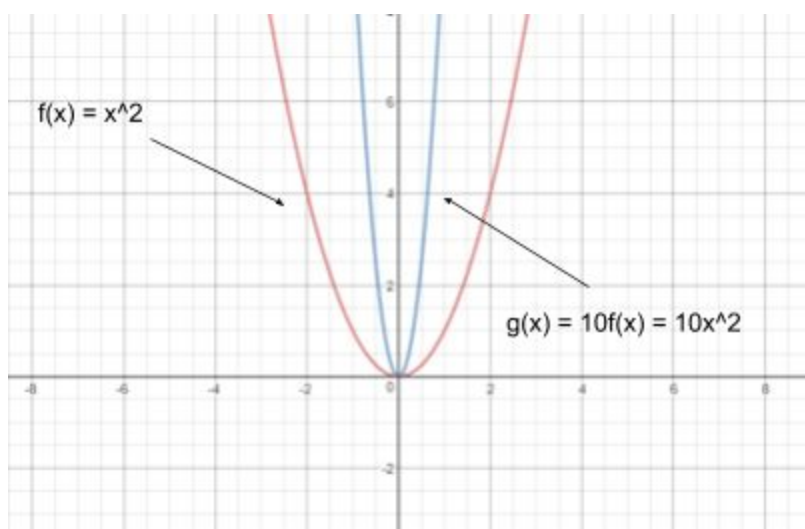
<https://www.desmos.com/calculator/eoll16teio>. Try putting in different functions to get interesting results.

3. Dilations (stretches and compressions)

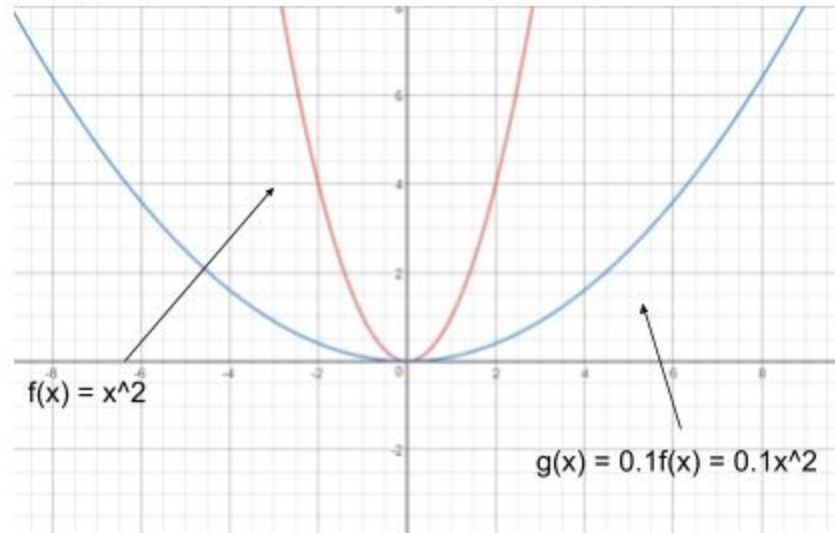
There is one final way which functions can be transformed. When you dilate a function vertically, you multiply the whole function (each term) by a constant positive real number. You cannot multiply the function by 0. If the original function is $f(x)$, then the dilated function would be $df(x)$, where d is a real nonzero number that $f(x)$ is being multiplied by. There are a few important things to know about vertical Dilations:

- The function cannot be multiplied by 0, and multiplying it by 1 will keep it the same.
- If the function $f(x)$ is being multiplied by a number greater than 1, the function will compress.
- If the function $f(x)$ is being multiplied by a number greater than 0 but less than 1, the function will stretch.
- If the function has relative maxima or minima, it will be at the same x values.

When you reflect a function across the x axis, you multiply the function by -1. With dilations, however, you can multiply the function by almost any number you want. The best way to understand what Dilations do is to graph these functions, especially with familiar functions such as $f(x) = x^2$.



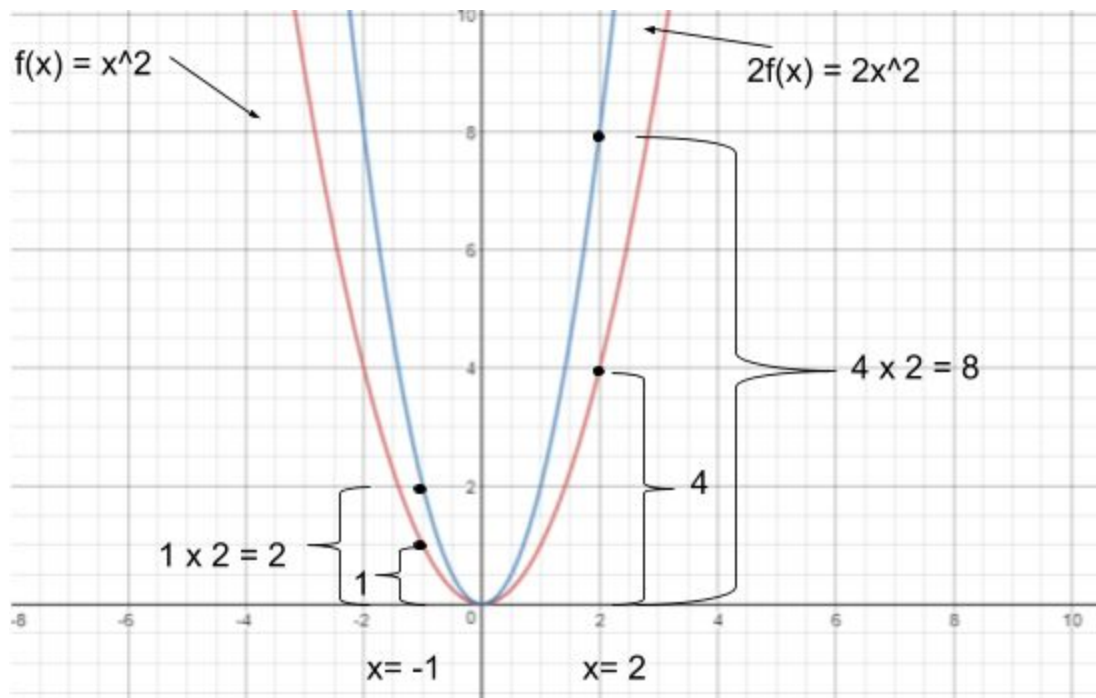
Since the function was multiplied by a number greater than 1, the resulting function got compressed, and looks skinnier than the first function.



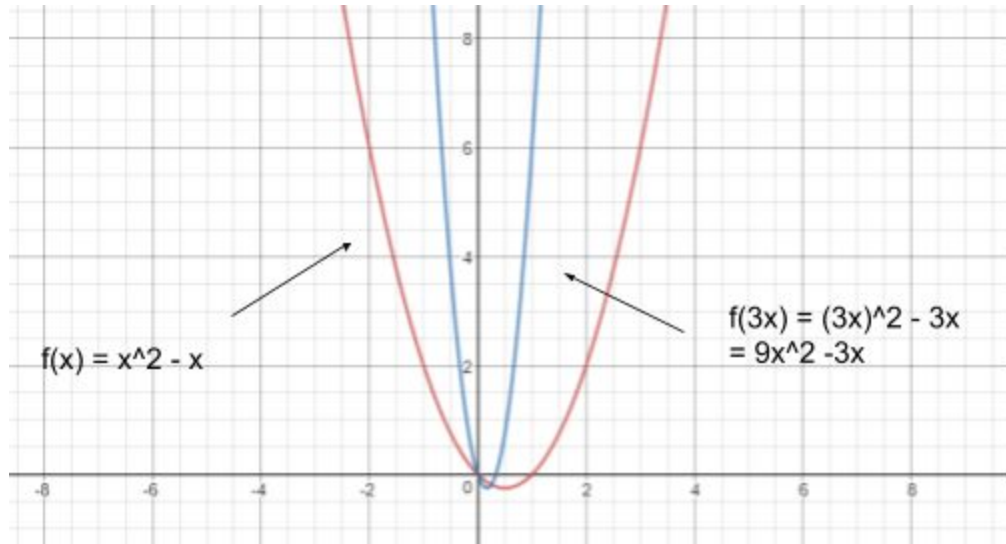
This function was multiplied by a number that was less than 1, so it is stretched, and looks wider than the original function. In general, if $f(x)$ is the original function, and the transformed function is $df(x)$, where d is the scale factor:

- a) When $d > 1$, the function is stretched vertically.
- b) When $0 < d < 1$, the function is compressed vertically.

Interestingly enough, if a function has a scale factor d , the y of the transformed function is the y -value of the original function times the d at the same x value.

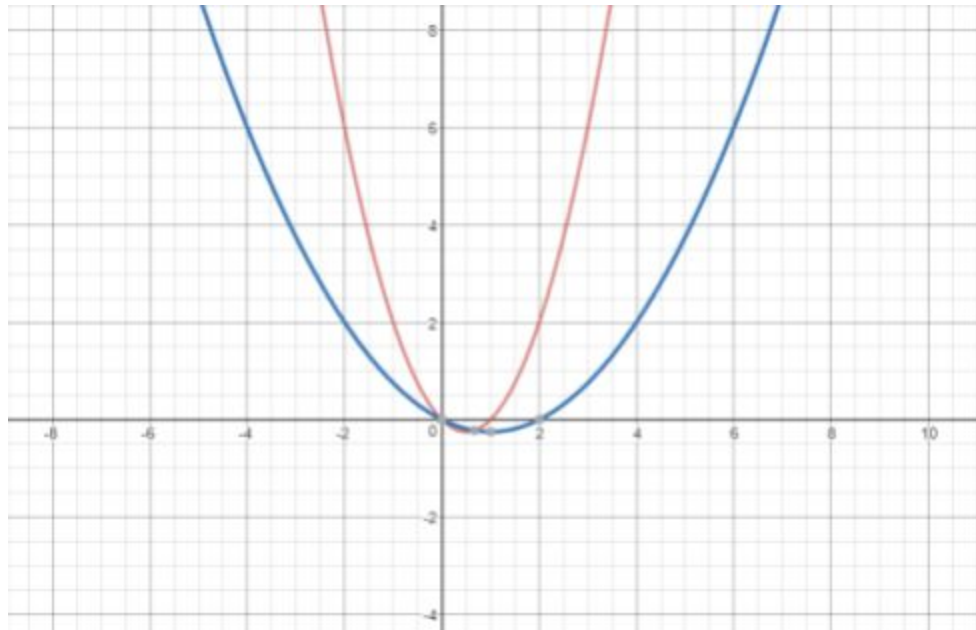


In addition, functions can be stretched or compressed horizontally, meaning that the function be changed to become closer to the x-axis or farther from it. If $f(x)$ is the original function, $f(hx)$ is the equation of the horizontally dilated function. If we used the function $f(x) = x^2 - x$ as our original function, then $f(3x) = (3x)^2 - 3x = 9x^2 - 3x$.



The function $g(x)$ is $f(x)$ horizontally compressed, since it is skinnier, but still has the same minimum value.

If we plugged in $0.5x$ instead of $3x$, this is what the function would look like.

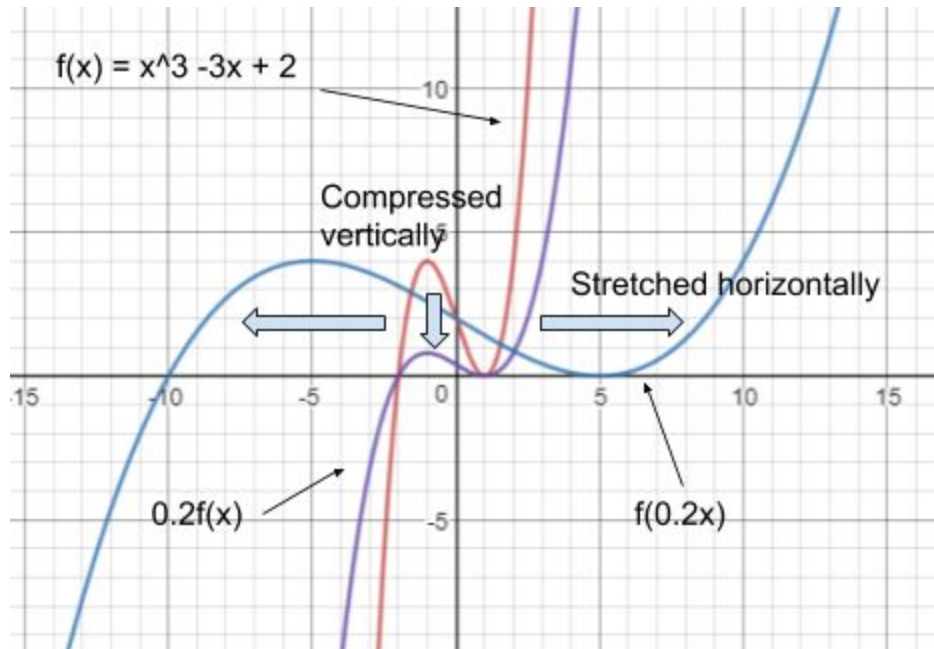


$g(x)$ is a horizontally stretched version from $f(x)$. Also, the y value of the relative minima and maxima will remain the same, though not at the same x values. In general, if $f(hx)$ is a dilated version of $f(x)$, and $\frac{1}{h}$ is the scale factor:

- a) If $0 < h < 1$, the function is stretched horizontally.

b) If $h > 1$, the function is compressed horizontally.

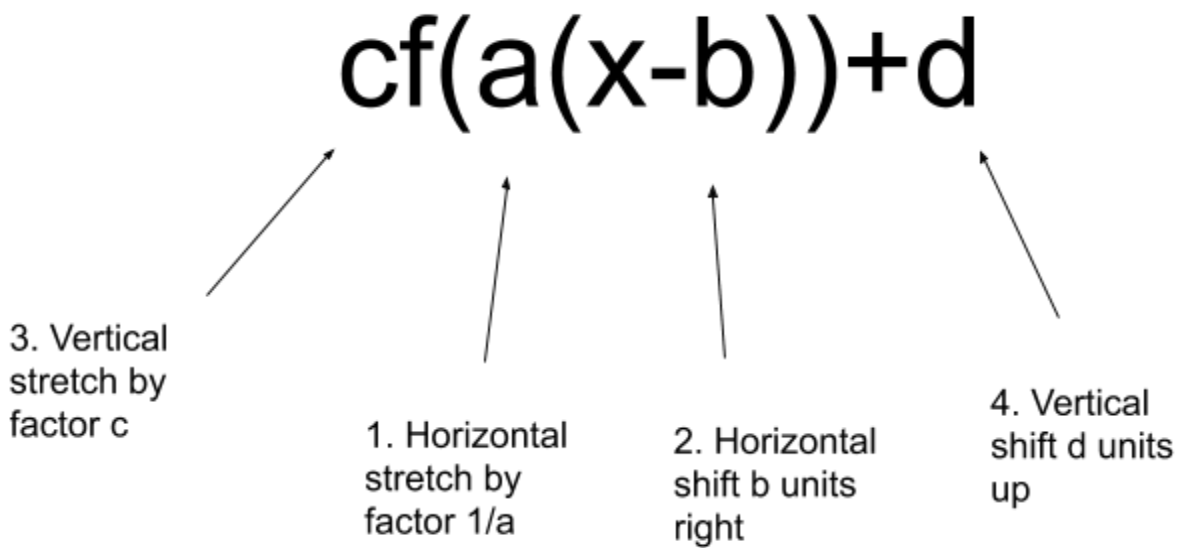
To make the horizontal and vertical dilation differences clear, look at the following dilations of the function $f(x) = x^3 - 3x + 2$.



If you would like to see the dilations of different functions, use this link: <https://www.desmos.com/calculator/gr3dfzo5jc>. Just change the function and drag the slider to see how dilation transformations work.

4. Conclusion

In general, functions can be transformed from $f(x)$ to the following:

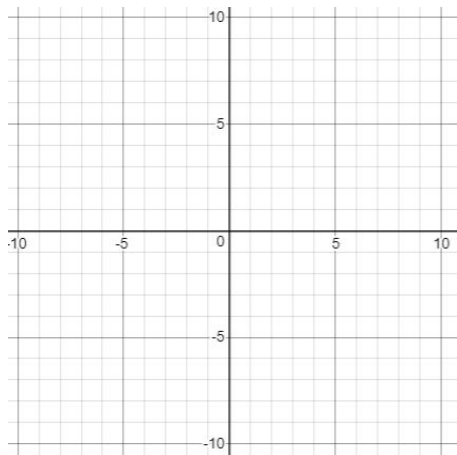


The order of the transformations matters. Make sure you do it correctly.

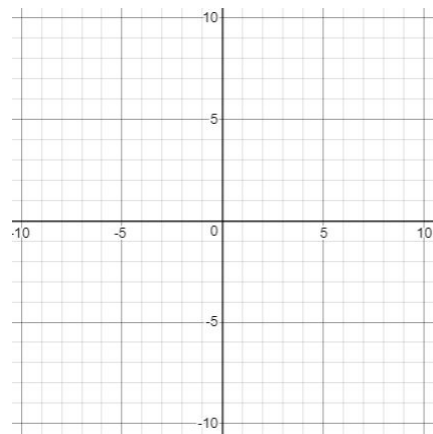
That is all there is to know about transformations. If you want to play around with every type of transformation more, use this link: <https://www.desmos.com/calculator/ljtdwzqjtz> You can use the sliders to see how the different transformations work, as well as change the function or remove the negative signs. Also playing the animation may help- just click the play button to the left of the sliders.

Chapter 7 test

1. Graph $y = 3\left(\frac{1}{2}\right)^{x-5}$.

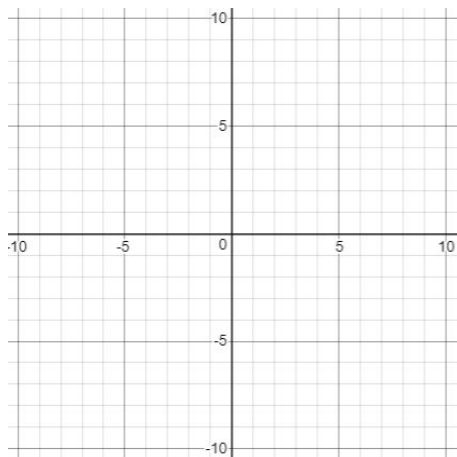


2. Graph $y = -\sqrt{4x} + 6$

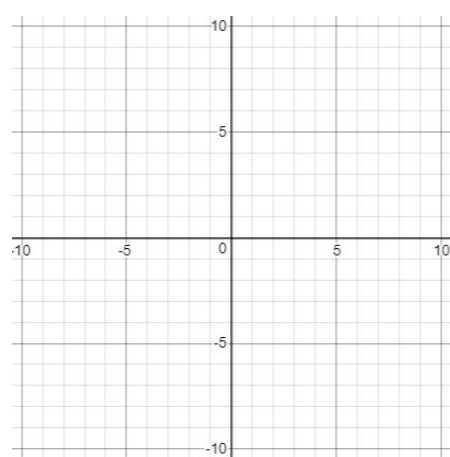


3. What is the domain and range of $y = \sqrt{2x+6}$?

4. Graph $y = 2(2x-2)(x+3)(x-3)$.



5. Graph $f^{-1}(x)$ if $f(x) = 2x + 3$.



6. If $g(x) = 2x^2 + 3$, for what value of a would $g(x-a) = 2(x+2)^2 + 3$?

7. How many units is $f(x+2) - 10$ translated along the x-axis and the y-axis from $f(x)$?

8. Graph $y = \sqrt{2^x}$ (remember the rules of exponents and radicals!)

